

Hahn–Tate Uniformization at Arbitrary Valuation Rank

Exact summability domains, integral node charts,
surcomplex elliptic curves, torsion ramification,
and multiscale theta series

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merged from three manuscripts

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Abstract

Let $K = \mathbb{C}((t^\Gamma))$ be a Hahn field, with no restriction on the rank or divisibility of the set-sized ordered abelian group Γ . For $q \in K$ of positive valuation α , we identify the exact domain on which the classical bilateral theta and Tate coordinate families are strongly Hahn-summable: it is the multiplicative subgroup U_q whose valuations belong to the principal convex subgroup generated by α . Outside this domain, leading exponents themselves contain a strictly decreasing sequence. Nevertheless, the resulting map gives an isomorphism $U_q/q^{\mathbb{Z}} \simeq E_q(K)$ onto *all* rational points of the Tate elliptic curve. Two independent proofs are given. The first combines a rank-one coarsening inside the principal convex subgroup with unrestricted formal evaluation at infinitesimals for a second, coarser valuation. The second uses no rank-one completion: an integral formal chart at the node, in which the smoothing becomes $UV = q$, together with a smooth-residue chart and a chart at infinity, inverts the parametrization explicitly, over any coefficient field of characteristic zero. Neither proof mistakes Hahn summation for convergence at every surreal scale.

We also prove compatibility with exponent embeddings, an exact reduction sequence, a valuation-circle quotient, equality of generated support monoids in the two coordinate systems of the node chart, exact valuation identities including a period-sensitivity formula, a kernel obstruction to extending the map beyond U_q , support-preserving recovery of q from j , and the formula $K(E_q[n]) = K_{\Gamma + \mathbb{Z}\alpha/n}$ for the full torsion field. Every elliptic curve over $\mathbf{No}[i]$ with negative valuation of its j -invariant is covered, using set-sized normal-form workspaces. The period-bounded quotient itself has a precedent in the logarithmic Picard theory of Molcho and Wise.

A second part treats several periods with monomial exponents. Every admissible lattice in a finite-dimensional higher-rank inner-product space has a finite positively generating set, which answers affirmatively, and constructively, Question 11.14 of Amini and Nicolussi as printed in their 2025 article. We characterize exactly when a quadratic lattice family is strongly Hahn-summable, determine the exact domain of multiscale theta series and their support order type ω^s , certify theta minima by finitely many inequalities, and lift smooth initial zeros.

Keywords. Surcomplex numbers; Hahn series; Tate curve; convex valuation subgroup; strong summability; formal groups; ramification; higher-rank lattices; theta series.

Status and scope. A merged report built from three manuscripts (section 1.3). Part I takes the *new-theorems* alternative, not a claimed solution of a named published open problem or conjecture. Its candidate original contribution is the exact Hahn-summability domain of the raw bilateral families, its surjectivity onto all rational points at arbitrary rank (proved twice, by independent routes), the explicit inverse charts, and the support and valuation refinements. The bounded-period quotient $U_q/q^{\mathbb{Z}}$ and its value sequence are *not* claimed as new shapes: Molcho–Wise’s bounded monodromy and their logarithmic Tate curve are a precedent [16]. Part II answers Amini–Nicolussi’s Question 11.14, a question of higher-rank lattice geometry, in its published formulation; no claim is made that nobody has answered it since, and Part II constructs no abelian variety and no higher-dimensional uniformization. Tate’s rank-one theorem and universal identities, theta identities, and the classical torsion construction are credited as inputs. Targeted literature and repository audits do not certify priority. The proofs have not been independently refereed or formalized in Lean. Included exact computations check finite identities, not the infinite arguments.

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1 The problem and the principal results

The classical multiplicative uniformization of an elliptic curve uses a parameter $u \in k^\times$ modulo powers of a small element q . Over a complete real-valued non-Archimedean field, Tate's theorem supplies the isomorphism $k^\times/q^\mathbb{Z} \simeq E_q(k)$ [1, 2]. Replacing k by a Hahn field of arbitrary rank is not a harmless change of coefficients. A positive value $\alpha = v(q)$ need not dominate all other values by its integer multiples. The same bilateral series can then have non-well-ordered support.

There are two different questions. Which parameters admit the literal Hahn formulas? And, after excluding the inadmissible parameters, do those formulas still reach every elliptic-curve point? The second question does not follow from a support calculation. It is the central issue addressed here.

Throughout, Γ is a set-sized ordered abelian group, written additively, and

$$K = K_\Gamma := \mathbb{C}((t^\Gamma)).$$

The coefficient field $\mathbb{C}$ has the trivial valuation. Every integer, degree, and index $n \in \mathbb{Z}$ in a series is an *ordinary finite* integer, not a surreal integer. We use $v(t^\gamma) = \gamma$ and $v(0) = +\infty$.

Definition 1.1 (The admissible group). For $q \in K^\times$ with $\alpha = v(q) > 0$, put

$$H_\alpha = \{\gamma \in \Gamma : -N\alpha \leq \gamma \leq N\alpha \text{ for some } N \in \mathbb{N}\}, \quad (1)$$

$$U_q = v^{-1}(H_\alpha) \subseteq K^\times. \quad (2)$$

Thus H_α is the smallest convex subgroup of Γ containing α . We call U_q the *Hahn-admissible multiplicative domain*.

Define the standard Tate coefficients

$$s_k(q) = \sum_{n \geq 1} \frac{n^k q^n}{1 - q^n}, \quad (3)$$

$$a_4(q) = -5s_3(q), \quad a_6(q) = -\frac{5s_3(q) + 7s_5(q)}{12}, \quad (4)$$

$$E_q : y^2 + xy = x^3 + a_4(q)x + a_6(q). \quad (5)$$

The point at infinity is denoted O . These definitions are legitimate strong Hahn sums; the discriminant is nonzero by proposition 4.3.

Theorem 1.2 (Exact-domain Hahn–Tate uniformization). *Let $q \in K_\Gamma$ satisfy $v(q) = \alpha > 0$. The bilateral family*

$$\Theta_q(u) = \sum_{n \in \mathbb{Z}} (-1)^n q^{n(n-1)/2} u^n \quad (6)$$

is strongly Hahn-summable if and only if $u \in U_q$. For $u \in K^\times \setminus q^\mathbb{Z}$, each of the two bilateral families defining

$$X_q(u) = \sum_{n \in \mathbb{Z}} \frac{q^n u}{(1 - q^n u)^2} - 2s_1(q), \quad (7)$$

$$Y_q(u) = \sum_{n \in \mathbb{Z}} \frac{(q^n u)^2}{(1 - q^n u)^3} + s_1(q) \quad (8)$$

is strongly Hahn-summable if and only if $u \in U_q$.

The map

$$\Phi_q(u) = \begin{cases} (X_q(u), Y_q(u)), & u \notin q^{\mathbb{Z}}, \\ O, & u \in q^{\mathbb{Z}}, \end{cases} \quad (9)$$

is a surjective group homomorphism $U_q \rightarrow E_q(K)$ with kernel $q^{\mathbb{Z}}$. In particular,

$$U_q/q^{\mathbb{Z}} \simeq E_q(K_{\Gamma}). \quad (10)$$

No divisibility, countability, finite-rank, or real-valuedness assumption on Γ is needed.

The summability assertions are proved in sections 3 and 4; the group and surjectivity assertions are proved in section 7. The qualification “the bilateral families” matters: this is an exact maximality theorem for specified expressions, not a claim that no different function can be assigned outside U_q . What is proved about extensions beyond U_q is an algebraic obstruction: no homomorphism $K^{\times} \rightarrow E_q(K)$ extending Φ_q has kernel $q^{\mathbb{Z}}$ unless $H_{\alpha} = \Gamma$ (theorem 10.5).

Theorem 1.2 is proved twice, by independent routes. The first route (sections 5 to 7) runs Tate’s complete rank-one theorem in the residue workspace $K_{H_{\alpha}}$ and lifts through a coarser valuation. The second route (section 8) uses neither Tate’s complete-field theorem nor any rank-one completion. An integral formal chart at the node, together with two elementary formal charts, inverts Φ_q point by point; the group law comes from Tate’s universal cleared secant identities. The second route works with $\mathbb{C}$ replaced by an arbitrary field of characteristic zero (theorem 8.13); in particular it gives the real surreal case (corollary 13.2). The first route yields in addition the coarse reduction diagram of corollary 7.2, which the second does not. The generality of the coefficient field applies to theorem 1.2, to the material of sections 8, 9 and 10.3, to the modular inversion in theorem 11.1, and to corollary 13.2. The torsion-field and negative- j classification statements below retain their hypotheses over $\mathbb{C}$; classification also assumes a divisible exponent group.

Theorem 1.3 (Exact arithmetic and scalar extension). *For an integer $n \geq 2$, let $\Lambda = \Gamma + \mathbb{Z}\alpha/n \subseteq \Gamma \otimes_{\mathbb{Z}} \mathbb{Q}$. Inside $K_{\Gamma \otimes \mathbb{Q}}$ one has*

$$K_{\Gamma}(E_q[n]) = K_{\Gamma}(q^{1/n}) = K_{\Lambda}, \quad [K_{\Lambda} : K_{\Gamma}] = \text{ord}(\alpha + n\Gamma) \leq n. \quad (11)$$

A single point $\Phi_q(q^{1/n})$ generates this field through its coordinates. The extension is cyclic and totally ramified.

An ordered embedding of exponent groups preserves the Hahn formulas and Φ_q . In a larger Hahn field the exact parameter domain is again the inverse image of the principal convex subgroup generated by the image of α ; it need not be the whole multiplicative group.

The torsion description is classical in its Kummer origin. The point of theorem 1.3 is the explicit arbitrary-group ramification formula and its compatibility with the domain theorem, not a claim to have discovered Tate torsion.

Corollary 1.4 (Surcomplex elliptic curves). *Fix the Conway normal-form valuation on $\mathbf{No}[i]$, with $v(\omega^{-\gamma}) = \gamma$. For every nonzero q with $\alpha = v(q) > 0$,*

$$\{u \in \mathbf{No}[i]^{\times} : |v(u)| \leq N\alpha \text{ for some } N \in \mathbb{N}\}/q^{\mathbb{Z}} \simeq E_q(\mathbf{No}[i]). \quad (12)$$

Every elliptic curve E over $\mathbf{No}[i]$ with $v(j(E)) < 0$ is isomorphic to an E_q of this kind. The parameter q is uniquely determined by $j(E)$. These are class statements whose every instance and proof is realized in a set-sized Hahn workspace.

1.1 Why the result is not an entire-function substitution

If $H_\alpha = \Gamma$, the parameter domain is all of $K^\times$. For a nonzero real-valued group this always happens. If $\Gamma = \mathbb{R} \oplus_{\text{lex}} \mathbb{R}$ and $\alpha = (0, 1)$, however, $H_\alpha = \{0\} \oplus \mathbb{R}$. Elements of valuation $(1, 0)$ or $(-1, 0)$ are excluded. Nevertheless, *all* K -points of E_q are represented. In particular the restriction does not mean that only points whose coordinates have q -bounded valuations are reached.

The accepted parameter $u = 1 + h$, where $v(h)$ dominates every $N\alpha$, already produces points with

$$X_q(1 + h) \sim h^{-2}, \quad Y_q(1 + h) \sim -h^{-3}.$$

Their coordinates can be vastly larger than every integral power of q^{-1} . Smallness of u and smallness of $u - 1$ are different questions.

1.2 Relation to the repository and to the literature

The repository audit of source 01 was targeted: its document map, the root description, the spectral report, and the rank-one analytic report were inspected, and code search was used for elliptic/Tate terminology. The pinned revision is a3124af79f66b8b9c196d76b4cbc5ac3938907c4 [20]; the pins of the other two sources are recorded in section 1.3. The closest existing report works over the fixed field $\mathbb{C}((t^{\mathbb{R}}))$ and develops Tate *algebras*, disks, annuli, and a partial-theta example [23]. A Tate algebra is not a Tate elliptic curve. The present paper neither reclaims that analytic infrastructure nor repeats the partial-theta zero calculation.

That report is emphatic that its base is the fixed rank-one field $\mathbb{C}((t^{\mathbb{R}}))$, and it does not treat higher rank as a cosmetic variant. Its `prop:rankobstruction` proves that the ordered group $\mathbb{Q} + \mathbb{Q}\omega \subset \mathbf{No}$ admits no order-preserving injective additive map to $\mathbb{R}$: since $n \cdot 1 < \omega$ for every positive integer n , an Archimedean target cannot receive both scales faithfully. A higher-rank valuation therefore cannot be silently scalarized; as that report puts it, a noninjective order-preserving homomorphism discards an entire layer of scale, and in its example it must kill 1. The present paper is the arbitrary-rank work that [23] explicitly declines to attempt, carried out on a different object. We take the obstruction as binding rather than as an obstacle to be argued away: the real-valued coarsening used below in lemma 5.3 is applied only inside K_{H_α} , where α is an order unit, and it is never presented as a faithful image of Γ itself. The rank-one valuation forgets H_α^- , but the base field K_{H_α} still contains series at those finer scales. The later coarse lifting uses the different quotient $w = v \bmod H_\alpha$: formal evaluation at w -positive inputs lifts from K_{H_α} to K_Γ . These two quotients have different kernels and roles.

Tate’s original proof explicitly starts with a nontrivial real-valued absolute value and proves uniformization over any complete field with that absolute value [1]. Modern treatments include [2, 4]; the 2026 work of Eterović–Vermeulen concerns p -adic definability and Tate uniformization [6]. Higher-rank analytic geometry already exists, notably the Hahn analytification of Foster–Ranganathan [8]. Thus neither “higher-rank geometry” nor “non-Archimedean elliptic functions” is new here. Tate’s own account already separates universal formal identities from their analytic specialization, and the second proof below uses exactly those identities [1]. Formal models of the nodal Tate degeneration are standard as well; Conrad’s account places the Tate curve over $\mathbb{Z}[[q]]$ in its modular context [5]. The Eterović–Vermeulen result is a definability theorem on $\mathbb{C}_p$, not a theorem about strong Hahn summability at arbitrary rank. Neumann’s support lemma, Gonshor’s normal forms and the summability conventions of Berarducci–Mantova supply the remaining foundations [9, 12, 15].

The bounded-period quotient has a precedent. Molcho and Wise impose *bounded monodromy* in the logarithmic and tropical Picard groups [16]. Their Definition 2.1.3.1 calls an element β of a partially ordered

abelian group *bounded* by $\delta \geq 0$ when $m\delta \leq \beta \leq n\delta$ for some integers m, n ; Definition 3.5.5 defines bounded monodromy, and Definition 3.6.1 defines the tropical Jacobian as the bounded part of $\text{Hom}(H_1, M^{\text{gp}})$ modulo H_1 . In their §5.1, on the Tate curve, the logarithmic Picard group LogPic^0 is the subfunctor of $\mathbf{G}_{\log}$ bounded by the loop length δ , modulo the lattice generated by a lift of δ , and their tropicalization sequence (5.1.4) exhibits it as an extension of the bounded tropical quotient by $\mathbf{G}_m$. For an ordered value group and the single length $\alpha = v(q)$, the bounded subgroup is exactly H_α of definition 1.1, the quotient LogPic^0 is the log-geometric form of $U_q/q^{\mathbb{Z}}$, and (5.1.4) has the shape of the value sequence of theorem 10.3. Neither the need for boundedness nor the abstract shape of the period-bounded quotient is therefore claimed as new here. Source 01 did not know this precedent; source 03 identified it, and it is credited throughout this report. No functorial comparison between the Hahn map of this report and the logarithmic Tate quotient is proved; that comparison is the most important remaining priority check (section 23).

The potentially new package is more specific: an *if and only if* Hahn-support domain for the raw bilateral families; a proof that this restricted domain nevertheless uniformizes every rational point, by explicit inverse formulas as well as by rank-one lifting; exact compatibility with coarse reduction, scalar extension, and exponent-group arithmetic; and the support-monoid and valuation identities of section 9. The extension obstruction of theorem 10.5 is a useful consequence, not an independent deep discovery. No assertion of bibliographic uniqueness follows from the targeted searches. Related formulations in formal, logarithmic, adic, or model-theoretic geometry may subsume parts of the result.

Several periods. Part II (source 02) takes up the multivariable theta series that a higher-dimensional analogue would need, with monomial periods $t^{\pi_{ab}}$. Its first theorem is purely lattice-geometric: every admissible lattice in a finite-dimensional higher-rank inner-product space, in the sense of Amini–Nicolussi [17], has a finite positively generating set (theorem 15.2). This answers their Question 11.14 affirmatively and constructively, in the formulation printed in [17, §11.5.3, p. 1181]. The remaining theorems give the exact strong-summability criterion for quadratic lattice families (theorem 16.3), the exact domain of the multiscale theta series and its unit robustness (theorem 18.1), its support order type (theorem 17.2), finite certificates for theta minima (theorems 19.1 and 19.3), and the lifting of smooth initial zeros (theorem 20.2). For one variable and a monomial period it recovers the domain of theorem 3.1 in a different normalization (corollary 18.3); the hypotheses of the two statements differ and are kept apart. Rank-one non-Archimedean theta functions and their tropicalization are due to Foster, Rabinoff, Shokrieh and Soto [18]; convergent higher-rank Hahn subrings are studied by Joswig and Smith [19]. Part II does not construct an abelian variety or a higher-dimensional uniformization.

1.3 Three source manuscripts, and where the merge chose

This report is built from three manuscripts written independently on 22 September 2026. The source manuscripts are not shipped with the report; their code and data are, under the file prefixes given here.

Source 01 *Hahn–Tate Uniformization at Arbitrary Valuation Rank*, repository audit pinned at a3124af.

It is the base text of Part I: sections 1 to 7 and 10 to 13 except where marked, with the first proof of theorem 1.2. Its files are the unprefix code/check_identities.py and data/verification.json, verification.log, requirements.txt, build_validation.json. Full pins are in Appendix A.

Source 02 *Finite Positive Generation and Multiscale Theta Series over the Surreals*, audit pinned at a3124af.

It supplies part II. Files prefixed 02-multiscale-theta-.

Source 03 *Tate Uniformization over Surcomplex Numbers: Sharp Hahn domains, support-controlled coordinates, and exact valuation geometry at arbitrary rank*, audit pinned at 0097304. It was written without knowledge of source 01, which entered the repository after that pin. It supplies lemma 2.2, the

second proof (section 8), section 9, section 10.3, corollary 13.2, example 13.4, three open directions in section 23, and the Molcho–Wise precedent. Files prefixed `03-tate-uniformization-`, including its proof-status file.

Sources 01 and 03 prove the same main theorem. Its domain half, including the decreasing-exponent witnesses, is printed once (theorem 4.1); the value sequence is printed once (theorem 10.3); the good coarse reduction is printed once (section 7); the example $q = \omega^{-1}$, $u = \omega^{-\omega}$ versus $u = 1 + \omega^{-\omega}$ is printed once (example 13.3). The surjectivity and group-law proofs are genuinely different and both are kept, the second as a marked second route. Source 03’s torsion consistency check (for $\mathbb{C}$ and divisible Γ) is subsumed by theorem 12.2 and is not repeated. Where notation collided, source 03 and source 02 were renamed, never source 01; the conventions are stated at the start of section 8 and of part II. The one choice of substance concerns framing: source 01’s claim that the exact-domain quotient was potentially new is corrected by source 03’s precedent, and source 03’s statement that elliptic uniformization was a direction absent from the repository, true at its pin `0097304`, no longer holds, since the repository now contains source 01. Likewise source 02 found no theta-lattice treatment at its pin `a3124af`; the collection now contains the one-variable bilateral theta domain (theorem 3.1), the partial-theta domain of the holonomic-rigidity report (`hol:thm:thetadomain` [21]) and the unilateral example $\sum t^{n^2} z^n$ of the foundations report (`found:ex:internal` [22]), but still no multivariable or lattice theta series outside part II.

Part I

One period: Hahn–Tate uniformization

2 Hahn summation and the relevant order structure

2.1 Strong summability

A Hahn series is a formal expression

$$a = \sum_{\gamma \in \Gamma} a_{\gamma} t^{\gamma}, \quad a_{\gamma} \in \mathbb{C},$$

whose support $\text{supp}(a) = \{\gamma : a_{\gamma} \neq 0\}$ is well ordered in the increasing order. For $a \neq 0$, its valuation is the least support element; put $v(0) = +\infty$. A family $(a_i)_{i \in I}$ is *strongly summable* when $\bigcup_i \text{supp}(a_i)$ is well ordered and each exponent occurs in the support of only finitely many a_i . Its sum is defined coefficientwise. Neither an order limit nor a limit in the full valuation topology is part of this definition.

Lemma 2.1 (Positive-support calculus). *Let $S \subseteq \Gamma_{>0}$ be well ordered. Its additive monoid S^* , including 0, is well ordered. Every element of Γ has only finitely many representations as a finite unordered sum of members of S , counting multiplicities. Consequently, for finitely many Hahn series $z_1, \dots, z_d$ of positive valuation, any formal power series $F \in \mathbb{C}[[Z_1, \dots, Z_d]]$ has a well-defined strong evaluation $F(z)$. Evaluation preserves sums, products, and composition when the inner formal series have zero constant coefficient.*

Proof. We recall the order-theoretic argument behind Neumann’s lemma [9]. The alphabet S , with its order, is well quasi-ordered. Higman’s lemma says that finite words over this alphabet are well quasi-ordered by order-preserving subsequence embedding with each letter allowed to increase [10]. If a word embeds into another, the sum of its letters does not exceed the sum of the other word. The inequality is strict if a letter increases or a positive letter is added. A descending sequence of sums would therefore contradict Higman’s

lemma. An infinite collection of distinct nondecreasing words of the same sum would also contradict it: an embedding between two such words with equal sum forces the words to coincide. Sorting a word removes the finitely many permutations of each finite multiset. This proves both assertions.

The union of the finitely many positive supports $\text{supp}(z_i)$ is well ordered. Expanding all monomials in $F(z)$ produces exponents in S^* . The finiteness assertion bounds the word lengths and the decompositions at a fixed exponent, and the number of variable labels is finite. Hence every coefficient is a finite sum. The same argument for the jointly indexed families proves multiplication and the stated composition rule. This is the usual Hahn–Neumann calculus, not a new summation principle. $\square$

In particular, $(1 - z)^{-1} = \sum_{n \geq 0} z^n$ and $(1 + z)^r = \sum_{n \geq 0} \binom{r}{n} z^n$ for $r \in \mathbb{Q}$ and $v(z) > 0$. These identities hold even when $nv(z)$ is bounded above in Γ . A series with leading term ct^γ is inverted by factoring it as $ct^\gamma(1 + \varepsilon)$ and applying this rule to ε .

2.2 Other coefficient fields and the formal implicit function

Nothing in lemma 2.1 or its proof uses a property of $\mathbb{C}$: coefficients enter only through finite sums and products. The lemma holds verbatim for $k((t^\Gamma))$ and $k[[Z_1, \dots, Z_d]]$ for any field k , which is how source 03 states it (its “positive-input specialization” lemma, printed here once). Its proof also gives the support control

$$\text{supp } F(z) \subseteq \langle \bigcup_j \text{supp } z_j \rangle_{\mathbb{N}}, \quad (13)$$

where $\langle S \rangle_{\mathbb{N}}$ is the additive monoid generated by S , including 0; if $F(0) = 0$, the value has positive valuation. The second proof of theorem 1.2 rests on the following lemma, taken from source 03.

Lemma 2.2 (Formal inverse and implicit function). *Let k be any field, and let $\mathfrak{m}_v$ be the elements of positive valuation in $k((t^\Gamma))$. Suppose $A(Z) \in k[[Z_1, \dots, Z_d]]^d$ has zero constant term and invertible linear part $L_A \in \text{GL}_d(k)$. It has a unique zero-constant formal inverse. Its evaluation gives a bijection of $\mathfrak{m}_v^d$. If A is defined over $\mathbb{Z}$ with $L_A \in \text{GL}_d(\mathbb{Z})$, its formal inverse is defined over $\mathbb{Z}$ as well, before any specialization to k .*

More generally, if $B(P, Z)$ has zero constant term and the matrix $\partial B / \partial Z$ at the origin is invertible, the equation $B(P, Z) = 0$ has a unique formal solution $Z = C(P)$ with $C(0) = 0$. Every solution obtained by substituting $P, Z \in \mathfrak{m}_v$ equals this solution.

Proof. Determine the homogeneous terms of an inverse successively. At total degree n , the unknown correction enters only through multiplication by L_A ; all other contributions were determined in earlier degrees. Solving this finite linear system constructs the inverse, proves uniqueness, and proves integrality when L_A^{-1} is integral. No division by the homogeneous degree occurs, so the construction works in every characteristic. Formal composition identities specialize by lemma 2.1. The inverse and the original map both have zero constant term, hence preserve $\mathfrak{m}_v^d$; their two composition identities give surjectivity and injectivity on that entire domain.

Apply the same inverse construction to the map $(P, Z) \mapsto (P, B(P, Z))$ for the implicit statement. Its inverse, specialized at $(P, 0)$, gives $C(P)$. Specializing the two-sided inverse identity proves uniqueness among actual infinitesimal inputs, not merely among formal series. $\square$

The characteristic-free formal inverse lemma does not remove the characteristic-zero hypotheses from the later Tate-curve statements; those retain their own coefficient assumptions.

The limits of these lemmas should be stated. If $\epsilon > 0$ lies in a proper convex subgroup of Γ , the sequence $n\epsilon$ need not be cofinal in Γ , so a formal remainder of degree N with valuation at least $N\epsilon$ need not tend to zero in the full valuation topology. The inverses are constructed as formal series and then specialized by

support arguments; they are not limits of Newton iterates in an assumed complete metric. Lemma 5.1 below is a different strengthening, with coefficients of unbounded valuation, and lemma 20.1 a third, with infinitely many monomial parameters.

2.3 Normalization and two convex subgroups

Lemma 2.3 (Finite normalization). *For $\beta \in \Gamma$, the following are equivalent:*

- (i) $\beta \in H_\alpha$;
- (ii) *there is an integer m with $0 \leq \beta + m\alpha < \alpha$.*

Such an m is unique. If $\beta \notin H_\alpha$, then either $\beta > N\alpha$ for every $N \in \mathbb{N}$, or $\beta < -N\alpha$ for every $N \in \mathbb{N}$.

Proof. If $|\beta| \leq N\alpha$, choose the largest integer $k \in \{-N, \dots, N\}$ with $k\alpha \leq \beta$; this set is nonempty. Then $k\alpha \leq \beta < (k+1)\alpha$: maximality gives the upper bound when $k < N$, and $\beta \leq N\alpha$ gives it when $k = N$. Taking $m = -k$ proves existence. If $m' > m$, then $\beta + m'\alpha \geq \beta + m\alpha + \alpha \geq \alpha$, proving uniqueness. The converse follows from $|\beta| \leq (|m|+1)\alpha$. Finally, if $\beta \notin H_\alpha$, then $|\beta| > N\alpha$ for every N ; its fixed sign gives the stated alternative. No real-valued floor function or divisibility of Γ is used. $\square$

There is another convex subgroup inside H_α :

$$H_\alpha^- = \{\gamma \in H_\alpha : n|\gamma| < \alpha \text{ for all } n \geq 1\}. \quad (14)$$

It consists of the values infinitesimal relative to α . To check it is a subgroup, take $\gamma, \delta \in H_\alpha^-$. For every $n \geq 1$, the inequalities $2n|\gamma| < \alpha$ and $2n|\delta| < \alpha$ imply $n|\gamma + \delta| < \alpha$; negation preserves the defining condition. Convexity follows from $|\xi| \leq |\gamma|$ with $\gamma \in H_\alpha^-$. It is proper because $\alpha \notin H_\alpha^-$.

The quotient H_α/H_α^- is nonzero and Archimedean. Indeed, for positive $\gamma \notin H_\alpha^-$ some $n \geq 1$ satisfies $n\gamma \geq \alpha$. Given positive $\delta \in H_\alpha$, choose N with $\delta \leq N\alpha$; then $(N+1)n\gamma \geq (N+1)\alpha > \delta$, with the strict inequality still true in the quotient since the difference is at least α . This gives the Archimedean comparison there. It also shows that H_α^- is the largest proper convex subgroup of H_α : a convex subgroup containing such a γ contains α , and then every element bounded by an integer multiple of α . Hölder's theorem embeds this quotient in $(\mathbb{R}, +)$; normalize the embedding so that α has image 1.

This construction is not new in the collection, and we do not present it as ours. By construction α is an *order unit* of H_α : its positive integer multiples are cofinal there. The companion report on entire Hahn functions states exactly the resulting passage as its `ent:lem:coarsening` [24]: for an order unit h of an exponent group, there is an order-preserving π to $\mathbb{R}$ with $\pi(h) = 1$ whose kernel is the convex subgroup of values infinitesimal relative to h , the formula $|a|_\pi = \exp(-\pi(v(a)))$ is a nontrivial real-valued non-Archimedean absolute value whose topology is the intrinsic valuation topology, and the Hahn field is complete for it. Applied with exponent group H_α and $h = \alpha$, that lemma is the statement we need, with kernel exactly (14). We cite it rather than claim it. One hypothesis must be carried across with it: that report's standing assumption is that its exponent group is *divisible*, and *we do not assume Γ , or H_α , divisible in the one-period uniformization theorem* — section 12 depends on keeping nondivisible groups. Its proof of that lemma uses only the order-unit property and no divisibility, so the statement does transfer; because the transfer is ours and not the cited report's, we retain the short direct argument in lemma 5.3 below, which uses no divisibility, rather than rest a step of the main theorem on a hypothesis we have dropped.

The two quotients serve different purposes. The quotient by H_α^- produces a rank-one valuation *inside* K_{H_α} . The quotient by H_α produces a coarser valuation on the *whole* field K_Γ , whose residue field is K_{H_α} . Interchanging them would destroy the proof.

3 The exact domain of the bilateral theta series

Theorem 3.1 (Theta domain). *The family in (6) is strongly summable exactly for $u \in U_q$. On that domain,*

$$\Theta_q(qu) = -u^{-1}\Theta_q(u). \quad (15)$$

Proof. Write $\beta = v(u)$. The leading exponent of its n th summand is

$$b_n = \frac{n(n-1)}{2}\alpha + n\beta. \quad (16)$$

Suppose first that $\beta \in H_\alpha$ and choose N with $|\beta| \leq N\alpha$. The differences $b_{n+1} - b_n = n\alpha + \beta$ show that the positive tail is eventually strictly increasing. Writing $n = -k$ gives the same property for the tail as $k \rightarrow +\infty$. Each tail eventually exceeds every prescribed element of H_α , by comparison with integer multiples of α . Together with the finitely many remaining indices, the set $B = \{b_n : n \in \mathbb{Z}\}$ is well ordered. Each value of b_n occurs at most twice: if $b_n = b_m$ and $n \neq m$, then $(n+m-1)\alpha + 2\beta = 0$.

Factor $q = c_q t^\alpha (1 + \varepsilon_q)$ and $u = c_u t^\beta (1 + \varepsilon_u)$ with positive-valuation errors. Let M be the monoid generated by their two supports. Every summand in (6) has support in $b_n + M$, by lemma 2.1, including negative powers of u . The sum of two well-ordered subsets of an ordered abelian group is well ordered, with finitely many pairs giving any fixed sum. Thus $B + M$ is well ordered. There are finitely many choices of n at a fixed leading value and finitely many decompositions in M . This proves strong summability of the original family.

If $\beta > N\alpha$ for all N , put $n = -k$. Then

$$b_{-(k+1)} - b_{-k} = (k+1)\alpha - \beta < 0.$$

These are actual nonzero leading coefficients, so their decreasing exponents already obstruct well-ordering of the union of supports. If $\beta < -N\alpha$ for all N , the sequence b_n decreases for $n \geq 0$. This proves necessity without assuming that possible cancellations between summands can repair an inadmissible family.

Finally, multiplication by q shifts the index in a strongly summable family. The coefficient equality $q^n (-1)^n q^{n(n-1)/2} = -(-1)^{n+1} q^{n(n+1)/2}$ proves (15). $\square$

Proposition 3.2 (Product and zeros). *For $0 \leq v(u) < \alpha$, the Jacobi product is a Hahn identity:*

$$\Theta_q(u) = (1-u) \prod_{n \geq 1} (1-q^n)(1-uv^n)(1-u^{-1}q^n). \quad (17)$$

The product means the strong sum over its finite subproducts after expansion. The zeros on U_q are exactly $q^{\mathbb{Z}}$, and all are simple in the local external-variable sense.

Proof. All of q , qu , and q/u have positive valuation. Their support monoid licenses the expanded product and its regrouping. The classical Jacobi triple-product identity, in this normalization, is an identity of the corresponding formal Laurent families [14]; strong evaluation therefore yields (17). One can also compare coefficients of q after treating u as a Laurent indeterminate and then use the same support argument for specialization.

The product after the factor $1-u$ is 1 plus a positive-valuation series, hence a unit. Therefore the only zero in the normalized annulus is $u = 1$. The functional equation and lemma 2.3 give all zeros. At $u = 1$ the derivative of the product is

$$\Theta'_q(1) = - \prod_{n \geq 1} (1-q^n)^3 \neq 0.$$

Reindexing transports simplicity to every q^m . This is differentiation of a local power series in a variable; no derivation on the scalar field is being imposed. $\square$

4 Tate coordinates, support certificates, and precision

4.1 A form with positive tails

Put

$$f(z) = \frac{z}{(1-z)^2}, \quad g(z) = \frac{z^2}{(1-z)^3}, \quad h(z) = \frac{z}{(1-z)^3}.$$

The elementary identities $f(z^{-1}) = f(z)$ and $g(z^{-1}) = -h(z)$ give, after normalizing $0 \leq \beta = v(u) < \alpha$,

$$X_q(u) = f(u) + \sum_{n \geq 1} (f(q^n u) + f(q^n/u)) - 2s_1(q), \quad (18)$$

$$Y_q(u) = g(u) + \sum_{n \geq 1} (g(q^n u) - h(q^n/u)) + s_1(q). \quad (19)$$

The minus sign and the unsquared numerator in the last tail of (19) are important. In contrast, the numerator in the original bilateral formula (8) is squared.

At $v(u) = 0$ we do *not* expand the isolated expressions $f(u)$ and $g(u)$ geometrically in u . They are finite rational operations, defined unless $u = 1$. Their denominators may be arbitrarily small.

Theorem 4.1 (Coordinate domain and support certificate). *For $u \in K^\times \setminus q^{\mathbb{Z}}$, the families in (7) and (8) are strongly summable exactly when $u \in U_q$. For $0 \leq v(u) < \alpha$ and $u \neq 1$, let*

$$S = \text{supp}(q) \cup \text{supp}(qu) \cup \text{supp}(q/u) \subseteq \Gamma_{>0}. \quad (20)$$

Then

$$\text{supp}(X_q(u) - f(u)), \text{supp}(Y_q(u) - g(u)) \subseteq S^* \setminus \{0\}. \quad (21)$$

The bilateral and positive-tail formulas agree.

The restriction $u \neq 0$ is necessary for this exact-domain assertion. At $u = 0$ every term of each bilateral coordinate family is zero, so both families are strongly summable, although $0 \notin U_q$. This does not extend the multiplicative uniformization domain, and the theta family, which contains negative powers of u , is not defined there.

Proof. The series $q^n u = q^{n-1}(qu)$ and $q^n/u = q^{n-1}(q/u)$ have positive valuation for $n \geq 1$. Expanding f, g, h in those arguments and expanding each term of s_1 gives exponents in $S^* \setminus \{0\}$. The full family, including n and every geometric-expansion index, is strongly summable: at a fixed exponent, the Neumann finiteness theorem bounds the total word length, hence also n and the geometric index. This proves (21) and justifies every regrouping in (18)–(19). Multiplication of u by a power of q merely reindexes the bilateral sums. Finite normalization therefore proves sufficiency on U_q .

For necessity, suppose first $\beta > N\alpha$ for every N . All $q^n u$ have positive valuation. The leading exponents in the X family are $\beta + n\alpha$ and in the Y family are $2\beta + 2n\alpha$. Both decrease strictly as $n \rightarrow -\infty$. If $\beta < -N\alpha$ for every N , all $q^n u$ have negative valuation. The rational identities above show that the leading exponents of both families are $-\beta - n\alpha$, with a nonzero sign in the Y case. These decrease as $n \rightarrow +\infty$. In either case the original families fail the well-ordering requirement. $\square$

Proposition 4.2 (Exact finite-tail bound). *For normalized u , let $X^{[N]}, Y^{[N]}$ retain the isolated rational terms and the terms $1 \leq n \leq N$ in the positive-tail formulas, and replace s_1 by $\sum_{n=1}^N nq^n/(1-q^n)$. Then*

$$v(X_q - X^{[N]}) \geq (N+1)\alpha - \beta, \quad v(Y_q - Y^{[N]}) \geq (N+1)\alpha - \beta. \quad (22)$$

Here $v(0) = +\infty$, so cancellations cause no exception.

Proof. Each omitted $f(q^n/u)$ or $h(q^n/u)$ has valuation $n\alpha - \beta$; the other omitted coordinate terms have at least that valuation. The omitted s_1 terms have valuation $n\alpha$. Strong summation and the valuation inequality give the bounds. $\square$

If a requested precision δ lies in H_α , a finite N certifies (22) beyond δ . If $\delta > N\alpha$ for all N , this certificate never reaches it. A valid exact Hahn sum has therefore not been converted into a convergence theorem at all scales.

4.2 The elliptic curve and its invariant differential

Proposition 4.3 (Discriminant and modular invariant). *The coefficients in (4) are Hahn evaluations of integer power series in q . The Weierstrass discriminant and j -invariant are*

$$\Delta(q) = q \prod_{n \geq 1} (1 - q^n)^{24}, \quad v(\Delta(q)) = \alpha, \quad (23)$$

$$j(q) = q^{-1} + 744 + 196884q + 21493760q^2 + \cdots, \quad v(j(q)) = -\alpha. \quad (24)$$

In particular E_q is a nonsingular elliptic curve over K_Γ .

Proof. The formulas for s_k are licensed by positive-support calculus. The integer divisibility of $(5n^3 + 7n^5)/12$ follows by checking its factors modulo 3 and 4, or from the standard Tate coefficient identity. With $a_1 = 1$, $a_2 = a_3 = 0$, the usual Weierstrass formulas give

$$b_2 = 1, \quad b_4 = 2a_4, \quad b_6 = 4a_6, \quad b_8 = a_6 - a_4^2, \quad c_4 = 1 - 48a_4.$$

The formal identities for Δ and c_4^3/Δ are classical [1, 2]. They are identities in integer formal power series (Laurent series for j), so positive-support evaluation proves them here. The product multiplying q is a unit of leading coefficient 1. Hence $\Delta \neq 0$, and the displayed valuations follow. $\square$

The coordinate functions satisfy the curve equation and the group law; a proof valid at every rank is given in section 7. The following local identity can already be checked term by term.

Proposition 4.4 (Differential and inversion identities). *On the local external-variable germs in U_q , away from poles,*

$$u \frac{dX_q}{du} = X_q + 2Y_q, \quad X_q(u^{-1}) = X_q(u), \quad Y_q(u^{-1}) = -X_q(u) - Y_q(u). \quad (25)$$

Consequently the invariant differential pulls back as

$$\Phi_q^* \left(\frac{dx}{2y + x} \right) = \frac{du}{u}. \quad (26)$$

The equation for differentials is interpreted in regular local charts also where the displayed affine denominator vanishes.

Proof. The rational identity $zf'(z) = f(z) + 2g(z)$ and the constants $-2s_1 + 2s_1 = 0$ prove the first formula. Differentiating the positive tails preserves their support estimates locally, so the interchange is licensed. The inversion formulas follow from $f(z^{-1}) = f(z)$ and $g(z^{-1}) = -f(z) - g(z)$, followed by reindexing. Equation (26) is the first identity rewritten as a differential. It extends in regular charts since an invariant differential has no pole on a smooth elliptic curve. $\square$

Nothing in this proposition says that a chosen scalar derivation $\partial : K \rightarrow K$ satisfies these formulas without the chain rule's q -dependence. In particular, it makes no assertion about the Berarducci–Mantova derivation of surreal scalars.

5 Coarse infinitesimals admit unrestricted formal evaluation

This section supplies the step that a bare appeal to “completeness of Hahn fields” would miss. We shall evaluate formal series whose coefficients have no uniform bound for the original valuation. A coarser valuation makes this possible.

5.1 The coarse residue field

Let $H \subseteq \Gamma$ be any convex subgroup. The quotient $\bar{\Gamma} = \Gamma/H$ is ordered. Define

$$w(a) = v(a) + H \quad (a \neq 0), \quad O_w = \{a : w(a) \geq 0\}, \quad \mathfrak{m}_w = \{a : w(a) > 0\}.$$

For $a \in O_w$, retaining exactly the terms with exponents in H gives a residue homomorphism

$$\text{res}_H : O_w \longrightarrow K_H, \quad \sum_{\gamma} a_{\gamma} t^{\gamma} \longmapsto \sum_{\gamma \in H} a_{\gamma} t^{\gamma}. \quad (27)$$

Its kernel is $\mathfrak{m}_w$, and the inclusion $K_H \subseteq O_w$ is a coefficient-field section. No section of the ordered group quotient $\Gamma \rightarrow \Gamma/H$ is required.

To verify multiplicativity, observe that the support of an element of O_w contains only nonnegative quotient values. A product exponent can belong to H only if both summand exponents belong to H . Thus its H -part is exactly the product of the H -parts.

Lemma 5.1 (Unrestricted coarse formal evaluation). *Let $z_1, \dots, z_d \in \mathfrak{m}_w$. For every formal series*

$$F(Z) = \sum_{\nu \in \mathbb{N}^d} a_{\nu} Z^{\nu} \in K_H[[Z_1, \dots, Z_d]],$$

the family $(a_{\nu} z^{\nu})_{\nu}$ is strongly Hahn-summable in K_{Γ} . There is no boundedness assumption on $v(a_{\nu})$ in H . Evaluation is a K_H -algebra homomorphism. It commutes with formal composition when the inner series have zero constant coefficient.

Proof. Project the supports of the nonzero z_i into Γ/H and let S be their union. A monotone image of a well-ordered set is well ordered: the least preimage of any nonempty chosen subset maps to its least image. Since $w(z_i) > 0$, every element of S is strictly positive. Apply lemma 2.1 in the quotient group. The possible quotient values form the well-ordered monoid S^* . At a fixed quotient value $\bar{\gamma}$, there are only finitely many decompositions into members of S , including word lengths. In particular, there are only finitely many multiindices ν that can contribute there.

For each of these finitely many choices, taking the indicated quotient fibers of the supports of the z_i and multiplying by a_{ν} involves ordinary finite Hahn products. These are well-defined and have well-ordered support within the fiber, with finite coefficient contributions. The finite union over the choices has the same properties. A subset of an ordered group with well-ordered quotient image and well-ordered support in every quotient fiber is well ordered: minimize first the quotient and then the fiber. Hence the union of all supports is well ordered in Γ , and each actual exponent receives finitely many contributions.

For addition and multiplication, use the same argument on the jointly indexed families. For composition, the zero-constant hypothesis ensures that at any bounded word length only finitely many formal substitutions can contribute. The quotient Neumann bound again supplies this word length bound. Thus the coefficientwise identities survive evaluation. $\square$

Example 5.2 (Why coefficient bounds are unnecessary). Let $\Gamma = \mathbb{Z} \oplus_{\text{lex}} \mathbb{Z}$, $H = \{0\} \oplus \mathbb{Z}$, and $z = t^{(1,0)}$. Then

$$\sum_{n \geq 0} t^{(0, -n^2)} z^n = \sum_{n \geq 0} t^{(n, -n^2)}$$

is strongly summable, although its coefficient valuations $(0, -n^2)$ are unbounded below in H . The first coordinate separates the degrees. This is precisely the type of separation used in the uniformization proof.

The coarse-positive condition cannot be replaced by fine positivity. In the same field, take $z = t^{(0,1)}$ and $F(Z) = \sum_{n \geq 0} t^{(0, -n)} Z^n \in K_H[[Z]]$. Now $v(z) > 0$ but $w(z) = 0$, and every evaluated term is 1. Infinitely many terms contribute at exponent zero, so this family is not strongly summable. A lower bound on the fine valuation of the input alone cannot compensate for unrestricted coefficient valuations.

5.2 The rank-one field at the q -scale

Now set $H = H_\alpha$. Use the embedding $H/H_\alpha^- \hookrightarrow \mathbb{R}$ from section 2 to define a real-valued valuation v_1 on K_H , with $v_1(t^\alpha) = 1$.

Lemma 5.3 (Completeness of the residue workspace). *The field K_H , equipped with the absolute value $|a|_1 = e^{-v_1(a)}$, is complete. Every $q_0 \in K_H$ with $v(q_0) = \alpha$ has $|q_0|_1 < 1$.*

This is `ent:lem:coarsening` of [24], instantiated at the exponent group H_α with the order unit α , and is cited rather than claimed as new; see section 2. The proof below is recorded only because that report states its lemma under a standing divisibility hypothesis on the exponent group which this one-period argument does not make, and because the package should read standalone. It is a second proof of a statement already in the collection, not an additional result.

Proof. Let (a_n) be a Cauchy sequence. Choose a subsequence (a_{n_k}) whose successive differences $d_k = a_{n_{k+1}} - a_{n_k}$ satisfy $v_1(d_k) > k$. Then $v(d_k) > k\alpha$. The sequence $k\alpha$ is cofinal in H : for $\gamma \in H$, choose an integer bound for $|\gamma|$ and then a strictly larger integer.

It follows that (d_k) is strongly summable in K_H . Indeed, below any fixed exponent only finitely many of its supports can occur; any hypothetical descending sequence of support exponents would therefore lie in a finite union of well-ordered supports. For each m , every term of the tail $\sum_{k \geq m} d_k$ has fine valuation greater than $m\alpha$, so the strong sum has fine valuation greater than $m\alpha$ as well, or is zero. Its v_1 -value is therefore at least m . Thus $a_{n_1} + \sum_k d_k$ is the v_1 -limit of the subsequence. The original Cauchy sequence has the same limit. The assertion about q_0 is $v_1(q_0) = 1$. $\square$

The coefficient field of this rank-one valuation need not be $\mathbb{C}$; it can itself retain infinitesimal scales. Tate’s theorem does not require it to be $\mathbb{C}$ or require the value group to be discrete. Its complete-field version is exactly the classical input needed here.

6 Formal germs and the passage from rank one to arbitrary rank

We record carefully how classical analytic identities can be used without illegally specializing a non-summable family.

6.1 The classical input, stated with its hypotheses

Theorem 6.1 (Tate; external classical input). *Let F be complete for a nontrivial real-valued non-Archimedean absolute value, and let $q_0 \in F$ satisfy $0 < |q_0| < 1$. The rational bilateral series (7)–(8), interpreted by their convergent tails, define an analytic group homomorphism*

$$\phi_{q_0} : F^\times \longrightarrow E_{q_0}(F)$$

which is onto and has kernel $q_0^{\mathbb{Z}}$. It extends analytically through those parameters as a map to the projective curve. The same formulas are locally analytic in the parameter q_0 as well as in u .

The surjective group statement is Tate’s Theorem 1 [1, author manuscript, p. 6], under the complete real-valued field assumptions stated on its first page. Neither discreteness nor algebraic closedness is required. Local analyticity in both variables follows directly from the rational tails: on sufficiently small neighborhoods of any non-pole pair, the finitely many relevant denominators remain nonzero and the remaining terms tend uniformly to zero geometrically. At a pole one uses the projective parameter below. This last observation requires no new uniformization theorem.

The proof of theorem 1.2 uses theorem 6.1 as an input, not as a theorem claimed to have been independently reproved here. All additional all-rank summability and lifting steps are supplied. The second proof in section 8 does not use theorem 6.1 at all.

6.2 The formal neighborhood of the identity

Lemma 6.2 (A universal formal parameter). *With an external variable h , the formal expression*

$$B(q, h) = -\frac{X_q(1+h)}{Y_q(1+h)} \tag{28}$$

after cancellation of its poles belongs to $h\mathbb{C}[[q, h]]$ and has linear coefficient 1 in h :

$$B(q, h) = h + h^2 C(q, h), \quad C \in \mathbb{C}[[q, h]]. \tag{29}$$

It has a unique inverse under composition in the second variable, $A(q, z) = z + z^2 D(q, z) \in z\mathbb{C}[[q, z]]$. For any $v(q) > 0$ and $v(h) > 0$, both series evaluate strongly, and

$$v(B(q, h)) = v(h), \quad \text{lc}(B(q, h)) = \text{lc}(h). \tag{30}$$

The germ $h \mapsto \Phi_q(1+h)$ identifies the multiplicative formal group with the formal group of E_q at O .

Proof. Expand the positive tails with $u = 1+h$, treating q, h as formal variables. They belong to $q\mathbb{C}[[q, h]]$. Hence

$$\begin{aligned} h^2 X_q(1+h) &= 1+h+h^2 R_X(q, h), \\ h^3 Y_q(1+h) &= -(1+h)^2 + h^3 R_Y(q, h), \end{aligned}$$

where $R_X, R_Y \in q\mathbb{C}[[q, h]]$. The denominator obtained after cancelling poles has constant coefficient 1. This proves (29), including the exact coefficient 1 for every q . Formal coefficient recursion gives the inverse A , since this linear coefficient is a unit.

Positive-support calculus evaluates B , A , and their composition identities at the specified arguments. The remainder $h^2 C(q, h)$ has valuation at least $2v(h) > v(h)$, which proves (30).

The formal-group identity is a universal identity of formal series in q, h_1, h_2 . For ordinary complex q and sufficiently small h_1, h_2 it follows from classical multiplicative uniformization. Uniqueness of Taylor coefficients gives the formal identity; alternatively it is the standard formal Tate-curve construction [1, 5]. Strong evaluation of the jointly indexed positive-support family proves it in K_Γ . $\square$

Here is a direct explanation of why this formal germ parametrizes all points in a coarse identity fiber, rather than just some of them. Use projective coordinates $[X : Y : Z]$ near $O = [0 : 1 : 0]$ and set

$$z = -X/Y = -x/y, \quad r = -Z/Y = -1/y.$$

The curve equation becomes

$$r(1 - z) = z^3 + a_4 z r^2 + a_6 r^3. \quad (31)$$

It has a unique solution $r \in z^3 \mathbb{C}[[q, z]]$ by coefficient recursion. For $z \in \mathfrak{m}_w$, positive-support evaluation gives such a solution. Uniqueness also holds among actual $r \in \mathfrak{m}_w$: subtracting two solutions factors their difference times

$$1 - z - a_4 z(r_1 + r_2) - a_6(r_1^2 + r_1 r_2 + r_2^2),$$

which is a w -unit because its residue is 1. Thus two points reducing to O and having the same z coincide. Evaluation of the inverse $A(q, z)$ supplies the unique corresponding $h \in \mathfrak{m}_w$.

6.3 Taylor prolongation through coarse infinitesimals

Let $w = v \bmod H_\alpha$, set $F = K_{H_\alpha}$, and put $q_0 = \text{res}_{H_\alpha}(q)$. Then

$$q = q_0(1 + \rho), \quad \rho \in \mathfrak{m}_w, \quad u = u_0(1 + \sigma), \quad \sigma \in \mathfrak{m}_w, \quad u_0 = \text{res}_{H_\alpha}(u) \in F^\times$$

for every $u \in U_q$.

Lemma 6.3 (Agreement of Taylor and Hahn evaluation). *The classical local Taylor germ of the projective map $(Q, U) \mapsto \phi_Q(U)$ at (q_0, u_0) , evaluated at (ρ, σ) by lemma 5.1, is well defined and equals $\Phi_q(u)$ as defined by the literal Hahn formulas. This remains true when $u_0 \in q_0^{\mathbb{Z}}$, using a projective local chart.*

Proof. First suppose $u_0 \notin q_0^{\mathbb{Z}}$. Replace (u, u_0) by $(uq^m, u_0q_0^m)$ so that $0 \leq v(u_0) < \alpha$. This is a finite reindexing of the bilateral formulas, and its relative error is still in $\mathfrak{m}_w$.

The isolated rational terms have nonzero constant denominators in F . Their Taylor series in ρ, σ evaluate to the actual rational terms by lemma 5.1. For the n th positive rational tail, expand first in the relative variables. At every fixed Taylor multiindex, the coefficient lies in F and its fine valuation tends to $+\infty$ cofinally in H_α as $n \rightarrow \infty$. More explicitly, after discarding finitely many n , the denominators $1 - q_0^n u_0$ and $1 - q_0^n / u_0$ are fine units, and a coefficient of any fixed relative Taylor degree has valuation at least $n\alpha - v(u_0)$, or a larger positive multiple for the g -tail. Differentiating relative powers only contributes ordinary integer coefficients and additional factors of these small rational arguments. The finitely many discarded denominators affect no tail conclusion.

It follows that the sum over n defining every Taylor coefficient is a strong sum in F and agrees with its rank-one analytic sum, by lemma 5.3. Next fix a quotient exponent for w . The quotient Neumann bound in lemma 5.1 permits only finitely many Taylor multiindices and quotient-support decompositions at this value. For each such choice, the remaining sum over n is the cofinally bounded sum just described. Thus the family jointly indexed by Taylor degree, tail index, and support decompositions is strongly summable. Fubini regrouping for this joint family proves that Taylor evaluation and the original Hahn-tail evaluation coincide. The s_1 family has the same, simpler estimates.

If $u_0 = q_0^m$, divide u by q^m . The result is $1 + h$ with $h \in \mathfrak{m}_w$. Apply lemma 6.2 and the regular projective chart (31). The poles of X, Y disappear in this chart, and the same formal expressions are the Taylor germs of the classical projective map. This proves the claim, including $h = 0$. $\square$

Remark 6.4 (The interchange that must not be omitted). Formal coefficients in F can have arbitrarily negative fine valuations, as example 5.2 shows. Ordinary positive-valuation substitution over $\mathbb{C}$ alone does not justify their evaluation. Also, a sequence cofinal in H_α need not be cofinal in Γ . The proof above uses two separate finiteness statements: quotient Neumann finiteness controls the coarse Taylor degrees, while cofinality inside H_α controls the rational tail index. Neither statement substitutes for the other.

7 Proof of arbitrary-rank uniformization

7.1 The curve equation and the group law

Take $F = K_{H_\alpha}$, with its complete rank-one coarsening, as in lemma 5.3. Since $v(q_0) = \alpha$, the classical theorem applies to q_0 . The projective Taylor germs in lemma 6.3 satisfy the curve equation: it is an identity of classical analytic germs, hence an identity of their Taylor series. Evaluation preserves these polynomial identities. Thus Φ_q maps into $E_q(K)$.

The same argument proves

$$\Phi_q(uv) = \Phi_q(u) + \Phi_q(v) \quad (u, v \in U_q). \quad (32)$$

Here one must use regular local charts on the elliptic curve, not the chord-slope formula at a vanishing denominator. The group operation on a smooth elliptic curve is a regular morphism. At the residue points $\phi_{q_0}(u_0), \phi_{q_0}(v_0)$ and their sum, choose affine neighborhoods for which it is given by rational expressions with nonzero residue denominators. Its local Taylor identity with $\phi_Q(UV)$ follows from theorem 6.1. All denominators remain w -units after coarse evaluation, and lemma 5.1 preserves the identity. At the identity, use (31). This includes $u_0v_0 \in q_0^{\mathbb{Z}}$, opposite points, equal points, and every other case where a particular affine addition formula would be undefined. By lemma 6.3, these evaluated germs are the Hahn-defined map, proving (32) everywhere on U_q .

For clarity, this is not an appeal to a new “identity principle on all surcomplex numbers.” It is an ordinary identity of rank-one analytic germs, followed by an explicitly proved algebra homomorphism on their formal Taylor rings.

7.2 Good reduction for the coarse valuation

By (23), $v(\Delta(q)) = \alpha \in H_\alpha$, so $w(\Delta(q)) = 0$. All coefficients of the projective Weierstrass equation belong to $\mathcal{O}_w$, and its discriminant is a unit there. It is therefore a smooth proper elliptic curve over $\mathcal{O}_w$, with residue curve E_{q_0} over F . The identification of the residue curve is the statement $\text{res}_{H_\alpha}(a_k(q)) = a_k(q_0)$ for $k = 4, 6$. It holds because res_{H_α} is a ring homomorphism on $\mathcal{O}_w$ that is computed coefficientwise, hence commutes with the strong sums (3): each term $n^k q^n / (1 - q^n)$ lies in $\mathcal{O}_w$, since $v(q^n) = n\alpha > 0$ forces $v(1 - q^n) = 0$, and $\text{res}_{H_\alpha}(q) = q_0$.

Every K -point has a well-defined reduction to that curve. In the present projective model, one can see existence directly: scale its three homogeneous coordinates so that their minimum w -value is zero, and reduce the resulting integral coordinates. The equation survives reduction, and at least one reduced coordinate is nonzero. Smooth properness also implies that reduction respects the group law. We write

$$\text{red} : E_q(K) \longrightarrow E_{q_0}(F), \quad E_1(K) = \ker(\text{red}).$$

The reduction compatibility supplied by lemma 6.3 is

$$\text{red}(\Phi_q(u)) = \phi_{q_0}(u_0). \quad (33)$$

Proposition 7.1 (The whole coarse identity fiber). *The homomorphism $h \mapsto \Phi_q(1+h)$ induces an isomorphism*

$$1 + \mathfrak{m}_w \simeq E_1(K). \quad (34)$$

For the local coordinate $z = -x/y$, it has the explicit inverse $h = A(q, z)$ and satisfies $v(z) = v(h)$ for $h \neq 0$.

Proof. A point reducing to $O = [0 : 1 : 0]$ lies in the chart of (31), with $z, r \in \mathfrak{m}_w$. The discussion following lemma 6.2 proves that z uniquely determines the point and that the inverse formal parameter supplies a unique $h \in \mathfrak{m}_w$. Formal composition identities show that these maps are mutual inverses. The formal-group identity gives the group law; the valuation assertion is (30). $\square$

7.3 Surjectivity

Let $P \in E_q(K)$ be arbitrary, and reduce it to $P_0 \in E_{q_0}(F)$. By the classical Tate theorem there is $u_0 \in F^\times$ with $\phi_{q_0}(u_0) = P_0$. Regard u_0 as an element of K . Its valuation belongs to H_α , so $\Phi_q(u_0)$ is defined. By (33) the point

$$P - \Phi_q(u_0)$$

belongs to $E_1(K)$. By proposition 7.1 it equals $\Phi_q(1+h)$ for a unique $h \in \mathfrak{m}_w$. The group law now gives

$$P = \Phi_q(u_0(1+h)).$$

The product parameter is in U_q . This proves surjectivity, including points with coordinates outside every scale bounded by powers of q .

When $H_\alpha = \Gamma$, the valuation w is trivial and $\mathfrak{m}_w = 0$. The proof reduces to the classical theorem on $K_H = K$ with its rank-one coarsening. No separate nontrivial-coarse-valuation hypothesis was smuggled into the argument.

7.4 The kernel

Suppose $\Phi_q(u) = O$. Reducing and using the classical kernel shows $u_0 = q_0^m$ for an integer m . Therefore $u/q^m \in 1 + \mathfrak{m}_w$. Reindexing already gives $\Phi_q(q^m) = O$; the group law and proposition 7.1 imply $u/q^m = 1$. Conversely every q^m is mapped to O by definition. Thus $\ker \Phi_q = q^\mathbb{Z}$. Together with the support theorems and surjectivity, this completes the proof of theorem 1.2.

7.5 The exact reduction diagram

The proof gives more than an abstract isomorphism.

Corollary 7.2 (Reduction and infinitesimal lifting). *There is a commutative diagram of exact sequences*

$$\begin{array}{ccccccc} 1 & \longrightarrow & 1 + \mathfrak{m}_w & \longrightarrow & U_q/q^\mathbb{Z} & \xrightarrow{\text{res}_H} & F^\times/q_0^\mathbb{Z} \longrightarrow 1 \\ & & \downarrow \sim & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & E_1(K) & \longrightarrow & E_q(K) & \xrightarrow{\text{red}} & E_{q_0}(F) \longrightarrow 0. \end{array}$$

The left-hand isomorphism is given by a support-controlled universal formal series and preserves the original valuation of the local identity parameter.

Proof. Surjectivity of the top residue map follows by taking representatives in F . Its kernel consists of those classes whose residue is a power of q_0 ; division by the corresponding power of q gives a unique representative in $1 + \mathfrak{m}_w$. Uniqueness uses $v(q_0) > 0$. The lower row and all vertical maps were established above. $\square$

8 A second proof: the integral node chart and three formal charts

This section and the next come from source 03. They prove theorem 1.2 again, independently of sections 5 to 7: the argument uses neither Tate’s complete-field theorem (theorem 6.1), nor the rank-one completion (lemma 5.3), nor unrestricted coarse evaluation (lemma 5.1), nor the Taylor/Hahn agreement of lemma 6.3. Its only external inputs are Tate’s universal formal identities (the curve equation and the cleared secant identities) and Tate’s elementary group lemma, which is proved below. Those four results of the first route stay in the report: the reduction compatibility (33), the coarse identity fibre (proposition 7.1) and the reduction diagram (corollary 7.2) are proved only by the first route.

Standing hypotheses and notation. Throughout this section and section 9, k is an arbitrary field of characteristic zero, not necessarily $\mathbb{C}$ and not necessarily algebraically closed; Γ is an arbitrary set-sized ordered abelian group; and

$$\mathbb{K} = k((t^\Gamma)),$$

with the valuation v given by the least exponent, $\mathcal{O}_v = \{a : v(a) \geq 0\}$, its unit group $\mathcal{O}_v^\times$, and $\mathfrak{m}_v = \{a : v(a) > 0\}$. When $k = \mathbb{C}$, $\mathbb{K} = K_\Gamma$. Fix $q \in \mathfrak{m}_v \setminus \{0\}$ and $\alpha = v(q)$, and define H_α and $U_q = v^{-1}(H_\alpha) \subseteq \mathbb{K}^\times$ as in definition 1.1. The Tate sums (3), the coefficients (4), the curve (5), the functions f, g, h of section 4, the bilateral families (7)–(8) and the map (9) are defined over $\mathbb{K}$ by the same formulas. To avoid collisions with the rest of the report, source 03’s notation is changed as follows: its v, δ, D_q, ϕ_q are v, α, U_q, Φ_q here; the Weierstrass polynomial is

$$\mathcal{F}(Q, x, y) = y^2 + xy - x^3 - a_4(Q)x - a_6(Q),$$

which is unrelated to the residue workspace $F = K_{H_\alpha}$ of sections 5 to 7; Q is a formal period variable, unrelated to the series $Q(J)$ of section 11; the linear part of the node chart is written $\mathbf{L}$; the valuation of a vector is written $\mathbf{v}$; and the fine filtration of section 10.3 is written $E_0^\vee \supseteq E_1^\vee$, never E_1 , which denotes the coarse kernel of section 7.

8.1 The Tate cubic over an arbitrary coefficient field

The coefficients of a_6 are integers: for each m , $5m^3 + 7m^5$ is divisible by 12, since $m^3 \equiv m^5 \equiv m$ modulo 3, and divisibility by 4 follows by distinguishing even and odd m . Thus s_k, a_4 and a_6 are evaluations at q of integral power series, and $a_4(q) = -5q - 45q^2 - \dots$, $a_6(q) = -q - 23q^2 - \dots$. The discriminant identity (23) is an identity of integral formal power series [1], so it holds in $\mathbb{K}$ and $v(\Delta(q)) = \alpha$; alternatively, nonvanishing alone follows by substituting the coefficients into the finite discriminant polynomial and reading its linear term. Hence E_q is an elliptic curve over $\mathbb{K}$. We also use

$$s_1(q) = \sum_{n \geq 1} f(q^n), \tag{35}$$

since both sides have coefficient $\sigma_1(d) = \sum_{m|d} m$ at q^d .

The proofs of theorem 3.1 and theorem 4.1 use only strong summation, lemma 2.1, the rational identities $f(z^{-1}) = f(z)$, $g(z^{-1}) = -h(z)$, and leading exponents with nonzero leading coefficients; they use no property of $\mathbb{C}$. Both theorems therefore hold verbatim over $\mathbb{K}$. For the coordinate families this is also source 03’s own theorem (its “sharp strong-summability domain”), whose necessity proof uses exactly the witnesses $\gamma - n\alpha$, $2\gamma - 2n\alpha$ and $-\gamma - n\alpha$ of the proof of theorem 4.1; it is printed once, there. The theorem concerns the specified raw bilateral families. A procedure that first assigns values to a nonsummable family by a different regularization is a different definition and is not justified by it; theorem 10.5 imposes an additional restriction on any homomorphic extension of the canonical values.

8.2 Universal identities

With Q formal and u an indeterminate, the coordinates have expansions in $R[[Q]]$, where $R = \mathbb{Z}[u, u^{-1}, (1 - u)^{-1}]$: the degree-zero terms are $f(u)$ and $g(u)$, and the coefficient of Q^d for $d \geq 1$ is given by (104)–(105). (Sources 01 and 03 display the same formulas; they are printed once.) In that ring one has

$$\mathcal{F}(Q, X_Q(u), Y_Q(u)) = 0. \quad (36)$$

For three parameters u, u', uu' , put $(X_1, Y_1) = (X_Q(u), Y_Q(u))$, $(X_2, Y_2) = (X_Q(u'), Y_Q(u'))$ and $(X_3, Y_3) = (X_Q(uu'), Y_Q(uu'))$. The cleared secant identities are

$$(X_1 - X_2)^2 X_3 = (Y_1 - Y_2)^2 + (Y_1 - Y_2)(X_1 - X_2) - (X_1 - X_2)^2(X_1 + X_2), \quad (37)$$

$$(X_1 - X_2)Y_3 = -(X_1 - X_2)(Y_1 + X_3) + (Y_1 - Y_2)(X_1 - X_3). \quad (38)$$

These are identities in the analogous two-variable coefficient ring, with $1 - u$, $1 - u'$ and $1 - uu'$ inverted. They are established classical identities, not inferred from any finite computation: Tate proves them as universal formal identities by comparison with the complex elliptic functions [1, Theorem 1, pp. 6–7, especially (25')]. The new obligation is to justify their *Hahn* specialization.

8.3 The wide annulus and Hahn specialization

Proposition 8.1 (Summability on the wide annulus). *If $-\alpha < v(u) < \alpha$ and $u \neq 1$, both families in (7)–(8) are strongly summable, with values*

$$X_q(u) = f(u) + \sum_{n \geq 1} (f(q^n u) + f(q^n / u) - 2f(q^n)), \quad (39)$$

$$Y_q(u) = g(u) + \sum_{n \geq 1} (g(q^n u) - h(q^n / u) + f(q^n)). \quad (40)$$

Proof. All three inputs q , $a = qu$ and $b = q/u$ have positive valuation. For example, the fully expanded positive tail of f is $\sum_{n,m \geq 1} m a^m q^{(n-1)m}$. It is a formal series in a, q : for a fixed pair of nonnegative powers there are finitely many contributing (n, m) , and its monomials have positive total degree. The positive-input form of lemma 2.1 therefore proves joint summability after substitution. The other tails have the same support pattern, with different constant coefficients. The negative indices of the bilateral families are converted by $f(z^{-1}) = f(z)$ and $g(z^{-1}) = -h(z)$, and (35) supplies the constants. The central rational terms $f(u), g(u)$ are single Hahn elements and may be added to the summable tails; when u is extremely close to 1, they are not expanded as geometric series in u . $\square$

Formulas (39)–(40) are (18)–(19) with s_1 absorbed into the tails by (35), extended from the normalized annulus $0 \leq v(u) < \alpha$ to the wide annulus $-\alpha < v(u) < \alpha$. The wider range is what lets the secant identities specialize.

On the wide annulus, expanding (39)–(40) and regrouping by $d = nm$ yields the Q -expansions above. The preceding proof establishes joint summability *before* regrouping. Finite products of these jointly summable families remain jointly summable, by Hahn multiplication and finite coefficient convolution, so a polynomial identity among the formal coordinate series can be evaluated either by grouping its formal Q -coefficients first or by summing its expanded Hahn terms first. The finitely many central denominators, such as $(1 - u)^{-1}$, are fixed Hahn elements and cause no infinite coefficient-fibre issue. Hence (36) specializes to an actual point of $E_q(\mathbb{K})$. For an addition identity, normalize one argument to $0 \leq v(u) < \alpha$ and the other to $-\alpha < v(u') \leq 0$; then all three of u, u', uu' lie in the wide annulus. Excluding the central poles for the moment, the same joint-summability argument applies to (37)–(38). This closes the specialization gap that would remain after merely invoking the classical real-valued convergence theorem.

8.4 An integral chart for the whole nodal smoothing

The special fibre of E_q is the nodal cubic $y^2 + xy = x^3$. The Tate parameter has two branches near its node. Instead of choosing one branch by a square root of q , introduce independent variables U, V and set $Q = UV$.

Definition 8.2 (The node chart). In $\mathbb{Z}[[U, V]]$ define

$$\mathcal{X}(U, V) = f(U) + \sum_{n \geq 1} (f(U^{n+1}V^n) + f(U^{n-1}V^n) - 2f(U^nV^n)), \quad (41)$$

$$\mathcal{Y}(U, V) = g(U) + \sum_{n \geq 1} (g(U^{n+1}V^n) - h(U^{n-1}V^n) + f(U^nV^n)), \quad (42)$$

where every f, g, h is expanded at zero, as $f(z) = \sum_{m \geq 1} m z^m$, $g(z) = \sum_{m \geq 1} \frac{m(m-1)}{2} z^m$ and $h(z) = \sum_{m \geq 1} \frac{m(m+1)}{2} z^m$. We write $T = (\mathcal{X}, \mathcal{Y})$ for the node chart.

The series are well defined in the total-degree formal topology: the smallest degree in the n th bracket is at least $2n - 1$, so only finitely many brackets contribute below any fixed total degree. Their first terms are

$$\mathcal{X}(U, V) = U + V + 2U^2 - 2UV + 2V^2 + 3U^3 + U^2V + UV^2 + 3V^3 + O((U, V)^4), \quad (43)$$

$$\mathcal{Y}(U, V) = -V + U^2 + UV - 3V^2 + 3U^3 - UV^2 - 6V^3 + O((U, V)^4). \quad (44)$$

The linear part is

$$\mathbf{L} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{L}^{-1} = \mathbf{L}. \quad (45)$$

Under $U = u$, $V = q/u$ one has $q^n u = U^{n+1}V^n$, $q^n/u = U^{n-1}V^n$ and $q^n = U^nV^n$, so $T(u, q/u) = (X_q(u), Y_q(u))$ whenever $0 < v(u) < \alpha$, by (39)–(40) and lemma 2.1.

Theorem 8.3 (Explicit formal smoothing coordinates). *There are unique $\mathcal{U}, \mathcal{V} \in \mathbb{Z}[[x, y]]$ with zero constant terms which form the two-sided formal inverse of $(\mathcal{X}, \mathcal{Y})$. Put $Q(x, y) = \mathcal{U}(x, y)\mathcal{V}(x, y)$. Then*

$$\mathcal{F}(UV, \mathcal{X}(U, V), \mathcal{Y}(U, V)) = 0, \quad \mathcal{F}(Q(x, y), x, y) = 0. \quad (46)$$

Moreover, there is a formal unit $\mathcal{A} \in \mathbb{Z}[[Q, x, y]]$ with $\mathcal{A}(0, 0, 0) = 1$ such that

$$\mathcal{F}(Q, x, y) = (Q - Q(x, y)) \mathcal{A}(Q, x, y). \quad (47)$$

Thus the formal Tate smoothing is explicitly equivalent to $UV = Q$.

Proof. The integral inverse follows from (45) and lemma 2.2. Substitute $Q = UV$, $u = U$ into the formal Tate identity (36). By (104)–(105), a term $Q^d u^{-m}$ has $m \leq d$ and becomes $U^{d-m} V^d$, of nonnegative exponents and total degree at least d . The central denominator $1 - U$ is a formal unit. Therefore the substitution is legitimate in $\mathbb{Z}[[U, V]]$ and gives the first identity in (46). Composition with the formal inverse gives the second.

Finally, $\partial \mathcal{F} / \partial Q$ at the origin equals 1, because $a_6(Q) = -Q + O(Q^2)$. Subtract the value at $Q = Q(x, y)$ from $\mathcal{F}(Q, x, y)$ and use the formal difference quotient in Q . This yields (47); its constant coefficient is that same derivative, namely 1. $\square$

Proposition 8.4 (First inverse coefficients). *The inverse starts as follows:*

$$\mathcal{U}(x, y) = x + y - 3x^2 - 7xy - 3y^2 + 13x^3 + 46x^2y + 47xy^2 + 16y^3 + O((x, y)^4), \quad (48)$$

$$\mathcal{V}(x, y) = -y + x^2 + xy - 3y^2 - 2x^3 - xy^2 - 16y^3 + O((x, y)^4), \quad (49)$$

$$Q(x, y) = -xy - y^2 + x^3 + 5x^2y + 5xy^2 + O((x, y)^4). \quad (50)$$

Proof. Start with $(U, V) = (x + y, -y)$. At each degree substitute the already known coefficients into (41)–(42) and cancel the homogeneous error using $\mathbf{L}^{-1} = \mathbf{L}$. This gives the displayed coefficients by finite integer arithmetic. The program code/03-tate-uniformization-verify.py implements this recursion and checks it through total degree 12 (section 22). $\square$

8.5 Actual nodal points

Theorem 8.5 (Nodal inverse over a Hahn field). *The map*

$$\{u \in \mathbb{K}^\times : 0 < v(u) < \alpha\} \longrightarrow \{(x, y) \in \mathfrak{m}_v^2 : \mathcal{F}(q, x, y) = 0\}, \quad u \longmapsto (X_q(u), Y_q(u)), \quad (51)$$

is a bijection. Its inverse is $u = \mathcal{U}(x, y)$, and $\mathcal{V}(x, y) = q/u$.

Proof. For u in the node annulus $0 < v(u) < \alpha$, both u and q/u lie in $\mathfrak{m}_v$. Lemma 2.1 evaluates the node chart there, and (39)–(40) identify it with the Tate coordinates. The curve identity is (46).

Conversely let $x, y \in \mathfrak{m}_v$ satisfy the curve equation. Evaluate the formal inverse at (x, y) and write $U = \mathcal{U}(x, y)$, $V = \mathcal{V}(x, y)$. Both are infinitesimal. Evaluating (47) shows $(q - UV)\mathcal{A}(q, x, y) = 0$. The second factor has residue 1 and is a unit, so $q = UV$. Since $q \neq 0$, both factors are nonzero and $v(U) + v(V) = \alpha$. Hence $0 < v(U) < \alpha$, and the two-sided inverse identity gives $(x, y) = (X_q(U), Y_q(U))$. Uniqueness follows from the same identity. $\square$

8.6 Points with nonsingular affine reduction

Lemma 8.6 (Normalization of the special cubic). *Every k -point $(x_0, y_0) \neq (0, 0)$ on $y^2 + xy = x^3$ is uniquely of the form*

$$x_0 = \frac{c}{(1-c)^2}, \quad y_0 = \frac{c^2}{(1-c)^3}, \quad c \in k^\times \setminus \{1\}.$$

The parameter is $c = y_0/(x_0 + y_0)$.

Proof. If $x_0 = 0$, then $y_0 = 0$. If $x_0 + y_0 = 0$, the equation forces $x_0^3 = 0$ and again gives the node. Thus both denominators used above are nonzero. The value of c is neither 0 nor 1. Substituting $y_0 = cx_0/(1-c)$ into the equation and cancelling x_0^2 yields $x_0 = c/(1-c)^2$, and the y_0 formula follows. $\square$

Proposition 8.7 (Smooth-residue inverse). *Let $(x, y) \in E_q(\mathbb{K})$ have $x, y \in \mathcal{O}_v$ and residue pair $(x_0, y_0) \neq (0, 0)$. There is a unique $u \in \mathcal{O}_v^\times$ with residue $c = y_0/(x_0 + y_0)$ and $(x, y) = (X_q(u), Y_q(u))$. It is obtained by a formal series in q and $y/(x + y) - c$ with coefficients in k .*

Proof. Write $u = c + s$. The formulas for the tails and the fact that both c and $1 - c$ are nonzero make $X_Q(c + s)$ and $Y_Q(c + s)$ formal series in (Q, s) with coefficients in k . Their sum is a unit, with constant term $h(c) \neq 0$, so

$$R(Q, s) = \frac{Y_Q(c + s)}{X_Q(c + s) + Y_Q(c + s)} \in k[[Q, s]]$$

is defined. At $Q = 0$ the rational functions satisfy $g(u)/(f(u) + g(u)) = u$, whence $R(0, s) = c + s$. The equation $R(Q, s) = c + r_1$ therefore has a unique formal solution $s = S_c(Q, r_1)$ with zero constant term (lemma 2.2). Specialize at $r_1 = y/(x + y) - c \in \mathfrak{m}_v$.

Set $(x', y') = (X_q(c + s), Y_q(c + s))$. This point and (x, y) have the same ratio $r = y/(x + y)$ and the same residue pair. The ratio satisfies $r \equiv c$, and $1 - r$ is a unit. Any such point has $y = rx/(1 - r)$, and its x -coordinate solves

$$x^3 - \frac{r}{(1 - r)^2}x^2 + a_4(q)x + a_6(q) = 0. \quad (52)$$

At $(q, r, x) = (0, c, x_0)$ the derivative with respect to x is x_0^2 , which is nonzero. Uniqueness can also be seen directly among actual Hahn points. Put $d_r = r/(1 - r)^2$ and subtract the two polynomial equations:

$$0 = (x - x')(x^2 + xx' + x'^2 - d_r(x + x') + a_4(q)).$$

The second factor has residue $3x_0^2 - 2x_0^2 = x_0^2 \neq 0$, so it is a unit and $x = x'$. Then $y = y'$. Any other u with residue c would solve the same invertible equation $R(q, u - c) = r$, so it is equal to this one. All substitutions use the positive-valuation inputs $q, r - c$ or $q, u - c$; the possibly noninfinitesimal constants c, x_0 have first been translated into the coefficient field. $\square$

8.7 Points reducing to infinity

Lemma 8.8 (The valuation pattern at infinity). *If an affine point $(x, y) \in E_q(\mathbb{K})$ has either coordinate of negative valuation, then*

$$v(x) = -2\epsilon, \quad v(y) = -3\epsilon \quad \text{for some } \epsilon \in \Gamma_{>0}.$$

In particular $z = -x/y$ and $r = -1/y$ belong to $\mathfrak{m}_v$.

Proof. Put $a = v(x)$ and $b = v(y)$. If $b < 0$ and $a \geq 0$, the left side $y^2 + xy$ has valuation $2b < 0$, whereas the right side has nonnegative valuation. This is impossible. Thus a negative coordinate forces $a < 0$. The right side then has valuation $3a$, because $v(a_4(q)), v(a_6(q)) > 0$. If $b \geq a$, both left-side terms have valuation at least $2a > 3a$, also impossible. Hence $b < a$, so y^2 strictly dominates xy and $2b = 3a$. Set $\epsilon = a - b > 0$. The equality gives $a = -2\epsilon$ and $b = -3\epsilon$, without dividing an element of Γ by 2 or 3. $\square$

Proposition 8.9 (Formal inverse at infinity). *Every point of lemma 8.8 is $\Phi_q(u)$ for a unique $u \in 1 + \mathfrak{m}_v$, $u \neq 1$. The inverse $s = u - 1$ is a zero-constant formal series in q and $z = -x/y$. The identity O corresponds to $u = 1$.*

Proof. In the coordinates $z = -x/y, r = -1/y$ the curve equation is (31). Its derivative with respect to r at the origin is 1. It therefore has a unique formal solution $r = r_\infty(Q, z)$ with zero constant term, which specializes uniquely on $\mathfrak{m}_v^2$; it begins with z^3 .

For $u = 1 + s$ the central terms of the Tate coordinates give

$$X_Q(1 + s) = s^{-2}N_X(Q, s), \quad Y_Q(1 + s) = s^{-3}N_Y(Q, s),$$

where $N_X, N_Y \in k[[Q, s]]$ satisfy $N_X(0, 0) = 1$ and $N_Y(0, 0) = -1$: the tails are regular in (Q, s) , $f(1 + s) = (1 + s)/s^2$ and $g(1 + s) = -(1 + s)^2/s^3$. Thus

$$-\frac{X_Q(1 + s)}{Y_Q(1 + s)} = -s \frac{N_X(Q, s)}{N_Y(Q, s)}$$

is a formal series vanishing at $s = 0$ with derivative 1 in s at the origin. It is the series $B(Q, s)$ of (28), and its inverse in the second variable is the series $A(Q, z)$ of lemma 6.2; source 01 evaluates them at the coarse valuation, here they are evaluated at the fine one. Put $s = A(q, z)$.

Since $A(Q, 0) = 0$ and its linear coefficient in z is 1, A is z times a formal unit. Thus $z \neq 0$ implies $s \neq 0$ and $v(s) = v(z) > 0$. After specialization $N_X(q, s)$ and $N_Y(q, s)$ are units, so the resulting Tate point has coordinate valuations $-2v(s)$ and $-3v(s)$. In particular, its $r = -1/Y_q(1 + s)$ is infinitesimal. It has the same z and hence, by (31) and the actual-input uniqueness in lemma 2.2, the same $r = r_\infty(q, z)$ as the original point. Since $x = z/r$ and $y = -1/r$, the points agree. The formal inverse also proves uniqueness. $\square$

8.8 Constructive surjectivity

Theorem 8.10 (Constructive surjectivity). *The map $\Phi_q : U_q \rightarrow E_q(\mathbb{K})$ is surjective as a map of sets. An inverse modulo q^Z can be constructed in the same Hahn field from the input q, x, y by one of the three formal charts above.*

Proof. The point O is already in the image. For an affine point, a negative coordinate puts it in proposition 8.9. Otherwise both coordinates belong to O_v , and their residue lies on the special cubic. If the residue is the node, apply theorem 8.5; otherwise apply proposition 8.7. These cases exhaust all possibilities. The annular parameter has valuation in $(0, \alpha)$, while the other two parameters are units, so all belong to U_q . $\square$

The charts in fact give a bijection from the normalized annulus $\{u : 0 \leq v(u) < \alpha\}$ onto $E_q(\mathbb{K})$, before proving the group law. Their images are disjoint and can be read from the input point:

point	normalized parameter	inverse
O	$u = 1$	1
$x, y \in \mathfrak{m}_v$	$0 < v(u) < \alpha$	$\mathcal{U}(x, y)$
nonsingular affine residue	$v(u) = 0, \bar{u} = c \neq 1$	$c + S_c(q, y/(x + y) - c)$
a negative coordinate valuation	$u \in 1 + \mathfrak{m}_v, u \neq 1$	$1 + A(q, -x/y)$

For the node row, use theorem 8.5. For the smooth row, the positive tails vanish in the residue field, leaving $(f(c), g(c))$. For the last row, the unit numerators at infinity give the negative valuations just computed. Thus every normalized parameter lies in exactly one row, and the inverse in that row is unique.

The argument is constructive in the formal-series sense. It does not assume that k is algebraically closed or that the value group permits root extraction; even the valuation pattern at infinity was obtained from integer linear combinations, without division in Γ . The node row may be empty: when $\Gamma = \mathbb{Z}$ and $\alpha = 1$, there is no value strictly between 0 and α , so the nodal inverse theorem also says that no $\mathbb{K}$ -point reduces to the node.

8.9 The group law and the kernel

Strong reindexing of the bilateral families gives

$$X_q(qu) = X_q(u), \quad Y_q(qu) = Y_q(u), \quad (53)$$

and the rational identities $f(z^{-1}) = f(z)$, $g(z^{-1}) = -f(z) - g(z)$ give the inversion identities of (25), $X_q(u^{-1}) = X_q(u)$ and $Y_q(u^{-1}) = -X_q(u) - Y_q(u)$, over $\mathbb{K}$. This is negation on the Weierstrass model. For two affine points with distinct x -coordinates the group law is

$$\lambda = \frac{y_1 - y_2}{x_1 - x_2}, \quad x_3 = \lambda^2 + \lambda - x_1 - x_2, \quad y_3 = -(1 + \lambda)x_3 - y_1 + \lambda x_1,$$

and the specialized identities (37)–(38) say precisely that the third point is $\Phi_q(uu')$. The cases involving O or a product in $q^{\mathbb{Z}}$ follow from (53) and inversion. To avoid a topological limit argument at equal x -coordinates, use Tate's elementary group lemma [1, Lemma 1].

Lemma 8.11 (Tate's generic-pair lemma). *Let $\psi : G \rightarrow A'$ map a multiplicative group to an abelian group, with infinite image. If $\psi(ab) = \psi(a) + \psi(b)$ whenever $\psi(a) \neq \pm\psi(b)$, then ψ is a homomorphism.*

Proof. For fixed a, b , choose c whose image avoids the finite collection specified by

$$\psi(c) \neq \pm\psi(a), \quad \psi(c) + \psi(a) \neq \pm\psi(b), \quad \psi(c) \neq \pm\psi(ab).$$

The assumed rule gives $\psi(c) + \psi(a) + \psi(b) = \psi(ca) + \psi(b) = \psi(cab) = \psi(c) + \psi(ab)$. Cancel $\psi(c)$. $\square$

Theorem 8.12 (Group uniformization). *The map $\Phi_q : U_q \rightarrow E_q(\mathbb{K})$ is a group homomorphism, and its kernel is exactly $q^{\mathbb{Z}}$.*

Proof. Two affine points have the same x -coordinate exactly when they are equal or negatives, since their y -coordinates are the roots of the same quadratic. Thus the secant argument proves the hypothesis of lemma 8.11. The image is infinite: for $u_n = 1 + q^n$, $n \geq 1$, the central term of $X_q(u_n)$ has valuation $-2n\alpha$, whereas its tails have positive valuation, so these values are distinct. The lemma proves the group law everywhere.

By definition $\Phi_q(q^n) = O$. Every other argument in U_q has well-defined affine coordinates, so cannot map to O . This proves the kernel assertion. $\square$

Theorem 8.13 (Theorem 1.2 over any coefficient field of characteristic zero). *Let k be a field of characteristic zero, Γ a set-sized ordered abelian group, $\mathbb{K} = k((t^\Gamma))$, and $q \in \mathbb{K}$ with $v(q) = \alpha > 0$. The theta family (6) is strongly summable exactly for $u \in U_q$; for $u \in \mathbb{K}^\times \setminus q^{\mathbb{Z}}$ each of the two bilateral coordinate families is strongly summable exactly for $u \in U_q$; and Φ_q is a surjective group homomorphism $U_q \rightarrow E_q(\mathbb{K})$ with kernel $q^{\mathbb{Z}}$, so*

$$U_q/q^{\mathbb{Z}} \simeq E_q(\mathbb{K}).$$

Every coset has a unique representative whose valuation lies in $[0, \alpha)$. No divisibility of Γ , algebraic closedness of k , or rank-one hypothesis is required. For $k = \mathbb{C}$ this is theorem 1.2.

Proof. Summability: theorems 3.1 and 4.1, whose proofs hold over $\mathbb{K}$ as noted at the start of this section. Homomorphism and kernel: theorem 8.12. Surjectivity: theorem 8.10. Normal forms: lemma 2.3. $\square$

9 Support monoids and exact valuation geometry

The standing hypotheses of section 8 remain in force. For a vector $a = (a_1, \dots, a_d)$ of elements of $\mathbb{K}$ put

$$\mathbf{v}(a_1, \dots, a_d) = \min_j v(a_j).$$

This records closeness with values in $\Gamma \cup \{\infty\}$; it is not exponentiated into a real-valued distance.

9.1 Equality of generated support monoids

Theorem 9.1 (Equality of generated support monoids). *For any $U, V \in \mathfrak{m}_v$ and $(x, y) = (X(U, V), \mathcal{Y}(U, V))$,*

$$\langle \text{supp } U \cup \text{supp } V \rangle_{\mathbb{N}} = \langle \text{supp } x \cup \text{supp } y \rangle_{\mathbb{N}}. \quad (54)$$

In particular, for $0 < v(u) < \alpha$, $\langle \text{supp } u \cup \text{supp } (q/u) \rangle_{\mathbb{N}} = \langle \text{supp } X_q(u) \cup \text{supp } Y_q(u) \rangle_{\mathbb{N}}$.

Proof. Write $M_{U,V} = \langle \text{supp } U \cup \text{supp } V \rangle_{\mathbb{N}}$ and $M_{x,y} = \langle \text{supp } x \cup \text{supp } y \rangle_{\mathbb{N}}$. The support bound (13), applied to $X, \mathcal{Y}$, gives $M_{x,y} \subseteq M_{U,V}$. Applying it to the zero-constant formal inverse $\mathcal{U}, \mathcal{V}$ gives $M_{U,V} \subseteq M_{x,y}$. This proves equality without requiring equality of individual supports. $\square$

Corollary 9.2 (Exact monoidal descent). *Let $M \subseteq \Gamma_{\geq 0}$ be an additive submonoid, and let $\mathfrak{m}_M$ be the elements of $\mathfrak{m}_v$ whose supports lie in M . For $0 \neq q \in \mathfrak{m}_M$, the node chart restricts to a bijection*

$$\{(U, V) \in \mathfrak{m}_M^2 : UV = q\} \simeq \{(x, y) \in \mathfrak{m}_M^2 : \mathcal{F}(q, x, y) = 0\}.$$

No enlargement of the positive exponent monoid is required in either direction.

Proof. Every finite sum of exponents in M remains in M , so both formal maps preserve $\mathfrak{m}_M^2$. Theorem 8.5 and its proof then restrict verbatim. $\square$

This retains more information than saying that the answer stays in the original Hahn *field*: it identifies the positive monoid needed for the inverse. The individual supports need not be equal; substantial cancellation can occur in one coordinate, and the theorem is formulated so that this does not destroy the invariant. It refines, in the node tube, the one-directional certificate (21) of theorem 4.1, which bounds the supports of $X_q(u) - f(u)$ and $Y_q(u) - g(u)$ by the monoid of $\text{supp}(q) \cup \text{supp}(qu) \cup \text{supp}(q/u)$. The analogous statement for the modular parameter, with generated groups instead of monoids, is (68) of theorem 11.1.

9.2 An isometry principle for formal charts

Lemma 9.3 (Valuation isometry). *Let $A(Z) \in k[[Z_1, \dots, Z_d]]^d$ have zero constant term and linear part $L_A \in \text{GL}_d(k)$. Then for $a, b \in \mathfrak{m}_v^d$,*

$$\mathbf{v}(A(a) - A(b)) = \mathbf{v}(a - b). \quad (55)$$

Proof. The assertion is immediate if $a = b$. Otherwise put $\rho_0 = \mathbf{v}(a - b)$ and let $\epsilon > 0$ be the least valuation among all nonzero coordinates of the two finite vectors a, b . Write $A(Z) = L_A Z + N(Z)$, where N has degree at least two. The difference of any monomial of degree at least two is a sum of terms containing one coordinate of $a - b$ and at least one additional coordinate of a or b . Formally this gives $N(a) - N(b) = C(a, b)(a - b)$, where every entry of the formal matrix C has zero constant term. This factorization specializes by lemma 2.1;

every nonzero entry of $C(a, b)$ has valuation at least ϵ . Therefore $\mathbf{v}(N(a) - N(b)) \geq \rho_0 + \epsilon > \rho_0$. A constant matrix has entries of nonnegative valuation, and so does L_A^{-1} . Applying both matrices gives $\mathbf{v}(L_A(a - b)) = \mathbf{v}(a - b) = \rho_0$. The nonlinear term cannot cancel all coordinates attaining that least valuation. $\square$

For $d = 1$ and linear coefficient 1 this is the isometry (30) of lemma 6.2, which source 01 proves for the coarse identity parameter.

Corollary 9.4 (The node chart on the infinitesimal bidisk). *The map $T = (X, Y)$ is a bijection $\mathfrak{m}_v^2 \rightarrow \mathfrak{m}_v^2$, and $\mathbf{v}(T(a, b) - T(c, d)) = \mathbf{v}(a - c, b - d)$ for all $a, b, c, d \in \mathfrak{m}_v$.*

Proof. Lemma 2.2 with (45) and lemma 9.3. $\square$

Theorem 9.5 (Two-period comparison). *Let $q, q' \in \mathfrak{m}_v \setminus \{0\}$, and choose $u, u' \in \mathbb{K}^\times$ with $0 < v(u) < v(q)$ and $0 < v(u') < v(q')$. Then*

$$\min\{v(X_q(u) - X_{q'}(u')), v(Y_q(u) - Y_{q'}(u'))\} = \min\{v(u - u'), v(q/u - q'/u')\}. \quad (56)$$

This identity remains valid when one difference is zero, with $v(0) = \infty$.

Proof. Both points are values of the *same* formal map T , at $(u, q/u)$ and $(u', q'/u')$. Apply corollary 9.4. $\square$

Corollary 9.6 (Fixed-period distance and coordinate size). *For $q' = q$ and $u \neq u'$ in the node annulus,*

$$\min\{v(X_q(u) - X_q(u')), v(Y_q(u) - Y_q(u'))\} = v(u - u') + \min\{0, \alpha - v(u) - v(u')\}. \quad (57)$$

Also

$$\min\{v(X_q(u)), v(Y_q(u))\} = \min\{v(u), \alpha - v(u)\}. \quad (58)$$

Proof. The identity $q/u - q/u' = q(u' - u)/(uu')$ gives the second valuation in (56). For (58), compare $T(u, q/u)$ with $T(0, 0) = (0, 0)$ using corollary 9.4. $\square$

Corollary 9.7 (Exact period sensitivity). *Suppose $q \neq q'$ and $0 < v(u) < \min\{v(q), v(q')\}$. Then*

$$\min\{v(X_q(u) - X_{q'}(u)), v(Y_q(u) - Y_{q'}(u))\} = v(q - q') - v(u). \quad (59)$$

Proof. In (56) the first difference is zero and the second is $(q - q')/u$. $\square$

These are equalities, not lower estimates from a triangle inequality: at least one affine coordinate detects the full change of the branch coordinates. They include the balanced case $2v(u) = \alpha$, where the leading terms of $X_q(u)$ may cancel, and smaller scales inside a proper convex subgroup, which a rank-one quotient of Γ would erase. No noncancellation hypothesis on leading coefficients is imposed.

9.3 Sharpness and cancellation

For a balanced example take $k = \mathbb{C}$, $q = t^2$ and $u = it$. Then $q/u = -it$, and (43) shows cancellation of the linear X -terms:

$$X_{t^2}(it) = -6t^2 + O(t^4), \quad Y_{t^2}(it) = it + O(t^2).$$

The first coordinate alone would suggest depth 2; the vector depth is 1, as (58) requires. No general-position assumption can replace the use of both coordinates.

For period sensitivity take $q = t^3$, $q' = t^3 + t^N$ with an integer $N > 3$, and $u = t$. Then (59) gives exact depth $N - 1$: the formula identifies the valuation of the difference, not merely an upper bound for its size.

To see what a rank-one quotient loses, work over $\Gamma = \mathbb{Z} \oplus_{\text{lex}} \mathbb{Z}$ with

$$q = t^{(2,0)}, \quad u = t^{(1,0)}, \quad u' = t^{(1,0)}(1 + t^{(0,1)}).$$

Both parameters have value $(1, 0)$, whereas $v(u - u') = (1, 1)$. The fixed-period formula therefore gives vector difference value $(1, 1)$, while each point separately has vector value $(1, 0)$. The first-coordinate quotient assigns value 1 to all three: it detects the common leading depth but loses the extra closeness in the second coordinate.

10 Scalar extension and the valuation-circle quotient

10.1 Strong functoriality

Theorem 10.1 (Exponent embeddings preserve uniformization). *Let $\iota : \Gamma \hookrightarrow \Gamma'$ be an ordered group embedding. The coefficient-preserving map*

$$\iota_* : K_\Gamma \hookrightarrow K_{\Gamma'}, \quad \sum a_\gamma t^\gamma \mapsto \sum a_\gamma t^{\iota(\gamma)}$$

is a field embedding preserving strong sums. If $q' = \iota_(q)$, then*

$$\iota_*(\Phi_q(u)) = \Phi_{q'}(\iota_*(u)) \quad (u \in U_q), \quad (60)$$

with the corresponding identities for Θ, X, Y where defined. Moreover

$$\iota^{-1}(H_{\iota(\alpha)}) = H_\alpha. \quad (61)$$

In the larger field the full admissible domain is $v^{-1}(H_{\iota(\alpha)})$, not in general all of $K_{\Gamma'}^\times$.

Proof. An order embedding preserves and reflects well-ordering and preserves finite coefficient contributions. It therefore preserves the defining Hahn operations and every strongly summable family. Apply this to the bilateral formulas, or to their positive-tail forms. The point at infinity and the powers of q are also preserved. Finally, inequalities $|\iota(\gamma)| \leq N\iota(\alpha)$ are equivalent to $|\gamma| \leq N\alpha$, proving (61). The exact-domain theorems over Γ' give the last assertion. $\square$

This theorem has a useful asymmetric meaning. An old parameter is admissible after an ordered exponent embedding if and only if it was admissible before, by (61). But the assertion “every nonzero parameter is admissible” may fail after enlargement: new valuations can lie outside the old q -scale. The curve acquires new points, yet its correct, enlarged admissible subgroup still reaches all of them.

Example 10.2 (A rank-one base and a rank-two extension). Embed $\mathbb{R}$ as the second summand in $\Gamma' = \mathbb{R} \oplus_{\text{lex}} \mathbb{R}$, and take $q = t^{(0,1)}$. Over $\mathbb{C}((t^{\mathbb{R}}))$, every nonzero parameter is admitted. Over $K_{\Gamma'}$, a parameter of valuation $(1,0)$ is not admitted. A parameter $1 + t^{(1,0)}$ is admitted and yields a new point reducing to O for the coarse first-coordinate valuation. Thus excluding a new multiplicative scale does not exclude its infinitesimal displacement from a nonzero old parameter.

10.2 A canonical circularly ordered value quotient

Let $O_v^\times = \{u \in K^\times : v(u) = 0\}$ be the units of the original Hahn valuation ring.

Theorem 10.3 (The valuation quotient). *There is a canonical surjective group homomorphism*

$$\text{trop}_q : E_q(K) \longrightarrow H_\alpha/\mathbb{Z}\alpha, \quad \text{trop}_q(\Phi_q(u)) = v(u) + \mathbb{Z}\alpha, \quad (62)$$

and an exact sequence

$$1 \longrightarrow O_v^\times \xrightarrow{\Phi_q} E_q(K) \xrightarrow{\text{trop}_q} H_\alpha/\mathbb{Z}\alpha \longrightarrow 0. \quad (63)$$

Every value class has a unique representative in $[0, \alpha)$. The ordering of that interval defines a translation-invariant circular order on the quotient, with addition followed by reduction modulo α .

Proof. Two parameters represent the same point exactly when their quotient is a power of q , so their valuations differ by an integer multiple of α . This proves well-definedness. Every $\gamma \in H_\alpha$ is the valuation of t^γ , proving surjectivity. If $v(u) = m\alpha$, the same point is represented by u/q^m , which is a valuation-zero unit. Conversely units map to zero in the value quotient. Their map into $E_q(K)$ is injective, since a power of q has valuation zero only when it is 1. Finite normalization gives the unique representative. The usual cyclic order of three representatives in a fundamental half-open interval is independent of simultaneous translation, by cutting the interval at the translating point. $\square$

There is a second exact sequence of additive groups

$$0 \longrightarrow H_\alpha^- \longrightarrow H_\alpha/\mathbb{Z}\alpha \longrightarrow (H_\alpha/H_\alpha^-)/\mathbb{Z}\tilde{\alpha} \longrightarrow 0. \quad (64)$$

The first map sends h to $h + \mathbb{Z}\alpha$ and is injective because $H_\alpha^- \cap \mathbb{Z}\alpha = 0$. If the image of $h + \mathbb{Z}\alpha$ in the last term vanishes, then $h - m\alpha \in H_\alpha^-$ for some $m \in \mathbb{Z}$; hence that class comes from the first map. Surjectivity follows from the quotient map $H_\alpha \rightarrow H_\alpha/H_\alpha^-$. This proves exactness.

It is the *last* term of (64) that becomes the real circle $\mathbb{R}/v_1(q)\mathbb{Z}$ when the rank-one group H_α/H_α^- is all of $\mathbb{R}$ under the chosen embedding. The full quotient $H_\alpha/\mathbb{Z}\alpha$ still retains the kernel H_α^- . For example, with $\Gamma = \mathbb{R} \oplus_{\text{lex}} \mathbb{R}$ and $\alpha = (1, 0)$, the map

$$(\mathbb{R} \oplus_{\text{lex}} \mathbb{R})/\mathbb{Z}(1, 0) \longrightarrow \mathbb{R}/\mathbb{Z}, \quad [(a, b)] \longmapsto a + \mathbb{Z}$$

has kernel $\{(0, b) : b \in \mathbb{R}\}$. A positive class $[(0, b)]$ remains before $[(1/2, 0)]$ after every positive integer multiple, in the normalized circular order. An ordinary real circle has no such nonzero infinitesimal direction.

When $H_\alpha^- = 0$, the full quotient embeds in the real circle and equals it if the embedded value group is $\mathbb{R}$. Over a complete algebraically closed rank-one field, Tate's analytic skeleton is the whole real circle; the values of field-rational parameters give its value-group subgroup, which can be proper if the value group is smaller than $\mathbb{R}$ [7, Remark 4.24]. At arbitrary rank, (63) is an *algebraic and circular-order* assertion. It does not construct an ordinary real-metric Berkovich skeleton retaining all the infinitesimal value directions. Higher-rank analytifications require their own definitions [8].

The sequence (63) has a log-geometric precedent. In the Tate-curve example of Molcho and Wise [16, §5.1], the logarithmic Picard group is the part of $\mathbf{G}_{\log}$ bounded by the loop length in the sense of their Definition 2.1.3.1, modulo a lift of that length, and their tropicalization sequence (5.1.4) presents it as an extension of the bounded tropical quotient by $\mathbf{G}_m$. For an ordered value group and one length, the bounded subgroup is H_α , and (63) is the explicit valuative counterpart of that extension: its kernel is the unit group $O_v^\times$ and its quotient is $H_\alpha/\mathbb{Z}\alpha$. Theorem 10.3 is therefore not claimed as a new shape of result; what is added is its explicit Hahn realization. Source 03 proves the same sequence (its “value exact sequence”, with its value map τ in place of trop_q) over any field k of characteristic zero, by the same argument; by theorem 8.13 the statement and proof of theorem 10.3 hold verbatim over $k((t^\Gamma))$. It is printed once, here.

10.3 The fine filtration and the extension obstruction

This subsection comes from source 03, and the standing hypotheses of section 8 apply: k is any field of characteristic zero and $\mathbb{K} = k((t^\Gamma))$. Define the subgroups

$$E_0^v = \Phi_q(O_v^\times), \quad E_1^v = \Phi_q(1 + \mathfrak{m}_v)$$

of $E_q(\mathbb{K})$. The three charts of section 8 identify E_0^v with the points whose projective reduction for the fine valuation v , a point of the nodal cubic over k , is nonsingular, and E_1^v with the points reducing to O . More explicitly, E_1^v consists of O and the affine points with the negative coordinate valuations of lemma 8.8; the other points of E_0^v have nonsingular affine reduction. The *node tube* is the complement of E_0^v in $E_q(\mathbb{K})$.

The superscript v is essential. The group $E_1(K)$ of section 7 is the kernel of reduction for the *coarse* valuation $w = v \bmod H_\alpha$; by proposition 7.1 it equals $\Phi_q(1 + \mathfrak{m}_w)$. Since $\mathfrak{m}_w \subsetneq \mathfrak{m}_v$ (for instance $q \in \mathfrak{m}_v \setminus \mathfrak{m}_w$) and Φ_q is injective on $1 + \mathfrak{m}_v$, one has $E_1(K) \subsetneq E_1^v$ for $k = \mathbb{C}$: the two filtrations never coincide.

Proposition 10.4 (Fine residue sequence). *The sequence*

$$0 \longrightarrow E_1^v \longrightarrow E_0^v \xrightarrow{\text{res} \circ \Phi_q^{-1}} k^\times \longrightarrow 0 \quad (65)$$

is exact, and $E_1^v \simeq (1 + \mathfrak{m}_v, \cdot)$. Here 0 denotes the trivial group even when a term is written multiplicatively.

Proof. Φ_q is injective on $O_v^\times$, because a nontrivial power of q is not a unit, so $\text{res} \circ \Phi_q^{-1}$ is well defined on E_0^v . Residue on $O_v^\times$ is onto $k^\times$ with kernel $1 + \mathfrak{m}_v$. $\square$

Theorem 10.5 (No full-domain extension with the same kernel). *Suppose $H_\alpha \neq \Gamma$. There is no group homomorphism $\Psi : \mathbb{K}^\times \rightarrow E_q(\mathbb{K})$ extending Φ_q and having kernel $q^\mathbb{Z}$. More precisely, any homomorphic extension Ψ has an exact sequence*

$$0 \longrightarrow q^\mathbb{Z} \longrightarrow \ker \Psi \xrightarrow{v \bmod H_\alpha} \Gamma/H_\alpha \longrightarrow 0. \quad (66)$$

Proof. Given $a \in \mathbb{K}^\times$, surjectivity of Φ_q supplies $d \in U_q$ with $\Phi_q(d) = \Psi(a)$. Hence $ad^{-1} \in \ker \Psi$, and its valuation modulo H_α equals that of a . This proves surjectivity of the last map in (66). Its kernel is $\ker \Psi \cap U_q = \ker \Phi_q = q^\mathbb{Z}$. If Γ/H_α is nontrivial, the extension kernel is strictly larger than $q^\mathbb{Z}$. $\square$

This is not an assertion that every imaginable noncanonical extension is impossible. It says that an extension cannot preserve both the canonical values on U_q and the intended period kernel. It is independent of the failure of the raw sums outside U_q : one is a support obstruction, the other an algebraic one. Source 03

also records the good coarse reduction of section 7 (its “coarser scales see good reduction”), with the same proof; it is printed once, there. As source 03 puts it, the apparent multiplicative degeneration is confined to the scales bounded by its own period: quotienting away all those scales removes the degeneration rather than creating a larger parameter domain.

11 Recovering the Tate parameter without adding scales

11.1 The inverse modular series

Write $J = 1/j$. The formal series

$$J(q) = \frac{\Delta(q)}{c_4(q)^3} \in q\mathbb{Z}[[q]]$$

has linear coefficient 1, hence a unique compositional inverse $Q(J) \in J\mathbb{Z}[[J]]$. Its first terms are

$$Q(J) = J + 744J^2 + 750420J^3 + 872769632J^4 + 1102652742882J^5 + \cdots. \quad (67)$$

The notation Q here denotes a formal series, not the rational field.

Theorem 11.1 (Unique, support-preserving recovery). *Let k be a field of characteristic zero and $\mathbb{K} = k((t^\Gamma))$. Given $j \in \mathbb{K}$ with $v(j) < 0$, there is a unique $q \in \mathbb{K}$ with $v(q) > 0$ and $j(q) = j$. It is $q = Q(j^{-1})$ and satisfies $v(q) = -v(j)$. If $G(a)$ denotes the additive subgroup of Γ generated by $\text{supp}(a)$, then*

$$G(q) = G(j). \quad (68)$$

More precisely,

$$\text{supp}(q) \subseteq (\text{supp}(j^{-1}))^* \setminus \{0\}. \quad (69)$$

Proof. Since $v(j^{-1}) > 0$, positive-support calculus evaluates Q and both formal inverse identities. Thus $J(q) = j^{-1}$ and $j(q) = j$. The leading coefficient of Q is 1, so $v(q) = v(j^{-1})$. If another positive-valuation q' gives $j(q') = j$, evaluating the identity $Q(J(q')) = q'$ proves equality. This argument does not choose a branch of a transcendental function.

Formula (69) follows directly from the positive powers of j^{-1} . Inversion of a Hahn series uses only the additive subgroup of its support, so $G(j^{-1}) \subseteq G(j)$. Therefore $G(q) \subseteq G(j)$. Conversely $j(q) = q^{-1}$ times a power series in q with nonzero constant coefficient. Inversion and strong positive evaluation use only $G(q)$, giving $G(j) \subseteq G(q)$. $\square$

The inverse modular expansion is classical. The support equality is the useful extra statement in a symbolic or nondivisible Hahn workspace: one need not pass to the divisible hull merely to recover the elliptic period parameter. All the series used here have integral coefficients, so this recovery needs neither algebraic closedness of k nor complex coefficients. Those are separate issues in classifying curves with that j -invariant.

11.2 Which elliptic curves this covers

Corollary 11.2 (Negative- j uniformization). *Assume Γ is divisible. Every elliptic curve E/K_Γ with $v(j(E)) < 0$ is K_Γ -isomorphic to E_q , for the unique parameter of theorem 11.1. After any such choice of isomorphism, $E(K_\Gamma)$ is parametrized by $U_q/q^\mathbb{Z}$.*

Proof. A Hahn field with algebraically closed coefficient field of characteristic zero and divisible exponent group is algebraically closed [11, Corollary 4]. Elliptic curves over an algebraically closed field of characteristic zero with the same j -invariant are isomorphic [3]. Apply this to E and E_q . $\square$

The parameter q is canonical, but an isomorphism $E \simeq E_q$ need not be canonical. The hypothesis on j excludes $j = 0$ and 1728, but does not remove the usual sign ambiguity of an elliptic isomorphism. For nondivisible Γ , theorem 11.1 still holds, while a curve with the same j can be a nontrivial twist; we do not silently identify all such twists with the split Tate curve over the original field. For a concrete example take $\Gamma = \mathbb{Z}$ and $q = t$. Put E_t in short Weierstrass form $Y^2 = X^3 + AX + B$ over $\mathbb{C}((t))$. Since $v(j(E_t)) = -1$, its invariant is neither 0 nor 1728, so $A, B \neq 0$. Its quadratic twist

$$E^{(t)} : Y^2 = X^3 + t^2AX + t^3B$$

has the same j but is not isomorphic to E_t over $\mathbb{C}((t))$. Indeed, an isomorphism of these short Weierstrass models would give a scaling $a \in \mathbb{C}((t))^\times$ with $a^4 = t^2$ and $a^6 = t^3$, hence $a^2 = t$, contradicting $2v(a) = 1$. The recovered period is still t ; the missing datum is the twist. Curves with $v(j) \geq 0$ are outside this positive- q theorem.

12 Torsion fields and exact exponent ramification

This section retains nondivisible exponent groups deliberately. Replacing Γ by its divisible hull at the start would make every degree computed here equal to one after base extension.

12.1 Roots of a Hahn parameter

Lemma 12.1 (A single radical extension). *Let $n \geq 2$, $q \in K_\Gamma$ with $v(q) = \alpha > 0$, and $\Lambda = \Gamma + \mathbb{Z}\alpha/n$. Then for any choice of $q^{1/n}$ in $K_{\Gamma \otimes \mathbb{Q}}$,*

$$K_\Gamma(q^{1/n}) = K_\Lambda, \quad [K_\Lambda : K_\Gamma] = [\Lambda : \Gamma] = \text{ord}(\alpha + n\Gamma). \quad (70)$$

The extension is cyclic, with Galois group $\text{Hom}(\Lambda/\Gamma, \mathbb{C}^\times)$; it has unchanged residue field $\mathbb{C}$ and value group Λ .

Proof. Write $q = ct^\alpha(1+\varepsilon)$, $c \in \mathbb{C}^\times$, $v(\varepsilon) > 0$. Choose $c^{1/n} \in \mathbb{C}$. The binomial series gives $b = (1+\varepsilon)^{1/n} \in K_\Gamma$, with $b^n = 1+\varepsilon$. Consequently $q^{1/n} = c^{1/n}t^{\alpha/n}b$ up to a root of unity already in $\mathbb{C}$, so adjoining $q^{1/n}$ is the same as adjoining $t^{\alpha/n}$.

The group Λ/Γ is finite cyclic. If its order is d , partition any Hahn support in Λ among its d cosets. Factoring a fixed monomial from each part gives a finite sum of d monomials with coefficients in K_Γ . These monomials are linearly independent over K_Γ , since their supports lie in disjoint cosets. Thus the entire Hahn field K_Λ , not just a smaller subfield with finite supports, is a degree- d extension generated by $t^{\alpha/n}$.

Multiplying the monomial of exponent λ by a character of $\lambda + \Gamma$ defines a field automorphism, preserving strong sums. There are d such characters because $\mathbb{C}$ contains all roots of unity, and they give all automorphisms. The coefficient residue field remains $\mathbb{C}$ and the ramification index is d , equal to the field degree. Finally d is the least positive integer with $d\alpha \in n\Gamma$, which is the displayed order in $\Gamma/n\Gamma$. $\square$

12.2 A primitive torsion point

Theorem 12.2 (The complete torsion field). *Under the hypotheses of lemma 12.1, fix a primitive n th root of unity $\zeta_n \in \mathbb{C}$ and one root $r = q^{1/n}$, and write $q^{b/n} = r^b$. Then the n^2 points*

$$\Phi_q(\zeta_n^a q^{b/n}), \quad 0 \leq a, b < n, \quad (71)$$

are precisely $E_q[n]$. Their field of definition is K_Λ , and already

$$K_\Gamma(\Phi_q(q^{1/n})) = K_\Lambda. \quad (72)$$

Here adjoining a point means adjoining its affine coordinates.

Proof. Every parameter in (71) lies in the admissible group of K_Λ , since its valuation is $b\alpha/n$. Its n th power is q^b , so the group law makes its image n -torsion. If two of the points coincide, injectivity modulo $q^\mathbb{Z}$ gives

$$\zeta_n^{a-a'} q^{(b-b')/n} = q^m$$

for some integer m . Comparing valuations gives $b - b' = nm$; the chosen ranges force $m = 0$ and $b = b'$, and then $a = a'$. Thus there are n^2 distinct points, all with coordinates in K_Λ by strong evaluation.

For exhaustiveness, work in the algebraically closed Hahn field $K_{\Gamma \otimes \mathbb{Z}\mathbb{Q}}$ [11, Corollary 4], which contains an algebraic closure of K_Γ . Every geometric torsion point is represented there by some $u \in U_q$. If $n\Phi_q(u) = O$, the kernel theorem gives $u^n = q^m$. Write $m = n\ell + b$ with $0 \leq b < n$. Then $(u/(q^\ell q^{b/n}))^n = 1$, so this quotient is ζ_n^a for some $0 \leq a < n$. Periodicity gives exactly the list (71); no separate torsion-count theorem is needed.

To prove the stronger primitive assertion, let $L = K_\Gamma(q^{1/n}) = K_\Lambda$ and let $\sigma \in \text{Gal}(L/K_\Gamma)$ fix the point $P = \Phi_q(q^{1/n})$. The character description in lemma 12.1 implies that σ preserves every strong sum, so it commutes with the coordinate formulas. Therefore

$$\Phi_q(\sigma(q^{1/n})) = \Phi_q(q^{1/n}).$$

The quotient $\sigma(q^{1/n})/q^{1/n}$ is a root of unity. By the uniformization kernel it must also be a power of q . A root of unity has valuation zero, whereas $v(q^m) = m\alpha$; hence that power is 1. Thus σ fixes $q^{1/n}$ and is the identity. The stabilizer of P is trivial, so Galois theory gives (72). The full torsion field contains this primitive point field and is contained in L , proving equality. $\square$

Corollary 12.3 (A finite-lattice degree formula). *If $\Gamma \cong \mathbb{Z}^r$ as an abelian group, with any compatible total order, and α has coordinates $(a_1, \dots, a_r)$, then*

$$[K_\Gamma(E_q[n]) : K_\Gamma] = \frac{n}{\gcd(n, a_1, \dots, a_r)}. \quad (73)$$

Proof. Put $g = \gcd(n, a_1, \dots, a_r)$. The integer $d = n/g$ makes every da_i divisible by n . Conversely, if n divides every ea_i , a Bézout expression of g as an integer combination of $n, a_1, \dots, a_r$ shows that n divides eg . Hence n/g divides e , proving minimality. $\square$

For example, $n = 12$ and $\alpha = (6, 4)$ give degree 6; $n = 4$ and $\alpha = (0, 6)$ give degree 2 in the lexicographic group $\mathbb{Z}^2$. The entire unit part of q can have infinite support and does not alter these degrees. It is removed by a binomial root already in the base field. Over $K_\mathbb{Q}$ every n th root scale is already available, so the corresponding degrees are one. This explains why scalar extension can erase arithmetic information while preserving all uniformization identities.

13 Transport to surreal and surcomplex numbers

13.1 Set-sized normal-form workspaces

The Conway normal form writes a surreal as a reverse-well-ordered sum of real multiples of monomials ω^γ . With the convention $t^\gamma = \omega^{-\gamma}$, increasing well-ordered Hahn supports correspond to these normal forms. Combining the real and imaginary normal forms produces complex coefficients and gives the surcomplex version [12, 13].

If a set of surcomplex parameters and point coordinates is specified, the union of all of its normal-form exponent supports is a set. Its additive subgroup $\Gamma \subseteq (\mathbf{No}, +)$ is a set as well. The associated normal-form map embeds K_Γ into $\mathbf{No}[i]$ and contains all the specified elements. Enlarge to the divisible hull only when algebraic closedness is actually needed.

The proof of theorem 1.2 works in this set-sized workspace. If a larger workspace is chosen, theorem 10.1 shows that the formulas and their values agree. There is therefore no choice-dependent global summation rule and no need to regard a proper class as a set.

Theorem 13.1 (Surcomplex uniformization). *For every $q \in \mathbf{No}[i]^\times$ with $\alpha = v(q) > 0$, the literal Hahn Tate map has exactly the admissible domain in (12) and induces the group isomorphism stated there. This domain is always a proper subclass of $\mathbf{No}[i]^\times$. For every elliptic curve $E/\mathbf{No}[i]$ with $v(j(E)) < 0$, the unique parameter $q = Q(j(E)^{-1})$ yields an isomorphic Tate curve.*

Proof. The domain test for one parameter involves only α , its valuation, and ordinary integer inequalities. Its truth is unchanged in any workspace containing those data. The same holds for all the strong families in the formulas. For surjectivity, put q and a given point's coordinates in one workspace and apply theorem 1.2 there. For the kernel and the group law, place the finitely many relevant parameters in a common workspace. These are instances of theorem 10.1.

To see properness, take a positive surreal value $\gamma = \omega\alpha$. It exceeds $N\alpha$ for every ordinary N . The monomial $u = \omega^{-\gamma}$ lies outside the domain, by the exact support theorem.

For the final assertion put the Weierstrass coefficients of E in a workspace and take its divisible hull. The algebraically closed Hahn field there contains an isomorphism supplied by corollary 11.2. Its image in $\mathbf{No}[i]$ proves the assertion. The parameter obtained by the inverse series is independent of the workspace, again by strong functoriality. $\square$

The following transfer comes from source 03 and uses the second route. For a set-sized ordered subgroup $\Gamma \subset \mathbf{No}$ there are coherent Hahn embeddings $\mathbb{R}((t^\Gamma)) \hookrightarrow \mathbf{No}$ and $\mathbb{C}((t^\Gamma)) \hookrightarrow \mathbf{No}[i]$, $t^\gamma \mapsto \omega^{-\gamma}$, whose values are genuine surreal or surcomplex numbers [12, 15]; complex conjugation acts coefficientwise. Enlarging the exponent workspace preserves coefficient sums, rational operations and every formal specialization used in sections 8 and 9, by lemma 2.1, so these constructions agree on intersections of workspaces. No single set-sized Hahn field is claimed to contain all surreals.

Corollary 13.2 (Real surreal case and transferred refinements). *Let $q \in \mathbf{No}[i]$ be nonzero and infinitesimal, with $\alpha = v(q)$. The inverse charts of section 8, the support-monoid equality theorem 9.1, the fixed- and moving-period valuation formulas of section 9, and the kernel obstruction theorem 10.5 all hold for actual surcomplex numbers. If $q \in \mathbf{No}$ is real, the same assertions, and the isomorphism*

$$\{u \in \mathbf{No}^\times : |v(u)| \leq N\alpha \text{ for some } N \in \mathbb{N}\}/q^\mathbb{Z} \simeq E_q(\mathbf{No}),$$

hold over $\mathbf{No}$ with real parameters and real points. The quotient may be represented by the class of unique normalized parameters with $0 \leq v(u) < \alpha$, with multiplication followed by integer normalization; no collection of all equivalence classes as a set is required.

Proof. For evaluation, choose a common Hahn workspace containing q and u and apply theorem 8.13 with $k = \mathbb{C}$, or with $k = \mathbb{R}$ when q and u are real. Non-summability outside the domain is witnessed by a descending sequence of leading exponents, so it cannot be removed by enlarging a workspace. For surjectivity, choose a workspace containing q and both coordinates of the desired point; the three inverse charts construct a parameter in that same workspace, with real coefficients when the inputs are real. The group law and every finite comparison can be checked in a common workspace containing the finitely many inputs. Coherence makes the resulting class function independent of this choice. All support and valuation assertions transfer literally.

For the extension obstruction, the group argument can be applied directly to the class map. If a homomorphism Ψ extends the canonical values, choose $a = \omega^{-\omega\alpha}$ outside the admissible domain. Surjectivity supplies an admissible d with $\Phi_q(d) = \Psi(a)$. Then $ad^{-1} \in \ker \Psi$ still has valuation outside H_α , so it cannot belong to $q^{\mathbb{Z}}$. This is a finite witness to the obstruction and requires no set of all class-sized cosets. $\square$

The real case is not covered by the first route as written, which works over $\mathbb{C}$ throughout; the classification of curves with $v(j) < 0$ in theorem 13.1 still needs $\mathbb{C}$ and a divisible workspace.

13.2 A forbidden monomial and an admitted fine displacement

Example 13.3 (Two ways to use $\omega^{-\omega}$). Let $q = \omega^{-1}$, so $\alpha = 1$, and put $\varepsilon = \omega^{-\omega}$. The parameter $u = \varepsilon$ is inadmissible: $v(u) = \omega > N$ for all ordinary N . For the theta family the exponents along $n = -k$ are

$$\frac{k(k+1)}{2} - k\omega,$$

a strictly decreasing sequence. For the X family they include $\omega - k$, again strictly decreasing. No cancellation argument can make either original family strongly summable.

In contrast $u = 1 + \varepsilon$ has valuation zero and is admissible. The point has

$$\begin{aligned} X_q(1 + \varepsilon) &= \varepsilon^{-2} + \varepsilon^{-1} + \text{a positive-valuation tail}, \\ Y_q(1 + \varepsilon) &= -\varepsilon^{-3} - 2\varepsilon^{-2} - \varepsilon^{-1} + \text{a positive-valuation tail}, \\ -X_q(1 + \varepsilon)/Y_q(1 + \varepsilon) &= \varepsilon + O(\varepsilon^2). \end{aligned}$$

The last O means valuation at least 2ω , not an ordinary real asymptotic estimate. Thus an infinitesimal displacement far finer than the q -scale is retained exactly, and its image has correspondingly large elliptic coordinates.

Source 03 gives the same pair of tests (with $s = \omega^{-\omega}$, the coordinates $X_q(1 + s) \sim s^{-2} = \omega^{2\omega}$ and $Y_q(1 + s) \sim -s^{-3} = -\omega^{3\omega}$, where $a \sim b$ means that a/b is 1 plus an infinitesimal); they are printed once, above. It notes that every negative-index summand for $u = \omega^{-\omega}$ is infinitesimal, which alone does not make the family summable, and that the domain restriction is a restriction on the valuation of u , not on the full support of u or on the sizes of the output coordinates. Its rank-two test, $\Gamma = \mathbb{Z} \oplus \mathbb{Z}$ in lexicographic order with $q = t^{(0,1)}$, where $t^{(1,0)}$ is inadmissible and $1 + t^{(1,0)}$ is admissible and maps near infinity, is the phenomenon of example 10.2 and of the introduction: the distinction is visible in a set-sized rank-two example and is not an artifact of proper classes. The next example, also from source 03, is new.

Example 13.4 (An internal non-Archimedean scale). Let $q = \omega^{-\omega}$ and $u = \omega^{-1}$. Then $0 < v(u) = 1 < \alpha = \omega$, and q/u has valuation $\omega - 1$. The node chart gives

$$X_q(u) \sim \omega^{-1}, \quad Y_q(u) \sim \omega^{-2}.$$

All ordinary powers of u have valuations n , still below ω ; they nevertheless form admissible Hahn families. The example shows why cofinal convergence of powers is an inappropriate proof requirement.

13.3 Why fine-topological convergence is not being used

For $q = \omega^{-1}$, the powers q^n tend to zero in a rank-one workspace such as $\mathbb{C}((t^{\mathbb{R}}))$. They do not tend to zero when neighborhoods can test against every surreal positive scale: none of the ordinary q^n is smaller than $\omega^{-\omega}$. The repository already emphasizes this distinction between its fixed rank-one analysis and full fine topology [23].

The present proof uses ordinary convergence only in the named complete rank-one coarsening of K_{H_α} . Everywhere else it uses strong Hahn summation or finite algebra. Its finite-tail bound is a statement in the ordered value group; it is not relabeled as an all-scale limit. Neither a global surcomplex exponential nor a value of $\sin \omega$ is required. Multiplicative periods $q^{\mathbb{Z}}$ replace those additive analytic constructions. The second route uses no convergence at all: only strong Hahn summation, formal inversion and finite algebra.

Which notion of uniformization is asserted. The word refers here to a group parametrization by explicit Hahn sums and formal local charts. It does not assert that $\mathbf{No}[i]$ carries a classical complex-analytic covering space, that ordinary contours have been lifted, or that all set-indexed partial sums converge in its fine topology. No evaluation of an exponential at an arbitrary infinite imaginary argument is used, and no choice of a global surcomplex logarithm is required: the construction is multiplicative from the beginning. Nor is every elliptic curve over $\mathbf{No}[i]$ classified. Source 03 treats the explicit family E_q and curves supplied with an algebraic isomorphism to a member of it; source 01 adds, through theorem 11.1 and corollary 11.2, every curve with $v(j) < 0$. A general reduction-type classification, including $v(j) \geq 0$, would be a separate theorem (section 23).

Part II

Several periods: finite positive generation and multiscale theta series

14 Setting, hypotheses, and the relation to Part I

This part comes from source 02. It takes up the first obstacle to a higher-dimensional analogue of Part I: deciding exactly when a theta series in several variables, with several interacting scales, is a Hahn sum. The gap is not the formal act of writing a familiar theta formula with larger scalars. It is deciding exactly when that formula represents a Hahn sum, identifying obstructions that survive every scalar extension, and replacing infinite lattice comparisons by certified finite ones. These questions become genuinely different when the value group has more than one Archimedean level. The first theorem of the part (theorem 15.2) is a statement of higher-rank lattice geometry that does not mention Hahn series; it answers a question posed in print by Amini and Nicolussi. The named question is not originally a question of surreal analysis; its application here gives the proposed results about surreal-valued forms and surcomplex theta series.

Hypotheses of Part II, kept apart from Part I. The coefficient field is $\mathbb{C}$. The exponent group Γ is set-sized, and it is assumed divisible whenever halves of exponents are used (equivalently, one starts with a divisible Γ). The periods are *monomials*: a symmetric matrix $\Pi = (\pi_{ab})$ of *exponents* in Γ is given, and the

theta series is

$$\vartheta_{\Pi}(z) = \sum_{n \in \mathbb{Z}^g} t^{\pi(n)} z^n, \quad \pi(n) = \frac{1}{2} n^T \Pi n, \quad \mathcal{E}_{\beta}(n) = \pi(n) + \beta \cdot n, \quad (74)$$

where $z = (z_1, \dots, z_g)$ and $z^n = \prod_j z_j^{n_j}$. Unit tails are allowed in the arguments z_j (theorem 18.1), not in the periods. Part I, by contrast, allows any Γ and a period q with an arbitrary Hahn tail; the two sets of hypotheses are not merged anywhere. Source 02 itself makes no claim about Tate uniformization, a global surcomplex exponential, or class-sized analytic convergence; the uniformization theorems of this report are those of Part I, in one variable.

Notation. Source 02's notation has been changed where it collided with Part I: its exponent matrix Q , quadratic form $q(n)$ and theta series Θ_Q are Π , $\pi(n)$ and ϑ_{Π} here (so neither the Tate parameter q nor the series Θ_q of (6) is reused); its real space H , higher-rank form B and lattice energy F are $\mathcal{H}$, $\mathcal{B}$ and $\mathcal{E}$ (not H_{α} , the series $B(q, h)$ of (28), or the residue workspace F); its minimum value δ_{β} is μ_{β} . The word *admissible* has a fixed meaning in Part I (the Hahn-admissible domain U_q); here the Amini–Nicolussi notion is always called *AN-admissible*. The integer g is the number of variables, unrelated to the function $g(z)$ of section 4; the integer s is the active depth of definition 14.1, unrelated to the Tate sums $s_k(q)$. Order type is written ot; it is support length, never birthday.

Relation to theorem 3.1. For $g = 1$ and $\Pi = (\alpha)$ with $\alpha > 0$, the series is $\vartheta_{(\alpha)}(z) = \sum_n t^{\alpha n^2/2} z^n$. If the Tate parameter is the monomial $q = t^{\alpha}$ and Γ is divisible, then $(-1)^n q^{n(n-1)/2} u^n = t^{\alpha n^2/2} (-t^{-\alpha/2} u)^n$, so

$$\Theta_{t^{\alpha}}(u) = \vartheta_{(\alpha)}(-t^{-\alpha/2} u),$$

and the leading exponent b_n of (16) is the energy $\mathcal{E}_{\beta'}(n)$ with $\beta' = v(u) - \alpha/2$. Since $\alpha/2 \in H_{\alpha}$, the domain $\beta' \in H_{\alpha}$ of corollary 18.3 is exactly $v(u) \in H_{\alpha}$: in this case corollary 18.3 recovers the domain half of theorem 3.1. Theorem 3.1 is more general in two directions that Part II does not reach: it allows an arbitrary $q = c_q t^{\alpha} (1 + \varepsilon_q)$, whose leading coefficient and tail enter the n th summand through powers with exponent $n(n-1)/2$, which are not argument perturbations of the kind covered by theorem 18.1; and it assumes no divisibility. Conversely Part II treats g variables. The two results are stated separately and neither is presented as a special case of the other beyond this dictionary.

Neighbouring reports. The closest relative in the collection is the report on Wick summability certificates (docs/surcomplex/wick-summability-certificates/), placed in the same batch. Both characterize strong summability of a monomial family indexed by an integer lattice or semigroup by a finite certificate, both allow Hahn-unit tails and show that such tails cannot repair a failure, and both assume a divisible Γ . The mathematics differs: here a quadratic energy on a lattice and its radical flags; there a linear weight on an integer semigroup and its Hilbert basis. Neither report uses a theorem of the other. The word *flag* here always means the chain of radicals (75); the Markov-generator report (docs/surreal/markov-generators-at-every-scale/) uses it for a chain of plateau idempotents, which is unrelated.

14.1 Lattices and higher-rank forms

All vector-space dimensions and matrix sizes are ordinary finite integers. A lattice in a real vector space is a discrete free abelian subgroup; it is full in its real span. Initially let $\mathcal{H}$ have a full lattice L , so a lattice basis identifies $(\mathcal{H}, L)$ with $(\mathbb{R}^g, \mathbb{Z}^g)$. Let

$$\mathcal{B} = (\mathcal{B}_1, \dots, \mathcal{B}_r) : \mathcal{H} \times \mathcal{H} \longrightarrow \mathbb{R}_{\text{lex}}^r$$

be a symmetric real-bilinear map. Larger coordinate indices have lower priority in the lexicographic order. Set $V_0 = \mathcal{H}$ and, when the indicated restriction is positive semidefinite, define

$$V_j = \text{rad}(\mathcal{B}_j|_{V_{j-1}}). \quad (75)$$

The radical of a positive semidefinite form is its nullspace, including all vectors orthogonal to its entire restricted space.

Definition 14.1 (AN-admissible forms). We call $\mathcal{B}$ an *AN-admissible higher-rank form* on $(\mathcal{H}, L)$ if, for every j , $\mathcal{B}_j|_{V_{j-1}}$ is positive semidefinite,

$$\mathcal{B}_j(V_j, \mathcal{H}) = 0, \quad (76)$$

each V_j is rational with respect to L , and $V_r = 0$. Rationality means $\text{span}_{\mathbb{R}}(L \cap V_j) = V_j$; it is a condition on the subspace, not a demand that the matrix entries of $\mathcal{B}_j$ be rational. The *active depth* is $s = \#\{j : V_j \neq V_{j-1}\}$. Repeated stages are allowed and do not count toward s .

Under (76), V_j is exactly the common global radical of $\mathcal{B}_1, \dots, \mathcal{B}_j$. Thus this definition is the higher-rank inner-product condition of Amini–Nicolussi together with their full-lattice admissibility condition, written recursively [17, Definition 2.5 and Theorem 8.1]; it is their definition, not a new one. If $x \neq 0$ and j is its first active level, then

$$\mathcal{B}_i(x, \mathcal{H}) = 0 \ (i < j), \quad \mathcal{B}_j(x, x) > 0. \quad (77)$$

Consequently $\mathcal{B}(x, x) > 0$ lexicographically. The converse to this last sentence is false: pointwise lexicographic positivity does not force (76).

14.2 Hahn preliminaries

Strong summability is the notion of section 2. For nonzero constants c_i , the family $(c_i t^{\gamma_i})_i$ is strongly summable precisely when $\{\gamma_i\}$ is well ordered and $i \mapsto \gamma_i$ has finite fibres; cancellation after grouping is not a licence to omit the finite-fibre requirement. Finite unions and pairwise sums of well-ordered subsets of an ordered abelian group are well ordered, and for well-ordered A, C each exponent has only finitely many representations $a + c$ with $a \in A, c \in C$. Together with lemma 2.1, these are the support facts used below. They are algebraic support statements, not completeness assertions in the fine topology of $\mathbf{No}[i]$. Every $z_j \in K^\times$ has a unique decomposition

$$z_j = c_j t^{\beta_j} (1 + \varepsilon_j), \quad c_j \in \mathbb{C}^\times, \quad \beta_j \in \Gamma, \quad \varepsilon_j \in \mathfrak{m}_v, \quad (78)$$

using the leading coefficient and exponent, not any surcomplex logarithm; here $\mathfrak{m}_v$ is the set of elements of positive valuation.

14.3 Why finite-rank coordinates suffice

The entries of Π and β generate a finite-dimensional rational vector space inside the divisible hull of Γ . Such an ordered rational space has a finite chain of convex subspaces, since every strict inclusion increases rational dimension. Consecutive quotients are Archimedean, for otherwise an element infinitesimal relative to another would give an intermediate convex subspace. Each quotient order-embeds in $\mathbb{R}$ by the usual rational-cut construction: fix $u_0 > 0$ and send x to $\sup\{r \in \mathbb{Q} : ru_0 \leq x\}$; Archimedeaness makes the set nonempty and bounded, makes the map additive and rational-linear, and ensures that a positive element has positive image. Choosing rational-linear sections yields an order embedding of the original space into some $\mathbb{R}_{\text{lex}}'$, the first nonzero quotient coordinate determining the sign.

Thus every particular quadratic lattice family, even in an infinite-rank value group, can be tested in finitely many real lexicographic coordinates. For surreal exponents one can instead use the finite-dimensional ordered *real* span of the entries. We never assert that one fixed coordinate model contains every possible evaluation vector in a larger value group. This is an elementary fact about finite-dimensional ordered rational spaces, not a new result; the Markov-generator report of the collection uses the same kind of finite rational sign separation.

15 Finite positive generation

Definition 15.1 (Positive generation). A finite subset $S \subset L \setminus \{0\}$ *positively generates* L for $\mathcal{B}$ if every $x \in L$ can be written

$$x = \sum_{a \in A} m_a a, \quad A \subset S, \quad m_a \in \mathbb{Z}_{>0}, \quad \mathcal{B}(a, b) \geq 0 \quad (a, b \in A). \quad (79)$$

The zero vector has the empty representation.

This is [17, Definition 11.10], which requires a positive integer combination of a subset $A \subset S$ whose members have pairwise nonnegative inner products.

Theorem 15.2 (Finite positive generation). *Every admissible lattice in a finite-dimensional higher-rank inner-product space, in the sense of Amini–Nicolussi, has a finite positively generating set. In particular every form of definition 14.1 has such a set. The construction is finite and uses lattice lifts, ordinary acute cones in graded quotients, and bounded integer shifts. It gives an affirmative answer to Question 11.14 of [17] in its published formulation: “Does L have a finite, positively generating set S in the sense of Definition 11.10?” [17, §11.5.3, p. 1181].*

The essential feature is not that a lattice is finitely generated as a group: arbitrary signed basis decompositions do not preserve the nonnegativity in (79). The following lemmas provide the extra control. The claim concerning Question 11.14 is pinned to its published formulation, checked in the 2025 article and in the arXiv preprint; it is not a claim that nobody has answered the question since.

15.1 The positive-definite case

Lemma 15.3 (Finite acute generating families). *Let L' be a full lattice in a finite-dimensional real space with a positive-definite real form G . There are finitely many finite sets $T_1, \dots, T_h \subset L' \setminus \{0\}$ such that every lattice vector is a nonnegative integer combination of one T_v , and $G(a, b) > 0$ for every pair $a, b \in T_v$.*

Proof. If $L' = 0$, take $T_1 = \emptyset$. Otherwise, for each unit vector choose a narrow simplicial cone with rational rays, containing that vector in its interior, such that every two nonzero vectors in the cone have positive G -pairing. Such cones exist by continuity of G , positive definiteness, and density of rational directions. Compactness of the ordinary unit sphere gives a finite covering by these cones.

For one rational simplicial cone, choose independent lattice generators $v_1, \dots, v_d$ for its rays. Every lattice point in that cone is $\sum_i a_i v_i$ with $a_i \geq 0$. Subtracting $\sum_i \lfloor a_i \rfloor v_i$ leaves a lattice point in the bounded half-open parallelepiped $\{\sum_i u_i v_i : 0 \leq u_i < 1\}$, which contains finitely many lattice points. The ray generators and the nonzero parallelepiped points generate the lattice semigroup in the cone. They remain pairwise strictly acute because they lie in that cone. Do this for each cone. $\square$

This is the familiar rank-one ingredient, also described by Amini and Nicolussi immediately after their question (via Gordan’s lemma); it is credited to them. Compactness here is only compactness of an ordinary real finite-dimensional sphere, not of a surreal sphere.

15.2 A bounded allocation lemma

Lemma 15.4 (Distributing lower-level coefficients). *Let N, M be positive integers and $m_a \in \mathbb{N}$ for a in a finite set A . There are integers $c_{ia} \in \{0, \dots, M\}$ and $r_a \geq 0$ such that*

$$m_a = \sum_{i=1}^N c_{ia} + r_a, \quad r_a > 0 \implies c_{ia} = M \text{ for every } i.$$

Proof. An explicit choice, for $1 \leq i \leq N$, is

$$c_{ia} = \min\{M, \max\{0, m_a - (i-1)M\}\}, \quad r_a = \max\{0, m_a - NM\}.$$

The first N bins contain a total of $\min\{m_a, NM\}$ units. If $r_a > 0$, then $m_a > NM$ and every bin is full. If $m_a = NM$, every bin is also full but the remainder is zero, as allowed. The constructions for different a are independent; no bound on $\sum_a c_{ia}$ is required. $\square$

Lemma 15.5 (A shift dominates every fixed mixed term). *For an AN-admissible higher-rank form, $a \neq 0$ and $x \in \mathcal{H}$, there is an ordinary integer $M > 0$ such that $\mathcal{B}(x, a) + M\mathcal{B}(a, a) > 0$. For finite sets of x and a one M works for every pair.*

Proof. Let j be the first level of a . By (77), both terms have zero coordinates before j , and $\mathcal{B}_j(a, a) > 0$. Choose an integer $M > -\mathcal{B}_j(x, a)/\mathcal{B}_j(a, a)$. The j th coordinate is then positive, regardless of all later coordinates. Taking a maximum handles finite sets. $\square$

The global radical condition is exactly what excludes an adverse mixed term at a scale higher than the self-pairing of a .

15.3 Proof by lifting and allocation

Proof of theorem 15.2 for a full lattice. Induct on the number of active levels. The zero-dimensional case is empty, and the one-level case follows from lemma 15.3. Delete initial zero stages and call the first form $\mathcal{B}_1$. Let $V = \text{rad } \mathcal{B}_1$ and $L_0 = L \cap V$. The subgroup L_0 is primitive: if a nonzero integer multiple of a lattice vector lies in V , the vector itself lies in V . A basis of L_0 therefore extends to a basis of L . Rationality says that the first basis spans V over $\mathbb{R}$, so the remaining basis vectors project to a real basis of $\mathcal{H}/V$. Consequently $\bar{L} = L/L_0$ is a full lattice in that quotient, and the induced $\mathcal{B}_1$ is positive definite there.

Choose finite acute families in $\bar{L}$ by lemma 15.3. Write $\mathcal{T}$ for their union, and choose once and for all a lift $\tilde{t} \in L$ for every $t \in \mathcal{T}$. On V , the remaining forms and L_0 satisfy the inductive hypotheses: their restricted radicals are the later V_j , their global annihilation conditions restrict from $\mathcal{H}$ to V , and those subspaces are rational relative to L_0 . At least one active level has been removed. Let S_0 be a finite positively generating set for L_0 . If $V = 0$, the lifted acute sets suffice, so assume $S_0 \neq \emptyset$. By lemma 15.5, choose a positive integer M satisfying

$$\mathcal{B}(\tilde{t}, a) + M\mathcal{B}(a, a) > 0 \quad (t \in \mathcal{T}, a \in S_0). \quad (80)$$

Define the finite candidate set

$$S = S_0 \cup \left\{ \tilde{t} + \sum_{a \in S_0} c_a a : t \in \mathcal{T}, 0 \leq c_a \leq M, c_a \in \mathbb{Z} \right\}. \quad (81)$$

Every vector in the second set has nonzero quotient, so none is zero.

Take $x \in L$. If its quotient is zero, use S_0 . Otherwise represent its quotient using one acute family, and expand integer multiplicities into a list $t_1, \dots, t_N$; every pair t_i, t_j has strictly positive $\mathcal{B}_1$ -pairing. The residual $w = x - \sum_{i=1}^N \tilde{t}_i \in L_0$ has a representation $w = \sum_{a \in A} m_a a$ for some pairwise $\mathcal{B}$ -nonnegative $A \subset S_0$. Apply lemma 15.4 to these coefficients and put $u_i = \tilde{t}_i + \sum_{a \in A} c_{ia} a \in S$. Then $x = \sum_i u_i + \sum_a r_a a$.

It remains to check all pairs of displayed summands. The u_i are mutually positive at the first level, since adding vectors of V does not change $\mathcal{B}_1$. The residual a 's remain pairwise nonnegative. Finally, if $r_a > 0$, all $c_{ia} = M$, and

$$\mathcal{B}(u_i, a) = \mathcal{B}(\tilde{t}_i, a) + M\mathcal{B}(a, a) + \sum_{b \in A \setminus \{a\}} c_{ib} \mathcal{B}(b, a) > 0$$

by (80). Combining duplicate summands gives the required positive-integer representation. This closes the induction. $\square$

Remark 15.6 (Non-full lattices). The theorem also covers the non-full lattices of the published question. Let F^j be the global radical filtration, with $F^1 = \mathcal{H}$, and $L_j = L \cap F^j$. Admissibility says that the image $\overline{L}_j$ of L_j in F^j/F^{j+1} is discrete [17, Theorem 8.1]. Set $W_L = \text{span}_{\mathbb{R}} L$. We prove $W_L \cap F^j = \text{span}_{\mathbb{R}} L_j$ by induction, starting at $j = 1$. Under this identification the kernel of

$$\text{span}_{\mathbb{R}} L_j \longrightarrow F^j/F^{j+1}$$

is $W_L \cap F^{j+1}$. Its dimension is $\text{rank } L_j - \text{rank } \overline{L}_j = \text{rank } L_{j+1}$: discreteness identifies ranks with dimensions of real spans, and the last equality is the rank formula for $0 \rightarrow L_{j+1} \rightarrow L_j \rightarrow \overline{L}_j \rightarrow 0$. It contains $\text{span}_{\mathbb{R}} L_{j+1}$, which has the same dimension, so they are equal.

The restricted form on W_L has this same radical filtration: a vector in $W_L \cap F^j$ outside F^{j+1} has positive j th self-pairing, while vectors in F^{j+1} annihilate that coordinate on all of $\mathcal{H}$. Hence its successive radicals are lattice-rational and its graded forms are positive definite. Thus L is a full AN-admissible lattice in W_L , to which the proof applies.

15.4 Size and dependence

The construction gives

$$|S| \leq |S_0| + |\mathcal{T}|(M+1)^{|S_0|}, \quad (82)$$

with possible duplicates removed. This bounds the cardinality of the constructed generating set, not the number of steps in a later descent algorithm. No practical optimality is claimed. The set depends only on $\mathcal{B}$ and L , not on any theta evaluation point or linear term. Adjoining its negative makes it symmetric without invalidating positive generation. The proof allows irrational coefficients in the forms and irrational lifts of orthogonal decompositions; only the radical *subspaces* need be lattice-rational. No algorithm for deciding arbitrary real-number equalities is asserted. For exact algebraic inputs, the construction uses ordinary exact arithmetic, sign tests, rational cones, and lattice semigroup generators.

16 An exact quadratic Hahn-summability criterion

16.1 The real one-level obstruction

Lemma 16.1. *For a real linear map $R : \mathbb{R}^d \rightarrow \mathbb{R}^e$, the subgroup $R\mathbb{Z}^d$ is discrete in its real span if and only if $\ker R$ is rational relative to $\mathbb{Z}^d$.*

Proof. Put $L_0 = \ker R \cap \mathbb{Z}^d$. It is primitive, since $R(mx) = 0$ for a nonzero integer m implies $R(x) = 0$. If $\ker R$ is rational, a basis of L_0 spans it over $\mathbb{R}$ and extends to an integer basis of $\mathbb{Z}^d$. The remaining basis vectors project to a real basis of $\mathbb{R}^d/\ker R$, and their images therefore generate a full lattice in $\text{im } R$.

Conversely, a discrete subgroup spanning a k_0 -dimensional real space is a lattice of rank k_0 . The exact sequence

$$0 \longrightarrow L_0 \longrightarrow \mathbb{Z}^d \xrightarrow{R} R\mathbb{Z}^d \longrightarrow 0$$

then gives $\text{rank } L_0 = d - k_0 = \dim \ker R$. As a subgroup of the discrete lattice $\mathbb{Z}^d$, L_0 has real span of that same dimension. Its span is contained in $\ker R$, so equality follows. $\square$

The elementary assertion about discrete subgroups follows, for example, by choosing a shortest nonzero vector, projecting successively to its orthogonal complement, and applying discreteness in a bounded fundamental strip; it is the usual finite-dimensional lattice theorem.

Lemma 16.2 (One-level quadratic criterion). *Let A be a real symmetric $d \times d$ matrix and $b \in \mathbb{R}^d$. The set $\{f_A(n) = \frac{1}{2}n^\top A n + b \cdot n : n \in \mathbb{Z}^d\}$ is well ordered if and only if A is positive semidefinite, $V = \ker A$ is rational, and $b|_V = 0$. Under these conditions every bounded sublevel contains finitely many cosets of $\mathbb{Z}^d \cap V$, and f_A is constant on each such coset.*

Proof. If A has a negative direction, it has a rational negative direction by continuity. Multiplying gives an integer u with $u^\top A u < 0$, and $f_A(ku) \rightarrow -\infty$, contradicting well-ordering.

Now suppose A is positive semidefinite. If $b(v_0) < 0$ for some $v_0 \in V$, choose integer vectors n_k by rounding the coordinates of kv_0 . The errors $e_k = n_k - kv_0$ remain bounded. Since $Av_0 = 0$, $f_A(n_k) = \frac{1}{2}e_k^\top A e_k + kb(v_0) + b(e_k) \rightarrow -\infty$. Changing the sign of v_0 handles any nonzero restriction of b to V . Thus $b|_V = 0$ is necessary.

Factor $A = R^\top R$ with $\ker R = V$. The vanishing of b on V gives a vector c with $b(n) = c \cdot Rn$. If V is irrational, lemma 16.1 gives nonzero elements $\ell_k \in R\mathbb{Z}^d$ with $\ell_k \rightarrow 0$. Replace each ℓ_k by its negative when necessary so that $c \cdot \ell_k \geq 0$. Then $0 < \frac{1}{2}\|\ell_k\|^2 + c \cdot \ell_k \rightarrow 0$, and a strictly decreasing subsequence exists, so the value set is not well ordered. This proves the necessity of rationality, including the case $b = 0$.

Conversely, rationality makes $R\mathbb{Z}^d$ a lattice. On its real span, $\frac{1}{2}\|\ell\|^2 + c \cdot \ell$ is a positive-definite quadratic polynomial after completing the square, so its bounded sublevels contain finitely many lattice points. These correspond precisely to the cosets of $\mathbb{Z}^d \cap V$. In particular any nonempty subset of the value set has a least member: start with one value and take the minimum of the finitely many values below it. $\square$

16.2 The flag theorem

Write a quadratic lattice energy in lexicographic coordinates as

$$\mathcal{E}(n) = (\mathcal{E}_1(n), \dots, \mathcal{E}_r(n)), \quad \mathcal{E}_j(n) = \frac{1}{2}\mathcal{B}_j(n, n) + b_j(n), \quad (83)$$

with arbitrary real linear forms b_j .

Theorem 16.3 (Exact flag criterion). *For $g \geq 1$, the following are equivalent:*

- (i) $\mathcal{E}(L)$ is well ordered and $\mathcal{E} : L \rightarrow \mathbb{R}_{\text{lex}}^r$ has finite fibres.
- (ii) $\mathcal{B}$ is AN-admissible (definition 14.1), and

$$b_j(V_j) = 0 \quad (1 \leq j \leq r). \quad (84)$$

Equivalently, any family $(c_n t^{\mathcal{E}(n)})_{n \in L}$ with all $c_n \neq 0$ is strongly Hahn-summable exactly under these conditions.

Proof. Identify L with $\mathbb{Z}^g$. Suppose (i). Projection of a well-ordered lexicographic set onto its first coordinate is well ordered: a least preimage gives a least projected value for every nonempty projected subset. Apply lemma 16.2 to $\mathcal{E}_1$. This gives positive semidefiniteness, the rational kernel V_1 , and $b_1(V_1) = 0$. The global radical condition at the first level holds automatically.

Assume the conditions have been proved through level $j - 1$. The preceding coordinates are constant on every coset $n_0 + L_{j-1}$, where $L_{j-1} = L \cap V_{j-1}$. Restrict $\mathcal{E}$ to one such coset, discard its constant preceding coordinates, and project onto coordinate j . The resulting quadratic polynomial on the full lattice L_{j-1} has quadratic part $\mathcal{B}_j|_{V_{j-1}}$ and linear part

$$x \mapsto b_j(x) + \mathcal{B}_j(n_0, x). \quad (85)$$

The one-level lemma gives positive semidefiniteness and a rational radical V_j . Taking $n_0 = 0$ gives $b_j(V_j) = 0$. Taking every lattice basis vector as n_0 shows $\mathcal{B}_j(n_0, V_j) = 0$ for a real basis of $\mathcal{H}$, so $\mathcal{B}_j(\mathcal{H}, V_j) = 0$; testing only the coset through zero would miss this essential condition. At the end every coordinate of $\mathcal{E}$ is constant along $L \cap V_r$. If $V_r \neq 0$, its rationality gives a nonzero integer vector and hence an infinite fibre. Thus $V_r = 0$, proving (ii).

Conversely suppose (ii). The first-coordinate values are well ordered, and every such value is represented by finitely many cosets of L_1 , by the one-level lemma. On each coset the remaining forms on V_1 satisfy the same conditions: the shifted linear terms (85) still vanish on every subsequent radical, because of the global radical conditions. Induct on the number of remaining coordinates. Within each coset the tail support is well ordered with finite fibres, and finite unions preserve both properties. Taking the lexicographic union over the well-ordered first-coordinate values preserves well-ordering. A full exponent fixes one first-coordinate value and therefore only finitely many cosets, each with a finite tail fibre. This proves (i). The induction ends either with a zero-dimensional radical lattice, whose only vector is zero, or with the last coordinate, where $V_r = 0$ makes every remaining fibre finite. $\square$

The two parts of (i) do different work. Well-ordering forces the semidefinite restrictions, rational intermediate radicals, global mixed-term annihilation and compatible linear terms. Finite fibres additionally force the terminal radical to vanish. For instance, $\mathcal{E}(m, n) = (m^2, 0)$ has a well-ordered value set, but every exponent has infinitely many lattice preimages; its monomial family is therefore not strongly summable. Positive self-pairings alone also do not suffice: examples 21.1 and 21.2 display the distinct irrational-radical and mixed-term obstructions.

Corollary 16.4 (The exact linear domain). *For an AN-admissible form $\mathcal{B}$, let*

$$\mathcal{D}_{\mathcal{B}} = \{(b_1, \dots, b_r) : b_j \in V_j^\perp \text{ for all } j\}. \quad (86)$$

This is exactly the set of linear terms giving a strongly summable quadratic lattice family. It is also exactly the set giving an energy with a global lattice minimum. If $\mathcal{B}$ is not AN-admissible, no linear term gives a strongly summable family.

Proof. Only the converse concerning minima needs proof. If a compatibility condition first fails at level j , choose $a \in L \cap V_j$ with $b_j(a) < 0$; rationality of V_j ensures that one exists. For every $n \in L$ the coordinates before j of $\mathcal{E}(n+a) - \mathcal{E}(n)$ vanish, and its j th coordinate is $b_j(a)$, because $\mathcal{B}_j(\mathcal{H}, V_j) = 0$. Hence $\mathcal{E}(n+a) < \mathcal{E}(n)$ for every n , so there is no minimum. The other assertions follow from theorem 16.3. $\square$

Remark 16.5 (Minimum attainment is stronger than boundedness). In a non-Archimedean ordered group, the descending sequence $(0, -k)$ is bounded below by $(-1, 0)$. Thus “has a global lattice minimum” in corollary 16.4 cannot be replaced by “is bounded below”. The criteria use well-ordering and finite fibres, not an ambient lower bound.

17 The ordinal length of the exponent support

Lemma 17.1 (A finite interleaving bound). *If $A_1, \dots, A_N$ are well-ordered subsets of a common linear order, each of order type at most ω^k for an ordinary integer $k \geq 0$, then their union has order type at most $\omega^k N$.*

Proof. For a point x in the union, record for each i the order type of $A_i \cap (-\infty, x)$, and take the Hessenberg natural sum of these N ordinals, meaning addition of coefficients in their finite Cantor normal forms. This rank strictly increases as x increases, since at least one component strictly increases and none decreases. Each rank is smaller than $\omega^k N$: all components are at most ω^k , and at least one is strictly smaller. The rank map embeds the union in that ordinal. $\square$

Theorem 17.2 (Exact ordinal support law). *Under theorem 16.3, let s be the active depth. Then*

$$\text{ot } \mathcal{E}(L) = \omega^s. \quad (87)$$

For $g = 0$ the value set is a singleton and the formula means $1 = \omega^0$. For $g > 0$ one has $1 \leq s \leq g$. The integer s is an intrinsic invariant of the ordered quadratic exponent family, independent of the chosen finite-rank lexicographic coordinate model.

Proof. Use induction on active depth. Before the first active level every quadratic and compatible linear coordinate is zero. At the first active level the energy descends to a positive-definite quadratic polynomial on a positive-rank quotient lattice. Its distinct values form a locally finite, bounded-below, unbounded-above subset of $\mathbb{R}$, and hence have order type ω ; each value comes from finitely many cosets of the radical lattice.

Write V for this first active radical. If $V = 0$, each such fibre is finite, and subsequent coordinates only split it into finitely many points; the whole order type is ω . Otherwise every radical coset has a tail energy of active depth $s - 1$, with compatible shifted linear terms, whose value set has order type ω^{s-1} by induction. More explicitly, on a coset $n_0 + (L \cap V)$ write $n = n_0 + x$; the terms independent of x translate the value set and hence do not change its order type. The remaining quadratic and linear terms are exactly the restricted and shifted ones used in the flag theorem. A first-coordinate fibre is a finite union of such sets, so its order type lies between ω^{s-1} and $\omega^{s-1}N$ for some positive finite N , by lemma 17.1. There are ω first-coordinate values. The ordinal sum of their fibre types is exactly ω^s : the lower estimate gives ω^s , and the upper estimate is a countable sum of finite multiples of ω^{s-1} , which is again ω^s . Intermediate inactive coordinates merely order finitely many cosets and are included in this argument.

Strict drops lower real dimension, so $s \leq g$. An order embedding of value groups preserves the order type of the value set; since distinct finite powers of ω are unequal, it also preserves the integer identified by (87). $\square$

Corollary 17.3 (Actual surreal support lengths). *At monomial evaluation points with positive real residues, the theta sum has no coefficient cancellation, and its actual Hahn support has order type ω^s . Under a normal-form embedding into **No**, this is the ordinal length of its Conway normal-form support. For $g \geq 1$, every ω^s with $1 \leq s \leq g$ occurs in dimension g . The zero-dimensional case has singleton support $\omega^0 = 1$.*

Proof. Each exponent receives a finite sum of positive real coefficients, so it remains in the support. For $1 \leq s \leq g$, split $\mathbb{Z}^g$ into s nonzero coordinate blocks and take one positive-definite sum of squares in each successive lexicographic coordinate; the radical flag then has exactly s strict drops. $\square$

The invariant in this corollary is *support length*, not birthday or sign-sequence length; those are different surreal invariants. At arbitrary complex residues, cancellations may shorten the support or annihilate the sum entirely; theorem 17.2 then describes the exponent family before cancellation.

18 Multiscale theta series over arbitrary-rank Hahn fields

18.1 The full multiplicative evaluation domain

Let Π be a symmetric $g \times g$ matrix of exponents in the divisible group Γ , and write the argument as in (78).

Theorem 18.1 (Exact domain and unit robustness). *The theta family $(t^{\pi(n)} z^n)_{n \in \mathbb{Z}^g}$ is strongly Hahn-summable if and only if*

$$\{\pi(n) + \beta \cdot n : n \in \mathbb{Z}^g\} \text{ is well ordered with finite exponent fibres.} \quad (88)$$

Equivalently, any finite-rank envelope of Π, β satisfies the flag and linear compatibility conditions of theorem 16.3. The condition depends only on Π and β , not on c or the ε_j . On this domain all polynomially weighted lattice families and all multiplicative Taylor expansions at the point are strongly summable.

Proof. The leading term of the n th summand is $c^n t^{\pi(n) + \beta \cdot n}$, which is nonzero. Summability of the full family therefore implies (88): its leading exponents form a well-ordered subset of the common support, and infinitely many equal leading exponents would violate pointwise finiteness.

Conversely, let A be the well-ordered leading-exponent set and put $\Sigma = \bigcup_j \text{supp}(\varepsilon_j) \subset \Gamma_{>0}$, a finite union of positive well-ordered sets. For every integer m , $(1 + \varepsilon_j)^m = \sum_{k \geq 0} \binom{m}{k} \varepsilon_j^k$ is a valid Hahn identity by the support lemma, with support in $M = \langle \Sigma \rangle$. Thus the entire theta family has support in $A + M$, which is well ordered. A fixed exponent has only finitely many representations $a + \mu$ with $a \in A$, $\mu \in M$, and each a comes from finitely many n . This proves summability.

Multiplying the n th coefficient by a polynomial in n only removes some terms and changes complex coefficients; it cannot worsen the support. For a multiplicative perturbation $z_j \mapsto z_j(1 + \eta_j)$ with $\eta_j \in \mathfrak{m}_v$, enlarge Σ by their finitely many supports. The same monoid argument justifies the binomial expansions and every interchange of the lattice indices with Taylor indices. Explicitly, fix an output exponent γ . There are finitely many decompositions $\gamma = a + \mu$ with $a \in A$ and μ in the enlarged positive monoid, and finitely many lattice indices n for each a . For each μ , Neumann's lemma permits only finitely many multisets of positive input exponents. Each occurrence has only finitely many labels, one for each of the finitely many unit or Taylor inputs. Thus the fully expanded family, with lattice and Taylor indices kept separate, is strongly summable before regrouping. $\square$

This is an exact domain for the displayed Hahn family. It does not prohibit some different analytic continuation or a deliberately chosen regularization outside that domain; neither is constructed or identified with the sum here.

18.2 Extension and non-extension

Corollary 18.2 (No repair by units or ordered scalar extension). *An order embedding $\Gamma \hookrightarrow \Gamma'$ and an injective coefficient-field extension preserve summability and non-summability of a fixed theta family at a fixed monomial valuation point. Infinitesimal unit changes cannot repair a point outside the domain. New evaluation points in Γ' must, however, be tested anew.*

Proof. An order embedding preserves well-ordering, descending subsets, equality of exponents, and finite fibres. Injectivity preserves nonzero leading coefficients. Apply theorem 18.1. $\square$

This is the multivariable counterpart of the one-variable behaviour recorded in theorem 10.1.

Corollary 18.3 (One-dimensional and diagonal domains). *For $g = 1$ and $\pi(n) = \alpha n^2/2$ with $\alpha > 0$, the exact valuation domain is the principal convex subgroup*

$$H_\alpha = \{\beta \in \Gamma : |\beta| \leq N\alpha \text{ for some ordinary } N \in \mathbb{N}\} \quad (89)$$

of (1). For a diagonal $\Pi = \text{diag}(\alpha_1, \dots, \alpha_g)$ with positive entries, the domain is $\prod_j H_{\alpha_j}$.

Proof. In a finite-rank model containing α and β , the quadratic form vanishes at every level preceding the first nonzero coordinate of α . The flag criterion says exactly that β vanishes at those levels too, which is equivalent to (89). In the diagonal case, necessity follows by restricting to each coordinate line. For sufficiency, the whole exponent family is the finite Cartesian product of the one-dimensional families, and Hahn multiplication with the finite-representation support lemma gives summability. $\square$

For the relation with theorem 3.1 see section 14. A one-dimensional theta family summable on a rank-one value group need not remain summable at a new valuation β dominating $N\alpha$ for every integer N : with the adverse sign, successive exponents have difference $(n + \frac{1}{2})\alpha - |\beta| < 0$ for all $n \geq 0$. The failure is structural, not a matter of an insufficient number of computed terms.

18.3 Quasi-periodicity and formal logarithmic derivatives

For $m \in \mathbb{Z}^g$ write $t^{\Pi m}$ for the coordinatewise monomial vector. Reindexing a summable family gives

$$\vartheta_\Pi(t^{\Pi m} z) = t^{-\pi(m)} z^{-m} \vartheta_\Pi(z), \quad \vartheta_\Pi(z^{-1}) = \vartheta_\Pi(z). \quad (90)$$

For the first identity, use $\pi(n + m) = \pi(n) + n^\top \Pi m + \pi(m)$ and replace n by $n + m$; for the second, replace n by $-n$ and use $\pi(-n) = \pi(n)$. These are the familiar theta symmetries, licensed here by the exact support criterion. The domain is stable under $\beta \mapsto \beta + \Pi m$ and $\beta \mapsto -\beta$; the first statement follows also from global radical annihilation, the second from $n \mapsto -n$. For $g = 1$ and $\Pi = (\alpha)$ the first identity is the counterpart of (15) in the normalization of section 14.

Define the formal logarithmic operator $D_j = z_j \partial / \partial z_j$ on the lattice characters by $D_j z^n = n_j z^n$. Then $D_1^{a_1} \cdots D_g^{a_g} \vartheta_\Pi(z) = \sum_n n_1^{a_1} \cdots n_g^{a_g} t^{\pi(n)} z^n$ is strongly summable wherever ϑ_Π is. This is a coefficientwise Taylor calculus. It is not a claim that a fine-topological notion of holomorphy has been recovered, and it concerns no derivation on the scalars.

19 Finite certificates and the geometry of theta minima

19.1 A finite test valid for every linear term

Let $\mathcal{B}$ be AN-admissible and choose a finite positively generating set S from theorem 15.2. For a compatible linear term b , put $\mathcal{E}(n) = \mathcal{B}(n, n)/2 + b(n)$ and $I_m(h) = \mathcal{E}(m + h) - \mathcal{E}(m)$.

Theorem 19.1 (Finite minimum certificate). *A lattice vector m is a global minimizer of $\mathcal{E}$ if and only if*

$$I_m(s) = \frac{1}{2}\mathcal{B}(s, s) + \mathcal{B}(m, s) + b(s) \geq 0 \quad \text{for every } s \in S. \quad (91)$$

The equivalence of this finite test with global minimality holds even for an incompatible b ; in that case no point passes it. For compatible b , repeatedly taking any $s \in S$ with negative difference terminates after finitely many steps at a global minimum.

Proof. Necessity is immediate. For sufficiency, write any displacement as $h = \sum_{s \in A} a_s s$ with positive integers a_s and pairwise $\mathcal{B}$ -nonnegative s . Direct expansion gives

$$I_m(h) = \sum_s a_s I_m(s) + \sum_s \binom{a_s}{2} \mathcal{B}(s, s) + \sum_{s < s'} a_s a_{s'} \mathcal{B}(s, s'). \quad (92)$$

Every term is nonnegative under (91); hence $\mathcal{E}(m + h) \geq \mathcal{E}(m)$. No compatibility of b was needed in this calculation. Compatibility ensures that $\mathcal{E}(L)$ is well ordered, so an infinite strictly descending search is impossible. At termination all finite tests pass. $\square$

Well-ordering proves finite termination for each instance. It supplies neither a useful uniform bound on the number of descent steps nor an algorithm for comparing unspecified arbitrary real constants; those are separate computational questions.

Corollary 19.2 (Finite higher-rank Voronoi inequalities). *For every $x \in \mathcal{H}$,*

$$x \in \text{Vor}_L(0) \iff 2\mathcal{B}(x, s) \leq \mathcal{B}(s, s) \text{ for every } s \in S.$$

The inequalities are lexicographic. The conclusion concerns the actual higher-rank Voronoi cell, not merely its Euclidean closure.

Proof. Use $b(n) = -\mathcal{B}(x, n)$ in theorem 19.1; it is compatible by the global radical conditions. The finite-inequality implication is also the one proved by Amini–Nicolussi conditional on positive generation, [17, Lemma 11.11]; theorem 15.2 supplies the finite generating set that lemma assumes. $\square$

19.2 All minima and a short zero-cost graph

Let $\mathcal{M} = \{n : \mathcal{E}(n) = \min \mathcal{E}(L)\}$, which is finite by theorem 16.3. Join two elements by an edge whenever their difference or its negative belongs to S ; use the symmetric enlargement of S when convenient.

Theorem 19.3 (Finite minimizer geometry). *The set $\mathcal{M}$ has at most 2^g elements. Its zero-cost S -move graph is connected and has diameter at most g . More precisely, for any $m, n \in \mathcal{M}$, the displacement $n - m$ is a sum of at most g distinct members of S that are pairwise $\mathcal{B}$ -orthogonal, and every partial sum, added to m , is a minimizer.*

Proof. If distinct m, n were congruent modulo $2L$, their midpoint u_0 would be a lattice vector and $\mathcal{E}(u_0) = \frac{1}{2}(\mathcal{E}(m) + \mathcal{E}(n)) - \frac{1}{8}\mathcal{B}(m - n, m - n) < \mathcal{E}(m)$, contradicting minimality. Thus reduction modulo $2L$ is injective on $\mathcal{M}$, giving the bound $|L/2L| = 2^g$.

For the stronger assertion, positively decompose $h = n - m$ and apply (92). Its left side is zero and every summand on the right is nonnegative, so each is zero. Since $\mathcal{B}(s, s) > 0$, every multiplicity a_s equals one; moreover $I_m(s) = 0$ and $\mathcal{B}(s, s') = 0$ for distinct selected generators, and expanding any partial sum gives energy change zero. The selected nonzero pairwise orthogonal vectors are linearly independent: if $\sum c_s s = 0$, pairing with one s' gives $c_{s'} \mathcal{B}(s', s') = 0$ in $\mathbb{R}^r$, so $c_{s'} = 0$. There are at most g of them, and the path of their partial sums has at most g edges. $\square$

Both bounds are sharp. With $L = \mathbb{Z}^g$, $\mathcal{B}(x, y) = 2 \sum_j x_j y_j$, $b(n) = -\sum_j n_j$ and $S = \{\pm e_1, \dots, \pm e_g\}$, one has

$$\mathcal{E}(n) = \sum_{j=1}^g n_j(n_j - 1), \quad \mathcal{M} = \{0, 1\}^g.$$

Each integer term $n_j(n_j - 1)$ is nonnegative and vanishes exactly at 0, 1. The set S positively generates L by selecting the signed coordinate directions of a vector, which are pairwise orthogonal. The minimum graph is the g -cube: it has 2^g vertices and diameter g , attained between opposite corners.

The parity estimate is the familiar midpoint mechanism of lattice Delaunay geometry, credited as such. Its combination with one constructive positively generating set gives the short zero-cost move certificate.

Corollary 19.4 (Complete finite enumeration after one minimum). *Starting from one certified minimum, breadth-first search through the zero-cost S -moves lists every minimizer. It visits at most 2^g vertices and requires at most $|S|2^g$ energy-difference tests.*

Proof. At each newly visited vertex m , test $m + s$ for every $s \in S$ and enqueue it only when $I_m(s) = 0$ and it has not been visited. All discovered vertices are minimizers. The proof of theorem 19.3 reaches any target from m using forward steps belonging to S , so symmetry of S is unnecessary. Each of at most 2^g vertices is processed once, using $|S|$ tests. If S is first replaced by $S \cup (-S)$, the bound uses the size of that enlarged set. $\square$

19.3 The finite initial theta polynomial

For β in the theta domain, define

$$\mu_\beta = \min_n \mathcal{E}_\beta(n), \quad \mathcal{M}_\beta = \{n : \mathcal{E}_\beta(n) = \mu_\beta\}, \quad P_\beta(Z) = \sum_{n \in \mathcal{M}_\beta} Z^n. \quad (93)$$

This is a nonzero Laurent polynomial with at most 2^g monomials. For z as in (78), direct coefficient extraction gives

$$\text{st}(t^{-\mu_\beta} \vartheta_\Pi(z)) = P_\beta(c). \quad (94)$$

Consequently $P_\beta(c) \neq 0$ implies $v(\vartheta_\Pi(z)) = \mu_\beta$. When $P_\beta(c) = 0$, the valuation is strictly larger or the sum is zero; no equality of valuations is asserted in that case. This is the higher-rank version of the initial-form mechanism underlying tropical theta geometry [18, §1.1 and Theorem 4.10]. That source uses a complete real-valued non-Archimedean field; the arbitrary-rank coefficient extraction here follows from strong Hahn summation. The finite certificate identifies all its active lattice terms without an unproved truncation radius. The quasi-periodicity identity gives $\mu_{\beta+\Pi m} = \mu_\beta - \pi(m) - \beta \cdot m$ and $\mathcal{M}_{\beta+\Pi m} = \mathcal{M}_\beta - m$, so $P_{\beta+\Pi m}(Z) = Z^{-m} P_\beta(Z)$.

20 Lifting smooth initial zeros

The formal implicit mechanism is classical, but a topological Newton-limit argument is insufficient at arbitrary rank. The following support argument extends lemma 2.2 to one unknown with infinitely many monomial parameters.

Lemma 20.1 (Implicit substitution with a Hahn certificate). *Let $\Sigma \subset \Gamma_{>0}$ be well ordered. For each $\rho \in \Sigma \cup \{0\}$, let $R_\rho(U, W)$ be an ordinary formal power series over $\mathbb{C}$ in finitely many variables $U = (U_1, \dots, U_d)$ and one variable W . Assume $R_0(0, 0) = 0$ and $\partial R_0 / \partial W(0, 0) = a_0 \neq 0$. For every tuple $u \in \mathfrak{m}_v^d$ there is a unique $w \in \mathfrak{m}_v$ satisfying*

$$R_0(u, w) + \sum_{\rho \in \Sigma} t^\rho R_\rho(u, w) = 0, \quad (95)$$

the equation being interpreted by strong summation. Moreover,

$$\text{supp}(w) \subset \left\langle \Sigma \cup \bigcup_j \text{supp}(u_j) \right\rangle \setminus \{0\}. \quad (96)$$

Proof. Introduce one independent formal variable Y_ρ for each $\rho \in \Sigma$, and use the ring of all coefficient families on finite-support multiindices in the variables U, Y . Infinite collections of monomials of one total degree are allowed; multiplication is still defined since a fixed multiindex has only finitely many decompositions. The formal equation $R_0(U, W) + \sum_\rho Y_\rho R_\rho(U, W) = 0$ has a unique formal solution $W = C(U, Y)$ with zero constant term: at each total degree the coefficient of the new unknown term is a_0 , while all other contributions depend on lower-degree terms, and for each multiindex these are finite coefficient calculations. Indeed, for a fixed multiindex ν , the summand $Y_\rho R_\rho(U, W)$ can contribute only if Y_ρ occurs in ν ; there are finitely many such ρ . After substituting a zero-constant W , its relevant powers are bounded by the total degree of ν , and each product coefficient has finitely many decompositions. This proves the needed local finiteness of the recursion despite the infinite variable set.

Now substitute $Y_\rho = t^\rho$ and $U_j = u_j$. The union of all exponent supports used is the positive well-ordered set in (96). Neumann’s lemma makes its additive monoid well ordered and gives only finitely many representations of any exponent. An exponent occurring as a generator has at most one label Y_ρ , and at most d labels from the supports of the U_j . Thus each finite multiset of generator exponents has only finitely many labelled realizations. This proves strong summability of the formal substitution, including all sums used in the equation. Its zero-constant value w has positive valuation and satisfies the support certificate. The same argument shows that the displayed equation is strongly summable for every candidate $w \in \mathfrak{m}_v$, before imposing that it vanish.

For uniqueness even among infinitesimals not known to have this certificate, let $w, w' \in \mathfrak{m}_v$ both satisfy (95). Subtract and use $W^k - (W')^k = (W - W') \sum_{\ell=0}^{k-1} W^{k-1-\ell} (W')^\ell$. The same support argument, now including the supports of w, w' , factors the difference as $(w - w')(a_0 + \eta) = 0$ with $\eta \in \mathfrak{m}_v$. The second factor is a unit, so $w = w'$. $\square$

Theorem 20.2 (Smooth initial zeros lift). *Let β be in the exact theta domain and $c \in (\mathbb{C}^\times)^g$. Suppose*

$$P_\beta(c) = 0, \quad \left(Z_k \frac{\partial P_\beta}{\partial Z_k} \right)(c) \neq 0$$

for some k . Choose arbitrary infinitesimals ε_j for $j \neq k$. There is a unique $\varepsilon_k \in \mathfrak{m}_v$ such that

$$\vartheta_\Pi(c_1 t^{\beta_1} (1 + \varepsilon_1), \dots, c_g t^{\beta_g} (1 + \varepsilon_g)) = 0. \quad (97)$$

With $\Sigma_\beta = \{\varepsilon_\beta(n) - \mu_\beta : n \notin \mathcal{M}_\beta\}$,

$$\text{supp}(\varepsilon_k) \subset \left\langle \Sigma_\beta \cup \bigcup_{j \neq k} \text{supp}(\varepsilon_j) \right\rangle \setminus \{0\}. \quad (98)$$

Thus the smooth residue zero gives a genuine Hahn and surcomplex zero with a specified support monoid, not just a formally suggested leading term.

Proof. Normalize the theta series by $t^{-\mu_\beta}$ and group by the exponent gap; every group is finite. At gap zero one obtains $R_0(U, W) = P_\beta(c_1(1 + U_1), \dots, c_k(1 + W), \dots, c_g(1 + U_g))$, with the k th U omitted. At each positive gap the same construction gives a finite Laurent polynomial in the $1 + U_j$ and $1 + W$, hence an ordinary formal power series at the origin. The set Σ_β is positive and well ordered. The assumed logarithmic derivative is precisely $\partial_W R_0(0, 0)$. Apply lemma 20.1; evaluation of the original theta family is licensed by theorem 18.1. $\square$

No assertion is made that every singular zero of P_β has the same unique lifting behaviour: at singular residue zeros, branching, ramification or additional obstructions require a separate analysis. The theorem does not construct an abelian variety or prove a higher-rank version of analytic uniformization.

Here is a singular initial zero where uniqueness fails. Take $g = 2$, $\alpha > 0$, $\Pi = \text{diag}(2\alpha, 2\alpha)$ and $\beta = (-\alpha, -\alpha)$. The minimum set is $\{0, 1\}^2$, so $P_\beta(Z) = (1 + Z_1)(1 + Z_2)$. At $c = (-1, -1)$ this polynomial and both logarithmic partial derivatives vanish. Strong Hahn multiplication gives

$$\vartheta_\Pi(z_1, z_2) = T(z_1)T(z_2), \quad T(z) = \sum_{n \in \mathbb{Z}} t^{\alpha n^2} z^n$$

on the domain. At $z_1 = -t^{-\alpha}$, the involution $n \mapsto 1 - n$ pairs opposite terms of $T(z_1) = \sum_n (-1)^n t^{\alpha n(n-1)}$, with no fixed integer. Hence $T(z_1) = 0$. Keeping this first argument fixed, every $z_2 = -t^{-\alpha}(1 + \varepsilon_2)$ with $\varepsilon_2 \in \mathfrak{m}_v$ is a zero of ϑ_Π . The smoothness hypothesis is therefore essential to the asserted uniqueness.

21 Surreal realization and explicit examples

21.1 Working inside the surreals without class-sized sums

For a set-sized additive subgroup $\Gamma \subset \mathbf{No}$, the normal-form map

$$\sum_{\gamma \in A} c_\gamma t^\gamma \mapsto \sum_{\gamma \in A} c_\gamma \omega^{-\gamma} \tag{99}$$

is an ordered Hahn-field realization for real coefficients and a surcomplex realization for complex coefficients; the sums are normal-form sums, not limits of partial sums [13]. Given a finite surreal exponent matrix and a finite tuple of surcomplex arguments, take the divisible subgroup generated by the matrix entries, the leading exponents, and the supports of all unit parts. This subgroup is a set, and all the constructions above take place inside its Hahn field, so the theorems apply pointwise throughout $\mathbf{No}[i]$ without a class-sized Hahn sum or a set containing all surreals. For $\Gamma = \mathbb{R}_{\text{lex}}^r$ a concrete realization sends $(a_1, \dots, a_r) \mapsto a_1 \omega^{r-1} + \dots + a_r$ in the exponent group. In rank two we write

$$\tau_1 = t^{(1,0)} = \omega^{-\omega}, \quad \tau_2 = t^{(0,1)} = \omega^{-1}; \quad 0 < \tau_1 < \tau_2^N \quad (N \in \mathbb{N}).$$

These are genuinely distinct surreal scales, not two ordinary small real parameters taken to zero together.

21.2 Irrational radicals: positive energies without a Hahn sum

Example 21.1. Take $L = \mathbb{Z}^2$ and $\pi(m, n) = ((m - \sqrt{2}n)^2, m^2 + n^2) \in \mathbb{R}_{\text{lex}}^2$. This is positive for every nonzero real vector, and its leading coordinate is positive at every nonzero lattice point. Nevertheless $\sum_{m,n} t^{\pi(m,n)}$ is

not Hahn-summable. Define $p_k, q_k \in \mathbb{Z}$ by $p_k + q_k\sqrt{2} = (1 + \sqrt{2})^k$. Then $(p_k - \sqrt{2}q_k)^2 = (3 - 2\sqrt{2})^k$ is strictly positive and strictly decreasing, so the full lexicographic exponents along (p_k, q_k) strictly decrease. The first radical is the irrational line $m = \sqrt{2}n$, precisely the flag obstruction. A finite numerical scan can never certify its absence.

21.3 Mixed drift: rational flags are not enough

Example 21.2. Consider $\pi(m, n) = (m^2, mn, n^2) \in \mathbb{R}_{\text{lex}}^3$. It is again positive for every nonzero real vector, and the successive restricted radicals are rational, but at the second stage the radical is still the n -axis and its mixed pairing with the m -axis is nonzero. The exponents $\pi(1, -k) = (1, -k, k^2)$ strictly decrease. This is why theorem 16.3 requires the *global* annihilation condition, not merely a sequence of positive semidefinite restrictions.

21.4 A legitimate sum of support length ω^2

Example 21.3. The sum $\sum_{m,n \in \mathbb{Z}} \tau_1^{m^2} \tau_2^{n^2}$ is strongly summable and has actual support of order type ω^2 . Its first-coordinate values are $0, 1, 4, 9, \dots$, and above each is an entire ω -sequence of second-coordinate values; positive coefficients prevent cancellation. No enumeration of its lattice terms makes ordinary finite partial sums converge to it in the rank-two valuation topology: every finite partial sum omits terms in the row $m = 0$, so its error has first valuation coordinate zero and never has valuation greater than $(1, 0)$. The Hahn identity remains valid. This separates summability from sequential convergence without appealing to any proper-class topology.

21.5 Positive generation with an irrational shear

Example 21.4. Let $\mathcal{B}(x, y) = (2x_1y_1, 2(x_2 - \sqrt{2}x_1)(y_2 - \sqrt{2}y_1))$. The radical flag is the rational x_2 -axis followed by zero, although the lower-level coefficients are irrational. A finite positively generating set is

$$S = \{(0, 1), (0, -1)\} \cup \{(\epsilon, k) : \epsilon \in \{-1, 1\}, -2 \leq k \leq 2\},$$

with twelve elements. In the construction of theorem 15.2, take upper lifts $(\pm 1, 0)$, lower generators $(0, \pm 1)$, and $M = 2$. For instance $(3, 7) = 3(1, 2) + (0, 1)$ and $(3, 1) = (1, 1) + 2(1, 0)$. In the first representation the mixed lower-level pairing is $2(2 - \sqrt{2}) > 0$; in the second, all pairings are positive at the first level. The program code `/02-multiscale-theta-verify.py` checks the allocation construction at all $1,681$ integer points of $[-20, 20]^2$, using exact signs in $\mathbb{Q}(\sqrt{2})$ and no floating point.

21.6 A coupled two-scale theta zero

Take the AN-admissible quadratic energy and linear term $\pi(m, n) = (m^2, n^2 + mn - m^2)$ and $b(m, n) = (-m, m - n)$. Then

$$\mathcal{E}(m, n) = (m(m - 1), n^2 + (m - 1)n - m^2 + m). \quad (100)$$

The first coordinate is minimized exactly at $m = 0, 1$. At $m = 0$ the second coordinate is $n(n - 1)$, minimized at $n = 0, 1$; at $m = 1$ it is n^2 , minimized only at $n = 0$. Hence $\mu = (0, 0)$, $\mathcal{M} = \{(0, 0), (0, 1), (1, 0)\}$ and $P(Z_1, Z_2) = 1 + Z_1 + Z_2$. The finite generating set $S = \{(0, \pm 1)\} \cup \{(\epsilon, k) : \epsilon = \pm 1, k = -1, 0, 1\}$ is obtained with $M = 1$; it certifies all three minima by eight inequalities at each point.

Choose the residue $(Z_1, Z_2) = (1, -2)$, a smooth zero since $Z_2 \partial_{Z_2} P = -2 \neq 0$. Fix the first unit to 1 and write the second residue-unit parameter as $\xi \in -2(1 + \mathfrak{m}_v)$. The theta equation is

$$\sum_{m,n \in \mathbb{Z}} \tau_1^{m(m-1)} \tau_2^{n^2+(m-1)n-m^2+m} \xi^n = 0, \quad (101)$$

and theorem 20.2 gives a unique such ξ . Let ξ_0 be its reduction for the coarsening that retains only the first-coordinate-zero terms. The support certificate gives $\xi_0 \in -2 + \tau_2 \mathbb{C}[[\tau_2]]$: every zero-first-coordinate generator has a positive integer second coordinate. Taking this reduction in the theta equation gives

$$\sum_{n \in \mathbb{Z}} \tau_2^{n(n-1)} \xi_0^n + \sum_{n \in \mathbb{Z}} \tau_2^{n^2} \xi_0^n = 0. \quad (102)$$

Indeed, reduction commutes with products and inverses of units, and terms from other m have first coordinate at least 2. This equation is for ξ_0 , not for the full ξ . Higher rows are essential: substituting ξ_0 into the full left-hand side of (101) leaves leading term $-2\tau_1^2 \tau_2^{-3}$, from $(m, n) = (-1, 1)$; the other terms with first coordinate 2 have second coordinate at least -2 . Exact recursion for ξ_0 , and the fact that $\xi - \xi_0$ has positive first-coordinate valuation, give

$$\xi = -2 + \frac{5}{2}\tau_2 - \frac{43}{8}\tau_2^2 + \frac{247}{16}\tau_2^3 - \frac{5955}{128}\tau_2^4 + \frac{36493}{256}\tau_2^5 + \rho_\xi, \quad v(\rho_\xi) \geq (0, 6). \quad (103)$$

The remainder includes the later terms of the zero row and all higher first-coordinate rows; (103) is not a claim that the whole solution lies in $\mathbb{C}((\tau_2))$. The program verifies the zero row through degree eight in exact rational arithmetic; the existence of all later rows follows from the support-controlled theorem, not from those finite checks. Under the surreal realization the original arguments of ϑ_Π are $z_1 = \tau_1^{-1}\tau_2$ and $z_2 = \tau_2^{-1}\xi$, so the example is an actual surcomplex zero at separated surreal scales, not a zero of an unrelated ordinary complex truncation.

21.7 Why no nontrivial theta family is all-surcomplex entire

Corollary 21.5. *For $g \geq 1$, no fixed symmetric surreal exponent matrix gives a theta family strongly summable at every monomial point of $(\mathbf{No}[i]^\times)^g$.*

Proof. A matrix failing the form conditions has no point in its theta domain. Otherwise $\pi_{11} = \mathcal{B}(e_1, e_1) > 0$. Choose a surreal γ larger than $N\pi_{11}$ for every ordinary positive integer N . At the valuation vector $\beta = (\gamma, 0, \dots, 0)$, the exponents along $-ne_1$ have successive differences $(n + \frac{1}{2})\pi_{11} - \gamma < 0$. They strictly decrease, contradicting Hahn summability. $\square$

The conclusion concerns the lattice series defining the function. It is the multivariable counterpart of the properness statement of theorem 13.1, and it accords with, but does not reprove or strengthen in full generality, the collection's all-scale entire-function restrictions. It does not contradict the unilateral example $\sum_{n \geq 0} t^{n^2} z^n$ of the foundations report [22], which is summable on all of the fixed field $\mathbb{C}((t^\mathbb{Q}))$, not at every surcomplex point.

21.8 A theorem checklist

Before applying the theta conclusions to a concrete expression, verify that the indexing lattice is finite-rank and set-sized; that the exponent matrix is symmetric; that successive restricted forms are positive semidefinite; that their radicals are lattice-rational and globally annihilate the corresponding coordinate form; and that

the final radical is zero. For the chosen argument, its leading-exponent linear forms must annihilate the corresponding radicals. Only then are the Hahn family, its unit perturbations, and its weighted derivatives licensed. For minimum computation, retain the positively generating set and the final finite inequalities. For an asserted zero, record its initial polynomial, its residue point, and the nonzero logarithmic derivative used for lifting. For an asserted ordinal support length, record positivity of coefficients or otherwise account for cancellations. Every one of these distinctions has a role in a proof or a counterexample above.

Part III

Computation, limitations, and audit

22 Exact computation, proof dependencies, and implementation

22.1 Finite checks included with this report

Three independent programs accompany the report, one from each source. None uses floating point, and none proves an infinite identity, a summability statement, surjectivity, or novelty. All three were rerun on copies when the report was assembled, on Python 3.14.4. The numerical and symbolic results agreed with the delivered records; the source-01 Python-version field and source-03 informational runtime differed. Each program writes its own evidence file, and two of them write to paths that collide inside this directory, so they must be run on a copy (appendix A.4).

Source 01: `code/check_identities.py`. This program uses only exact rational symbolic arithmetic. Its recorded output is in `data/verification.json` and `data/verification.log`. The included JSON and console record are the delivered run on Python 3.13.5 and SymPy 1.14.0. The assembly rerun on a copy used Python 3.14.4 and SymPy 1.14.0 and reproduced the values, with a different Python-version string. The delivered files are preserved here; subsequent review runs and their environments are recorded in `README.md`. The checks do not require a numerical value of q or a realization of an arbitrary ordered group.

For clarity, the coordinate coefficients used by the script are

$$[q^d]X_q(u) = \sum_{m|d} m(u^m + u^{-m} - 2), \quad d \geq 1, \quad (104)$$

$$[q^d]Y_q(u) = \sum_{m|d} \left(\frac{m(m-1)}{2} u^m - \frac{m(m+1)}{2} u^{-m} + m \right), \quad d \geq 1. \quad (105)$$

The degree-zero coefficients are $f(u)$ and $g(u)$. These identities come directly from the positive rational tails; they are also recorded in the contemporary Tate formulas of [6]. The script checks the curve equation, (25), and the inversion identity through q^5 as rational functions of the indeterminate u . It checks the discriminant/product identity through q^9 , computes j through q^5 , and checks inverse modular composition through J^6 .

The first verified coefficient lists are

$$\begin{aligned} a_4(q) &= -5q - 45q^2 - 140q^3 - 365q^4 - 630q^5 + O(q^6), \\ a_6(q) &= -q - 23q^2 - 154q^3 - 647q^4 - 1876q^5 + O(q^6), \\ \Delta(q) &= q - 24q^2 + 252q^3 - 1472q^4 + 4830q^5 + O(q^6). \end{aligned}$$

The formal-parameter sign is checked at $q = 0$:

$$-\frac{X_0(1+h)}{Y_0(1+h)} = \frac{h}{1+h} = h - h^2 + h^3 - h^4 + \cdots.$$

The test suite also checks finite index-shift instances of the theta functional equation, explicit decreasing exponent witnesses in a rank-two lexicographic group, and sample torsion-degree calculations.

These checks are normalization and transcription guards. They do not prove strong summability, arbitrary-rank surjectivity, an identity at all orders, or a proper-class transport theorem. Those conclusions depend on the mathematical arguments above.

Source 03: `code/03-tate-uniformization-verify.py`. This program uses only Python’s standard library. It represents bivariate polynomials as finite dictionaries with integer coefficients and works modulo total degree 13 in the recorded run. It independently constructs (41)–(42), computes the homogeneous inverse, and checks eight identities through total degree 12. There are $\binom{14}{2} = 91$ bivariate monomials of total degree at most 12, and all $8 \cdot 91 = 728$ coefficient comparisons pass. Its record is `data/03-tate-uniformization-verification.json`, which also lists the inverse coefficients through degree 12.

Identity, modulo degree 13	Result
$\mathcal{F}(UV, \mathcal{X}(U, V), \mathcal{Y}(U, V)) = 0$	pass
$\mathcal{X}(\mathcal{U}(x, y), \mathcal{V}(x, y)) = x$ and $\mathcal{Y}(\mathcal{U}(x, y), \mathcal{V}(x, y)) = y$	pass
$\mathcal{U}(\mathcal{X}(U, V), \mathcal{Y}(U, V)) = U$ and $\mathcal{V}(\mathcal{X}(U, V), \mathcal{Y}(U, V)) = V$	pass
$\mathcal{F}(\mathcal{U}(x, y)\mathcal{V}(x, y), x, y) = 0$	pass
$\mathcal{X}(V, U) = \mathcal{X}(U, V)$ and $\mathcal{Y}(V, U) = -\mathcal{X}(U, V) - \mathcal{Y}(U, V)$	pass

The last two are the node-chart form of inversion modulo the period. It also asserts that 12 divides $5\sigma_3(d) + 7\sigma_5(d)$ for the degrees used. The degree-four terms of the inverse are

$$\begin{aligned} [\mathcal{U}]_4 &= -66x^4 - 314x^3y - 520x^2y^2 - 360xy^3 - 82y^4, \\ [\mathcal{V}]_4 &= 6x^4 + 26x^3y + 68x^2y^2 + 32xy^3 - 82y^4, \\ [\mathcal{Q}]_4 &= -5x^4 - 25x^3y - 48x^2y^2 - 46xy^3 - 23y^4, \end{aligned}$$

where brackets denote the homogeneous component of exactly that degree. The suite does not check the secant identities, the q -expansions or any summability statement. It overlaps the first suite only in the curve equation and inversion, and in a different form; the two are complementary.

Source 02: `code/02-multiscale-theta-verify.py`. This program uses only the standard library, with exact integers, fractions, and exact signs in $\mathbb{Q}(\sqrt{2})$ obtained by integer squaring with sign cases, so near-zero floating-point errors cannot hide negative pairings. Its record is `data/02-multiscale-theta-verification.json`; `data/02-multiscale-theta-verification.txt` is a byte-identical copy delivered with the source, which the program does not regenerate.

Exact check	Cases
Bounded allocation identities and residual saturation (lemma 15.4)	5,555
Positive decompositions for the irrational shear (example 21.4)	1,681
Compatible linear terms for the coupled rational energy (section 21.6)	441
Pell identities supplying descending-support witnesses (example 21.1)	12
Zero-row root expansion (102)	through degree 8

For the 441 linear terms, the program compares finite-certificate descent and zero-cost graph search with an independent exact quotientwise quadratic minimization, not with an arbitrary finite box, and verifies the graph-diameter bound in those cases. The largest observed descent length was 35 from the prescribed starting point; this is an experimental fact for the specified cases, not a bound asserted for the algorithm. The formal zero-row computations use exact fractions and truncated power-series arithmetic; coefficients outside the checked range are not claimed to have been computationally verified. These are finite example checks; they do not machine-prove the positive-generation, summability, ordinal or lifting theorems.

22.2 What an exact symbolic implementation must know

To represent a concrete instance, a symbolic system needs finite expressions for the input Hahn series or a reliable lazy coefficient interface, exact leading exponents and coefficients, and a specified ordered exponent group. The domain test additionally requires deciding whether $|\beta| \leq N\alpha$ for some ordinary N . Comparisons with individual $N\alpha$ are not, by themselves, a terminating decision procedure for that existential statement.

For a lexicographically ordered free finite-rank group, the test is finite. Inspect the first nonzero coordinate of α . Values whose first nonzero coordinate occurs earlier are outside H_α ; values supported at that coordinate or later are inside it. In rational coordinates the finite normalization in lemma 2.3 is then performed by ordinary floor arithmetic at that coordinate, with the lower coordinates determining boundary cases. For a general group, the needed principal-convex-subgroup operation must be part of the representation contract.

For normalized admissible u , an evaluator can return the two isolated rational terms, a lazy positive-tail object, the support certificate (21), and the tail bound (22). That is an exact expression with an explicit precision policy. A finite truncation is not an exact elliptic point unless the discarded tail is retained symbolically. Testing the curve equation numerically after truncation is not a replacement for the exact identity.

The torsion-degree calculation requires membership in $n\Gamma$ and finite cyclic order in $\Gamma/n\Gamma$, which is extra group data. When $\Gamma \cong \mathbb{Z}^r$, (73) supplies a direct algorithm. In a black-box ordered group, it need not be decidable from order comparisons alone.

For the inverse direction, the second route is explicit: inspect the coordinate valuations and residues of a point to select the node, smooth or infinity chart, and compute the formal coefficients of that chart's inverse by finite homogeneous recursion. There is an important distinction between such a construction and an algorithm on all surreal numbers. Arbitrary surreal exponents do not come with a universal effective comparison or boundedness oracle; even in an effective subgroup, an unbounded search for an integer N only semidecides $|\gamma| \leq N\alpha$ unless a decision procedure for convex-subgroup membership is available, and the theorems do not promise one. Formal total-degree truncation is fully effective but must be described honestly: for inputs of least positive valuation ϵ , omitting terms of degree above N bounds the omitted valuation below by $(N+1)\epsilon$, and at higher rank this may never exceed a requested cutoff. The input Hahn series themselves may need infinitely many coefficients below a cutoff; support finiteness at each exponent is not a finite algorithm for an arbitrary initial segment.

For Part II an implementation should keep four objects separate: an ordered exponent representation, a rational radical flag, a certificate for positive generation, and a support expression for the resulting Hahn series. A successful finite numerical sample replaces none of them. For real-algebraic matrix entries the flag computation can use exact linear algebra and real-algebraic sign tests; rationality of a subspace is a distinct step and cannot be inferred from a numerical nullspace. In a fixed algebraic number field, expand coordinates in a rational basis to solve the required rational linear equations. For arbitrary symbolic surreal constants these operations require explicit comparison and identity certificates from the chosen representation. After the form and valuation domain are certified, a finite positively generating set makes global-minimum

testing finite: a descent trace plus the final list of inequalities certifies a minimum; a zero-cost graph trace, the completeness theorem and the 2^8 vertex bound certify the initial Laurent polynomial; and a nonzero initial derivative with the support monoid of (98) certifies a smooth lifted branch. These are exact logical certificates; the report does not implement a general-purpose surreal computer algebra system.

22.3 A dependency-aware formalization plan

The repository already has a substantial Hahn-support layer, including positive-support evaluation and regrouping [20]. It should be reused. The new proof obligations naturally separate as follows.

Layer	Precise obligations
Ordered groups	Principal convex subgroup, H_{α}^- , finite annulus normalization, exact theta-leading-exponent obstruction.
Hahn calculus	Strong coordinate tails, exponent embedding functoriality, coarse coefficient field, unrestricted coarse formal evaluation.
Formal algebra	Tate coefficients, local parameter B and its inverse, projective identity chart, formal-group identities.
Rank-one bridge	Completeness of $K_{H_{\alpha}}$ under its rank-one coarsening and the full classical Tate theorem with explicit hypotheses.
All-rank geometry	Taylor/Hahn agreement, group law in regular charts, coarse reduction diagram, surjectivity and exact kernel.
Arithmetic	Finite coset decomposition of a Hahn extension, character Galois action, primitive torsion-point stabilizer.
Second route	Finite-variable evaluation and the formal implicit function; the universal integral Tate identities (curve and cleared secant identities) as formal identities; the unimodular node chart and its integral inverse; the three-chart surjectivity; the generic-pair lemma and the group law; then the support and valuation refinements. The node isometry needs only finite-dimensional linear algebra and formal difference factorization, so it can precede the global elliptic-curve theorem.
Part II	Positive generation over a finite-dimensional real space, a lattice and a finite tuple of symmetric bilinear forms, with the radical flag and global annihilation as explicit hypotheses; the allocation lemma (a statement about natural numbers); the energy decomposition and the midpoint bound (finite algebra); then well-ordering, the discrete-image lemma and finite fibres; finally strongly summable Hahn families with a typed positive-support monoid for the unit and implicit results. The ordinal theorem must be stated with support order type, not birthday.

This is a plan, not a supplied Lean formalization. In particular, proving the support lemmas while postulating classical Tate uniformization would only establish a theorem conditional on that input. A full formal audit must distinguish that situation from a checked construction of the rank-one theorem. The second route replaces that input by Tate’s universal formal identities, which a formalization must either prove or state as explicit hypotheses. The abstract Hahn development should precede the bridge to the actual surreal field, which requires normal-form evaluation, preservation of strong sums, and coherence under enlargement of

the exponent workspace; the repository’s formalization ledger distinguishes such prerequisites from their actual-surreal instantiations. Nothing here is represented as an already compiled repository theorem. No Lean source is supplied, no library declaration names are assumed, and no unchecked code skeleton is included as though it were a machine-verified proof.

23 Limitations and further mathematical questions

The present results settle the internal domain and surjectivity questions posed in section 1. They leave several broader tasks genuinely separate.

Beyond negative valuation of j . The Tate parameter $v(q) > 0$ produces $v(j) < 0$. Curves with nonnegative valuation of j require a different treatment, such as formal groups around a good-reduction model and a residue elliptic curve. The surcomplex field’s algebraic closedness does not by itself provide a single useful transcendental parametrization of that different regime. Constant (good-reduction) abelian varieties over $\mathbf{No}[i]$ are treated by formal groups in Part VI of the differential-equations report of this collection (`diff:aut:prop:abeliansplit`, `diff:aut:thm:abelianimage`), including their logarithmic-derivative image for every normalized derivation. That treatment concerns constant varieties and does not cover Hahn fields with a coarse valuation.

Higher-dimensional totally degenerate abelian varieties. A natural extension replaces $q^{\mathbb{Z}}$ by a finite-rank multiplicative period lattice and the quadratic theta exponent by a positive polarization form. One would need an exact ordered-group condition for all lattice directions and a replacement for the single principal convex subgroup H_α . The support proof in Part I suggests the right obstruction, but it is not a proof of a general multivariable uniformization theorem. Source 03 frames the same gap: for several periods, support admissibility is likely governed by a boundedness region associated with their valuation pairing rather than by one principal convex subgroup, and an adequate theorem must establish finite support fibres for multidimensional theta sums and prove that the admissible region still reaches the desired abelian-variety points; the one-node formal inverse of section 8 does not by itself prove this.

Part II (source 02) now supplies the theta-summability half of this programme, under its own hypotheses: monomial periods $t^{\pi_{ab}}$, coefficients in $\mathbb{C}$, and a divisible Γ . For those periods the exact ordered-group condition for all lattice directions is the radical-flag criterion of theorem 16.3 and corollary 16.4; it gives the exact domain of ϑ_{Π} , with finite support fibres and unit robustness (theorem 18.1), and it replaces the single subgroup H_α by the flag conditions, reducing to H_α for $g = 1$ (corollary 18.3). Finite certificates for theta minima and the lifting of smooth initial zeros are also proved. The paragraph remains open. Not proved are: a multivariable domain theorem for periods with Hahn tails, which theorem 3.1 handles only for $g = 1$; any nondivisible version of Part II; the analogue of the coordinate maps and the group structure; surjectivity onto the points of a totally degenerate abelian variety, or any uniformization theorem at all; the lifting of singular initial zeros; and geometric constructions of higher-rank theta line bundles.

Questions about positive generation and minimization. Efficient bounds for the size of a positively generating set, beyond the existence bound (82), and the bit complexity of the minimization algorithm for algebraic inputs remain open; well-ordering gives termination of each descent but no uniform bound on its length.

Comparison with logarithmic uniformization. A precise functorial comparison between the explicit Hahn map here and the bounded-monodromy logarithmic Tate quotient of Molcho–Wise would identify exactly which parts of the construction are realizations of existing formal and logarithmic geometry. The coincidence of the one-loop bounded group with H_α is already visible in [16]; a full comparison of structures is not proved here. This is the most important remaining priority check for the proposed refinements of Part I.

Intrinsic higher-rank analytic geometry. The algebraic quotient $H_\alpha/\mathbb{Z}\alpha$ retains infinitesimal value directions discarded by its Archimedean quotient. Relating it precisely to Hahn analytification, adic spaces, or a suitable skeleton is a separate geometric theorem. The exact sequence (64) is a starting invariant, not a substitute for that theorem.

Intrinsic invariants beyond the chosen Weierstrass chart. The exact formulas of section 9 concern the specified coordinate model. It would be useful to identify which support-monoid invariants and valutive distances admit a model-independent formulation, and which necessarily depend on the chosen Tate coordinates. Calling the current affine distance canonical on every elliptic model would be an unsupported strengthening, and it is not claimed.

Functions outside the literal Hahn domain. We proved that the displayed bilateral families fail strong summability outside U_q . We also proved an algebraic obstruction: when $H_\alpha \neq \Gamma$, no group homomorphism $K^\times \rightarrow E_q(K)$ extending Φ_q has kernel $q^\mathbb{Z}$, and for any homomorphic extension Ψ the excess kernel satisfies $\ker \Psi/q^\mathbb{Z} \simeq \Gamma/H_\alpha$ (theorem 10.5). We did not prove that every possible resummation, noncanonical assignment, or change of analytic category is impossible: an extension that gives up either the canonical values on U_q or the period kernel is not excluded. Any proposed extension must specify its summation rule and which identities it preserves; it cannot call the original families strongly Hahn-summable when their leading supports are decreasing. The same holds for the theta families of Part II outside their domain.

Independent verification and priority. For the first route, the most important independent proof checks are the joint-family interchange in lemma 6.3 and the use of regular projective charts in the group law. The second route proves theorem 1.2 without either, but those two steps remain load-bearing for corollary 7.2, (33) and proposition 7.1. For the second route, the review targets are the Hahn specialization of the secant identities on the wide annulus and the exhaustiveness of the three charts; for Part II, the induction in theorem 15.2 and the global-radical step of theorem 16.3. None of the three source searches was an exhaustive search of valuation theory, formal or logarithmic models, higher-rank lattice geometry, or the model theory of analytic structures. Finding an earlier theorem with these conclusions would change the priority assessment, not the support calculations. The report should be treated as a research draft with explicit proofs, not as an independently certified publication.

24 Conclusion

At arbitrary valuation rank, the correct multiplicative source of the Hahn Tate map is neither automatically all of $K^\times$ nor merely one rank-one coefficient subfield. It is

$$U_q = \{u : v(u) \in H_{v(q)}\}.$$

The restriction is forced by the bilateral support geometry, yet it loses no elliptic-curve points. Classical rank-one uniformization reaches the coarse residue curve; formal evaluation at coarse infinitesimals recovers

every remaining point. This proves

$$U_q/q^{\mathbb{Z}} \simeq E_q(K)$$

without claiming convergence at scales where the series do not converge.

The same analysis yields a canonical valuation quotient, exact behavior under scalar extension, support-preserving modular inversion, and an explicit torsion ramification law. In particular, the construction gives a coherent theory of multiplicatively uniformized elliptic curves over $\mathbf{No}[i]$ while retaining its genuinely non-Archimedean scale structure.

The second proof gives the constructive reason. The integral formal chart $(U, V) \mapsto (X(U, V), \mathcal{Y}(U, V))$ with $UV = q$ has unimodular linear part. It recovers the two branches from any nodal point, preserves the generated positive support monoid in both directions, and is a valuation isometry on the whole infinitesimal bidisk. Together with two elementary charts it gives global surjectivity at every value-group rank and over every coefficient field of characteristic zero, and its exact period-sensitivity and pairwise-distance formulas survive arbitrary leading-coefficient cancellation. The period-bounded quotient itself belongs to an established landscape, that of bounded monodromy in logarithmic Picard theory; what is proposed here is its support-explicit, valuation-exact realization in Hahn and surcomplex arithmetic.

For several monomial periods, Part II settles the summability side. Finite positive generation of admissible higher-rank lattices, which answers Question 11.14 of Amini–Nicolussi in its printed form, turns every theta minimum into a finite certificate; the radical flag decides exactly which quadratic lattice families are Hahn sums, with support order type ω^s ; and smooth initial zeros lift with explicit support monoids. These are support and finite-comparison foundations for a higher-dimensional theory, not that theory: no abelian variety is uniformized here, and no rank-one analytic geometry is silently imported into an arbitrary-rank setting.

A Source and verification audit

A.1 What is attributed, and what is proved here

The principal classical inputs of Part I are Tate’s complete rank-one uniformization theorem, the formal Tate curve and its discriminant, Jacobi’s product, algebraic closedness of divisible Hahn fields over $\mathbb{C}$, and the standard j -classification of elliptic curves. The torsion list is classical; its exhaustiveness here is derived directly from the uniformization kernel over the divisible Hahn extension. The second route replaces the first input by Tate’s universal integral identities (the curve equation and the cleared secant identities) and Tate’s generic-pair lemma, together with the finite algebraic group law on a nonsingular Weierstrass cubic. The bounded-period quotient and its value sequence are credited to the logarithmic Picard theory of Molcho–Wise as a precedent. Part II credits the higher-rank inner products, admissible lattices, Definition 11.10, Lemma 11.11 and the rank-one acute-cone argument to Amini–Nicolussi; rank-one theta functions and their tropicalization to Foster–Rabinoff–Shokrieh–Soto; convergent higher-rank Hahn subrings to Joswig–Smith; the midpoint parity argument to lattice Delaunay geometry; and initial forms and the formal implicit theorem to the classical literature. All of these are cited rather than presented as discoveries.

The proofs supplied here establish the exact strong-summability domains, the support and precision certificates, unrestricted formal evaluation across a convex coarsening, its agreement with the literal Tate sums, the resulting arbitrary-rank surjective quotient (twice), the node-chart inverse and its support and valuation identities, the fine filtration and the extension obstruction, the indicated consequences, and the results of Part II. The finite computations independently check selected identities and signs. They neither replace the classical inputs nor certify the all-rank proofs.

A.2 Repository provenance

The reference revision identified during the audit of source 01 was

`a3124af79f66b8b9c196d76b4cbc5ac3938907c4.`

The rank-one README was fetched explicitly at that revision. The root README, document map, formalization ledger, and spectral README were read from `main` during the same audit. Source 02 used the same revision `a3124af`; it inspected the documentation map and the spectral-theory report, with targeted searches for theta and elliptic material. Source 03 used revision

`0097304c7ae9d5de46d2ea342e2494f8fc126303,`

which predates the placement of source 01 in the repository; it consulted the root README, the documentation map and the opening scope and implementation discussion of the formalization ledger. Their statements about the repository have been corrected where the repository has since changed (section 1.3). No claim is made that all manuscripts in the repository were read line by line, and none of the sources treats the repository’s unrefereed manuscripts as mathematical axioms. The negative result of a keyword search is treated as a navigation aid, not a formal proof that a topic occurs nowhere.

The report does not claim that its theorems are part of the checked Lean build. Their implementation status is recorded in the repository’s formalization ledger; the supplied programs check finite examples.

A.3 Literature-search scope

Source 01’s audit checked Tate’s original complete-field formulation and subsequent treatments; the explicit formulas in the 2026 Eterović–Vermeulen paper; rank-one skeleton statements in Baker–Payne–Rabinoff; and the existence and scope of higher-rank Hahn analytification in Foster–Ranganathan. Searches combined Tate uniformization, Hahn fields, higher-rank valuations, and theta series. No matching exact-domain/all-points statement was identified in that limited search. Source 03’s search, which combined Tate uniformization, Hahn fields, surreal numbers, higher-rank valuations and bounded monodromy, found the Molcho–Wise precedent for the bounded-period quotient; it located no matching statement of the combined support-and-valuation refinements, and it did not establish that every equivalent formulation in the formal and logarithmic geometry literature has been excluded. Source 02 checked the named question both in the arXiv preprint and in the published 2025 article, where the numbering is Question 11.14 with Definition 11.10 and Lemma 11.11; the question, its subsection 11.5.3 and page 1181 were confirmed again against the published article when this report was assembled. Source 02’s closest theta source assumes a real-valued non-Archimedean absolute value and is not a source for the arbitrary-rank converse; its higher-rank Hahn source distinguishes formal fields from convergent subrings, and the construction here uses the former. None of these is a claim that such results are absent from all published or unpublished literature; research priority remains subject to specialist review.

A.4 Reproduction

The source uses standard TeX Live packages and requires no external figures, bibliography download, or font installation. From the report directory, run

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Recompiling the article does not execute the verification code. The three programs write their evidence files, and two of them would overwrite each other's: `code/check_identities.py` and `code/02-multiscale-theta-verify.py` both write `data/verification.json` relative to the directory above their own, and `code/03-tate-uniformization-verify.py` writes the file named by `--output` (default `verification.json` in the working directory). Run them on a copy, for example as `code/verify.py` in a scratch directory with its own `data/`, and compare the result with `data/verification.json`, `data/02-multiscale-theta-verification.json` and `data/03-tate-uniformization-verification.json` respectively. The first program requires SymPy (`data/requirements.txt`); the other two need only the Python standard library. The recorded outputs are included so that the finite results remain inspectable without running Python. The build scripts `code/02-multiscale-theta-build.sh` and `code/03-tate-uniformization-build.*` are kept as delivered; they build their source manuscripts in their original layouts and do not build this report.

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